

तो चलो शुरू करते हैं एक ऐसा chapter जिससे बच्चों को सबसे ज्यादा डर लगता है... bcz this chapter says

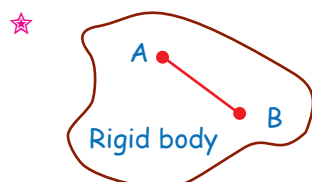


Jo Dar Gaya,
Samjho Mar
Gaya

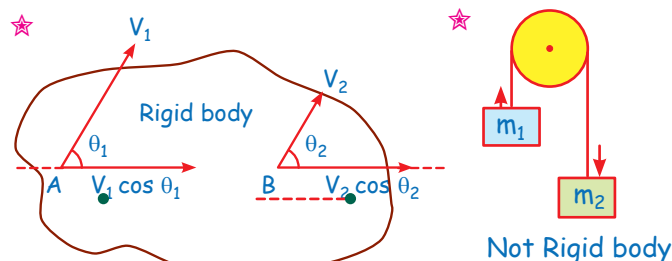
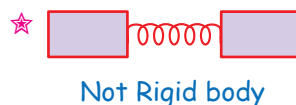
भाई सच बोलू तो ये chapter बहुत ही मजेदार है बस इससे डरना नहीं है... मैंने youtube channel JEE Wallah पर rotation की 12 hr non-stop एक class ले रखी है... For more better understanding you can watch it (or search saleem sir rotation motion on youtube) Mains और Advance दोनों के pov से ये बहुत important chapter है और जितना content मैं यहाँ cover कर रहा हूँ that sufficient for you to build your concept... almost हर प्रकार के question मैंने यहाँ cover कर दिए हैं चलो start करते हैं।

RIGID BODY

A rigid body is an object where the gap between any two points on the body do not change w.r.t time.

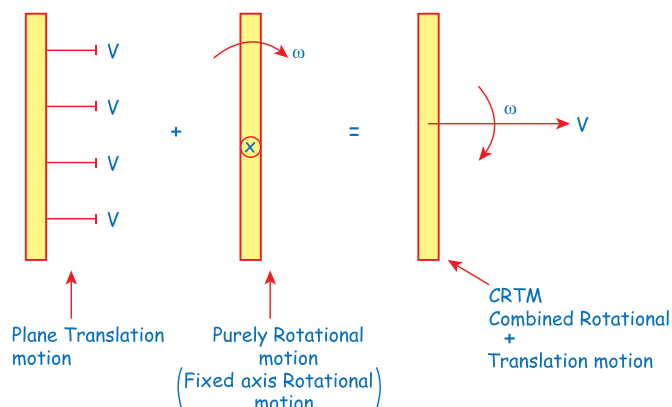


Here, Distance AB not changing with time.

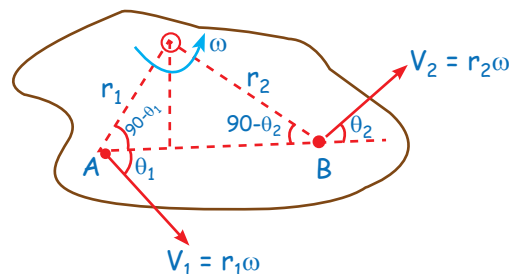


Component of velocity along line joining A & B is same.
In a Rigid Body $\rightarrow V_2 \cos \theta_2 = V_1 \cos \theta_1$

Types of Motion of a Rigid Body

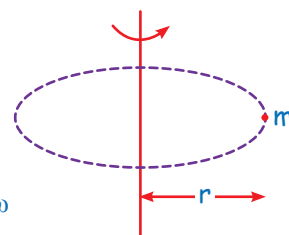


1. **Pure translation:** Each points have same velocity.
2. **Fixed axis rotation:** Body rotates about a fixed axis.
3. **CRTM:** Body rotates as well as translate.



$$V_1 \cos \theta_1 = V_2 \cos \theta_2$$

$$\begin{aligned} \omega_{B/A} &= \frac{V_2 \sin \theta_2 + V_1 \sin \theta_1}{AB} \\ &= \frac{V_2 \sin \theta_2 + V_1 \sin \theta_1}{r_1 \sin \theta_1 + r_2 \sin \theta_2} \\ &= \frac{r_2 \omega \sin \theta_2 + r_1 \omega \sin \theta_1}{r_1 \sin \theta_1 + r_2 \sin \theta_2} = \omega \end{aligned}$$



Hence, we can say that in rigid body ω of any point about any general point is same.

Rotation शुरू करने से पहले हमें एक article पढ़ना पड़ेगा Moment of inertia (MOI) जिसने आपको काफी maths देखने को मिलेगी.... जैसे पहले inertia होता था (tendency of an object to resist changes in its state of motion) उसी प्रकार rotation में property to oppose to change in rotation motion is called MOI.



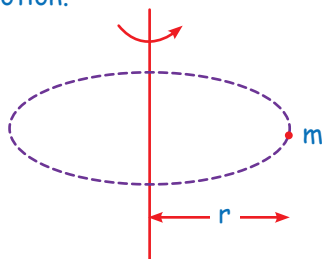
MOMENT OF INERTIA (MOI)

It is a property of a body by virtue of which it opposes to change in its rotation motion.

For a point mass

MOI of a point mass about an axis is calculated as

$I = mr^2$ where r is the $\perp^{\text{ar}}$ distance from axis.



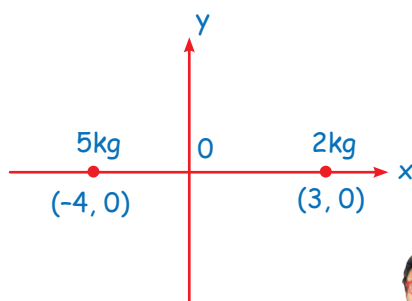
For system of discrete particle (point mass)

$$I = \sum m_i r_i^2$$

For a continuous body

$$I = \int dm \cdot x^2$$

Q. Find moment of inertia of following system of particles about y-axis.

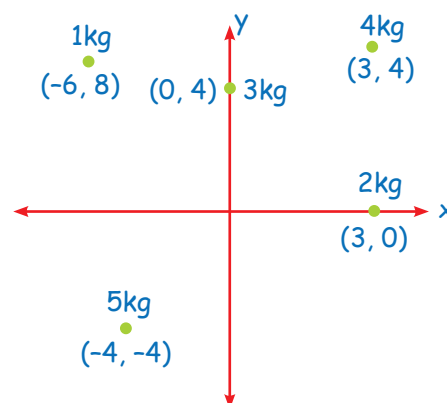


Sol. $I_y = 2 \times 3^2 + 5 \times 4^2$
 $= 18 + 80 = 98$



अब यहाँ 5 kg के MOI के लिए minus मत लगा देना हम MOI निकाल रहे हैं ना की centre of mass

Q. Find moment of inertia of following system of particles about x-axis, y-axis and z-axis.



Sol. $I_x = 0 + 4 \times 4^2 + 3 \times 4^2 + 5 \times 4^2 + 1 \times 8^2 = 256$
 $I_y = 2 \times 3^2 + 4 \times 3^2 + 0 + 5 \times 4^2 + 1 \times 6^2 = 170$
 $I_z = 2 \times 3^2 + 4 \times 5^2 + 3 \times 4^2 + 1 \times 10^2 + 5 \times (4\sqrt{2})^2$
 $= 18 + 100 + 48 + 100 + 160$
 $= 426$

Also, $I_z = I_x + I_y = 256 + 170 = 426$

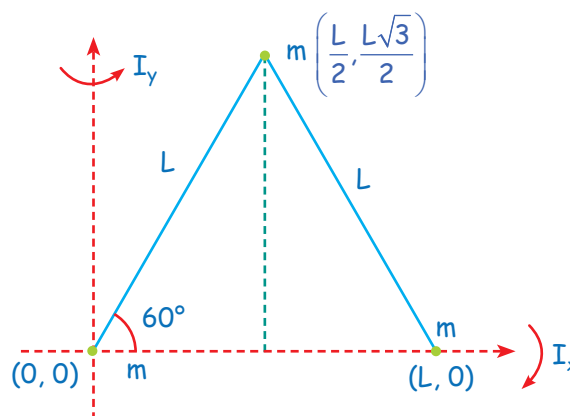
अरे ये देखा तुमने.... देखा की नहीं कि $I_z = I_x + I_y$ आया....



क्यों आया क्या हमेशा आएगा.... ?

JEE Mains Favorite Question

Q. Three particles of mass m each are kept at an equilateral triangle of side L . Find MOI about x, y and z axis.

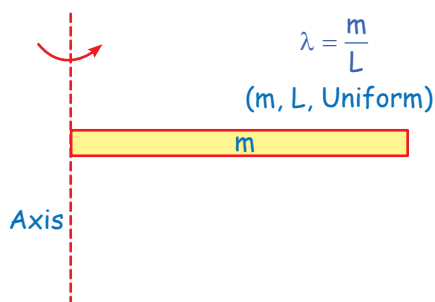


Sol. $I_x = 0 + 0 + m \left(\frac{L\sqrt{3}}{2} \right)^2 = \frac{3mL^2}{4}$

$I_y = 0 + mL^2 + m \left(\frac{L}{2} \right)^2 = \frac{5mL^2}{4}$

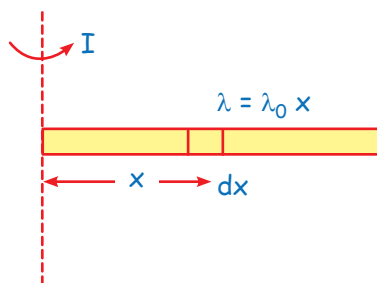
$I_z = 0 + mL^2 + m \left[\left(\frac{L}{2} \right)^2 + \left(\frac{L\sqrt{3}}{2} \right)^2 \right] = 2mL^2$

- Q.** Find moment of inertia of uniform rod about axis as shown in figure passing through one of its end ($\lambda = \frac{m}{L}$ = linear mass density).



Sol. $dm = \lambda dx$
 $dI = dm x^2$
 $\int dI = \int dm \cdot x^2$
 $I = \int_0^L \lambda dx \cdot x^2$
 $= \lambda \int_0^L x^2 \cdot dx$
 $I = \frac{\lambda L^3}{3} = \frac{m L^3}{L^3} = \boxed{\frac{mL^2}{3}}$

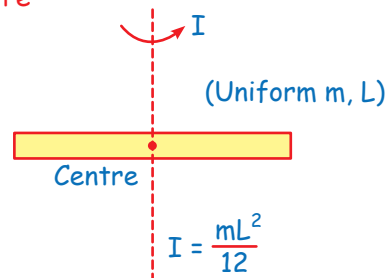
- Q.** Find moment of inertia of rod whose linear mass density is given by $\lambda = \lambda_0 x$ about axis passing through one end as shown in figure.



Sol. $dI = dm x^2$
 $\int dI = \int dm \cdot x^2$

$I = \int_0^L \lambda dx \cdot x^2$
 $I = \int_0^L \lambda_0 x \cdot x^2 dx$
 $= \boxed{\frac{\lambda_0 L^4}{4}}$

- Q.** Find moment of inertia of uniform rod about axis as shown in figure passing through its centre



- Sol.** We can consider the above rod as two rods of mass $\frac{m}{2}$ and length $\frac{L}{2}$.

$I = \left(\frac{m}{2} \right) \left(\frac{L}{2} \right)^2 \times 2 = \frac{mL^2}{12}$

भाई इसे integration से भी कर सकते हैं... तुम लोग try करो

Hint: $I = \int_{-L/2}^{+L/2} dm \cdot x^2 = \int_{-L/2}^{L/2} \lambda dx \cdot x^2$



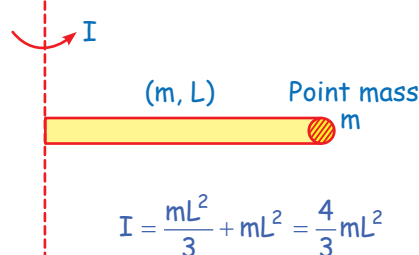
Moment of inertia of rod about the given axis will be 0.



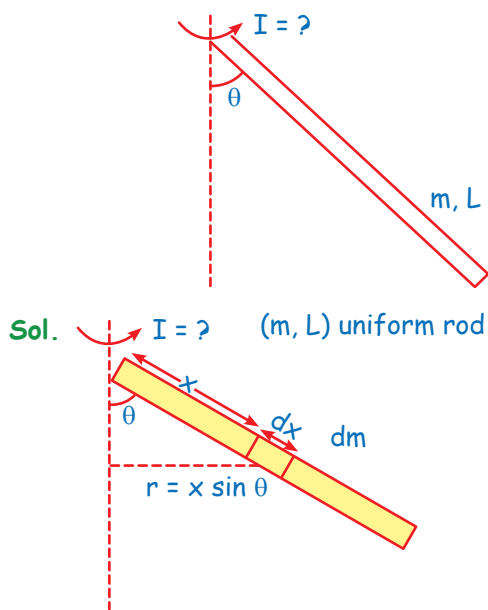
Axis is perpendicular to plane

Moment of inertia in the given case will be $I = \frac{mL^2}{3}$

- Q.** Find MOI for following where a particle of mass m is attached at the end of rod of mass m and length L .



Q. A rod makes an angle θ with the axis of rotation. Find MOI about the given axis.



Sol. $I = ?$ (m, L) uniform rod

$$dI = dm \cdot r^2 \quad (r: \text{perpendicular distance})$$

$$\int dI = \int \lambda dx (x \sin \theta)^2$$

$$= \lambda \sin^2 \theta \int_0^L x^2 dx$$

$$= \frac{m}{L} \sin^2 \theta \times \frac{L^3}{3}$$

$$= \frac{ML^2 \sin^2 \theta}{3}$$

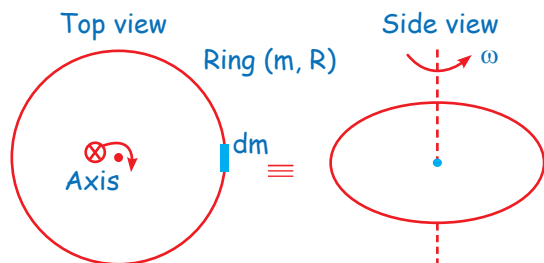
Method-2: SKC

अगर rod के सारे mass को मैं नीचे उतार लूँ तो $L \sin \theta$ length की एक rod बन जाएगी

$$\text{hence } I = \frac{m(L \sin \theta)^2}{3}$$



★ MOI of a uniform ring (m, R) about axis as shown in figure.



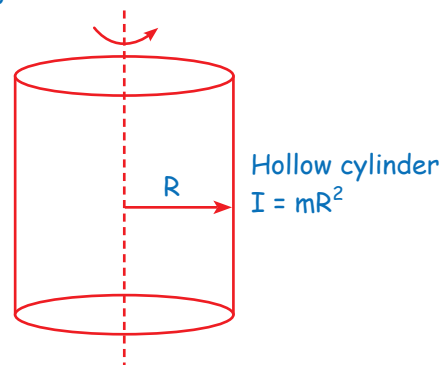
$$I = \int (dm)R^2 = R^2 \int dm = mR^2$$

$$I = mR^2$$

अगर uniform ring की जगह m mass का non-uniform ring होता या m mass का arc/semi ring होता तो भी $I = mR^2$ आता सोचो सोचो

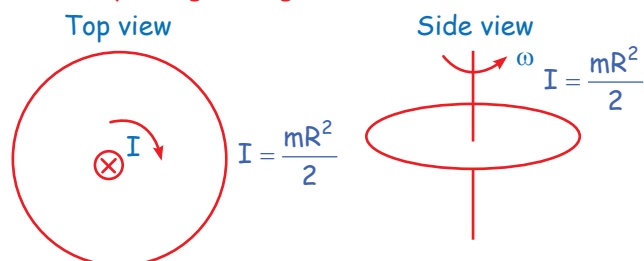


★ MOI of Hollow cylinder (m, R) about axis as shown in figure.



Hollow cylinder
 $I = mR^2$

Q. Find MOI of uniform Disc of mass m and radius R about axis perpendicular to the plane of disc and passing through centre.



Sol. Surface mass density $\sigma = \frac{m}{\pi R^2}$

Area of thin strip of width dr

$$dA = 2\pi r \cdot dr$$

Mass of element = $dm = \sigma dA$

$$dm = \sigma dA$$

$$MOI = dI = dm \cdot r^2$$

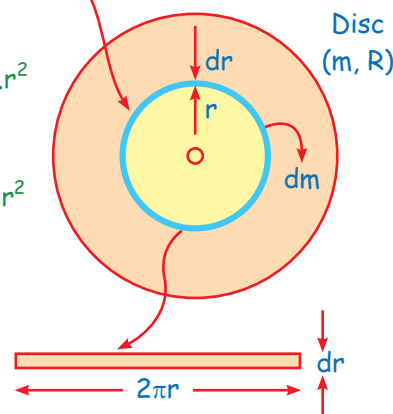
$$dI = \sigma dA \cdot r^2$$

$$\int dI = \int_0^R \sigma 2\pi r \cdot dr r^2$$

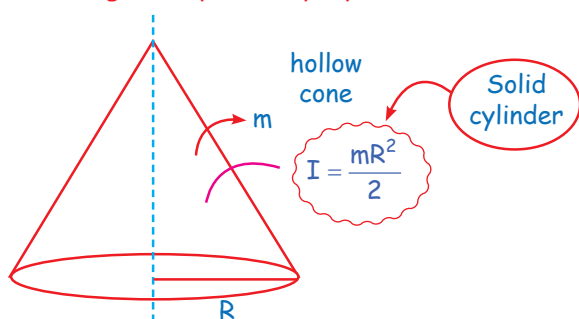
$$= \sigma 2\pi \int_0^R r^3 dr$$

$$= \frac{m}{\pi R^2} \cdot 2\pi \cdot \frac{R^4}{4}$$

$$= \frac{mR^2}{2}$$

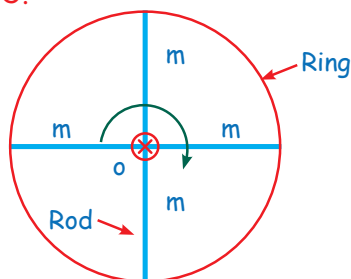


- Q. Find MOI of hollow cone about an axis passing through its apex and perpendicular to its base.



Sol. hollow cone & solid cylinder have same MOI.

- Q. Four rods of length R mass m each and a ring of mass m and radius R are arranged as shown. Find MOI about axis as shown in figure about O.



Sol. $MOI = I_0 = \left(\frac{mR^2}{3} \times 4 \right) + mR^2 = \frac{7mR^2}{3}$

Parallel Axis Theorem

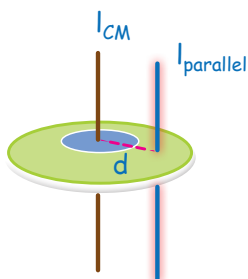
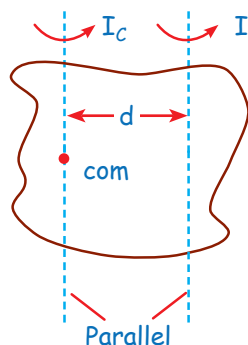
MOI (I_c) about an axis passing through COM is related to MOI (I) about an axis parallel to axis passing through COM and at a distance d from it by following relation.

$$I = I_c + md^2$$

where I and I_c Axis should be parallel

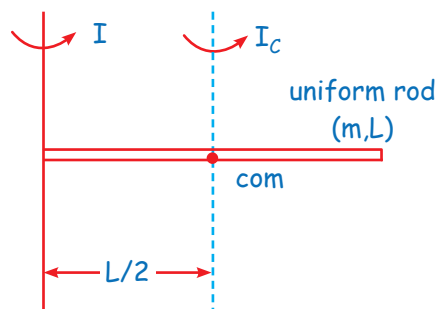
d is shortest distance between both axis.

I_c is MOI of axis passing through com.



$$I_{\text{parallel}} = I_{CM} + Md^2$$

- Q. Using parallel axis theorem calculate MOI of rod about axis passing through end as shown in figure.

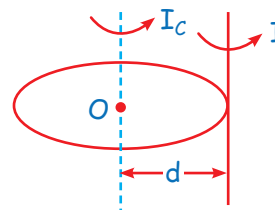


Sol. $I = I_c + md^2$

$$= \frac{mL^2}{12} + m(L/2)^2 \quad \left(d = \frac{L}{2} \right)$$

$$= \frac{mL^2}{12} + \frac{mL^2}{4} = (1+3) \frac{mL^2}{12} = \frac{mL^2}{3}$$

- Q. Find MOI (I) of ring of mass m and radius R about axis perpendicular to its plane as shown in figure



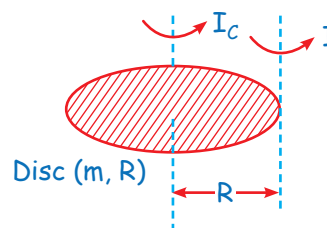
Sol. $I = I_c + md^2$

[Parallel axis theorem]

$$I = mR^2 + mR^2$$

$$I = 2mR^2$$

- Q. Find MOI (I) about axis as shown in figure.



Sol. $I = I_c + md^2$

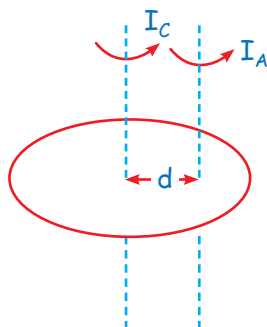
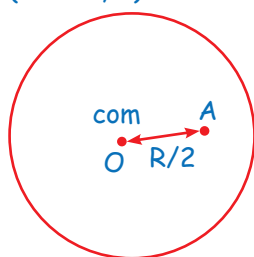
[From parallel axis theorem]

$$I = \frac{mR^2}{2} + mR^2$$

$$I = \frac{3mR^2}{2}$$

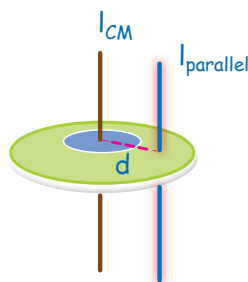
Q. Find the MOI of Disc. at an axis passes through A & $\perp^{\text{ar}}$ to the plane of Disc.

(Disc m, R)



Sol. $I_A = I_C + md^2$

$$= \frac{MR^2}{2} + m\left(\frac{R}{2}\right)^2 = \frac{3}{4}mR^2$$



$$I_{\text{parallel}} = I_{CM} + Md^2$$

Q. Find the MOI of a rod about an axis through its centre of mass and perpendicular to the length whose linear density varies as $\lambda = ax$ where a is a constant and x is the position of an element of the rod relative to its left end. The length of the rod is ℓ .

Sol. $I = \int dm x^2 = \int_0^\ell (\lambda dx) x^2 = \frac{a\ell^4}{4} \quad (\lambda = ax)$

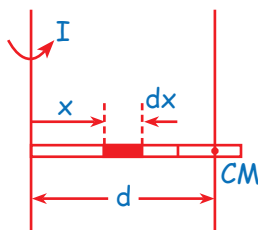
Location of CM from one end.

$$d = \frac{\int_0^\ell x \lambda dx}{\int_0^\ell \lambda dx} = \frac{2\ell}{3}$$

$$I = I_{cm} + md^2$$

$$\therefore m = \int \lambda dx = \frac{a\ell^2}{2}$$

$$\therefore I_{cm} = \frac{a\ell^4}{4} - \left(\frac{a\ell^2}{2}\right) \frac{4}{9} \ell^2 = \frac{a\ell^4}{36}$$



Q. For which value of x moment of inertia (I) is minimum? Where $I = 2x^2 - 12x + 27$ here x-coordinate of com

Sol. After differentiation we get

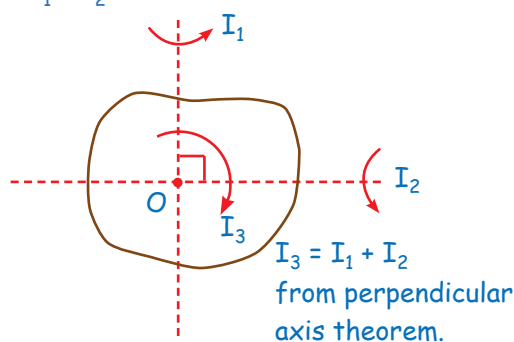
$$dI/dx = 4x - 12 = 0$$

$$x = 3$$

Perpendicular Axis Theorem

★ If I_1, I_2 are moment of inertia about axis perpendicular to each other in the plane of body and I_3 is moment of inertia about axis perpendicular to plane & passing through intersection of I_1 & I_2 axis then.

$$I_3 = I_1 + I_2$$



★ This theorem valid for planar, 2D body only



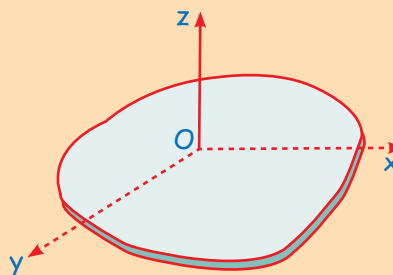
काम का डब्बा

★ This theorem only applicable in plane bodies (2D).

★ $I_3 = I_1 + I_2$ or $I_z = I_x + I_y$

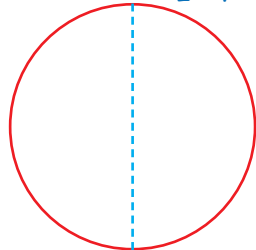
★ I_1 and I_2 का point of intersection जरूरी नहीं है कि COM ही हो।

★ $I_1 I_2$ body के plane में एक दूसरे के perpendicular axis है और I_3 body के plane के perpendicular $I_1 I_2$ के point of intersection से पास होने वाली axis है



Q. Find MOI about an axis in plane of a ring passing from the centre of ring as shown in figure.

(Ring m, R) Find $I = ?$



Sol. From perpendicular axis theorem

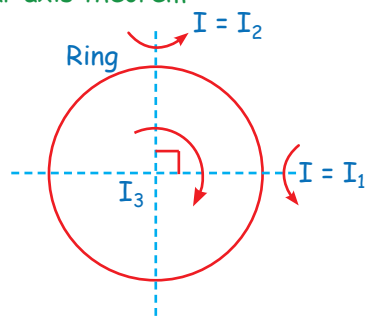
$$I_1 + I_2 = I_3$$

$$[I_1 = I_2 = I]$$

$$I + I = mR^2$$

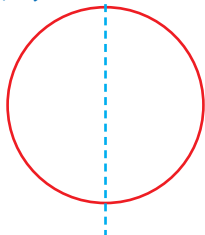
$$2I = mR^2$$

$$I = \frac{mR^2}{2}$$



Q. Find moment of inertia about an axis in the plane of disc (m, R) passing through centre along diameter.

(Disc, m, R)

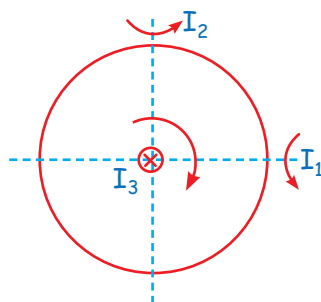


Sol. $I_1 + I_2 = I_3$

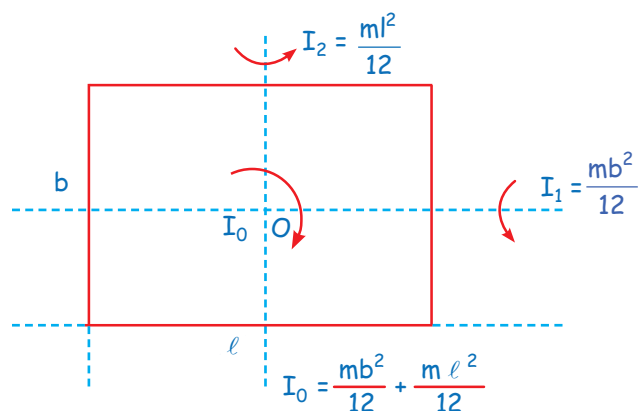
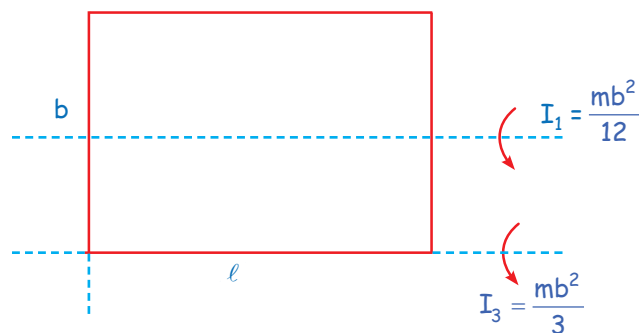
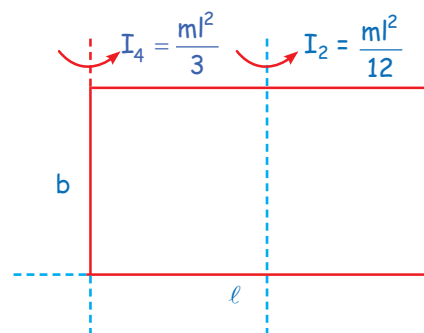
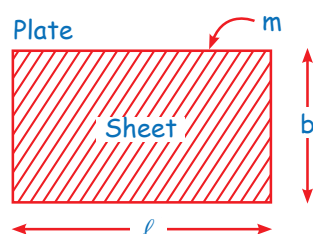
$$I_x = I_y = I$$

$$I + I = \frac{mR^2}{2}$$

$$I = \frac{mR^2}{4}$$

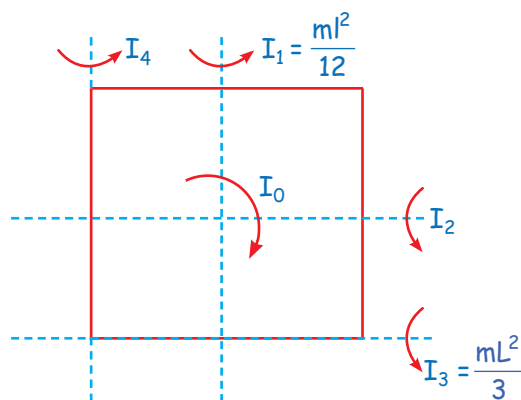


★ Suppose we have a sheet/plate of mass m and dimension $\ell \times b$ as shown in figure.



Q. Now, consider a square plate of mass m and side length L . Find MOI about all the axis you calculated in the previous question.

Sol. Put $L = b = L$ in above question

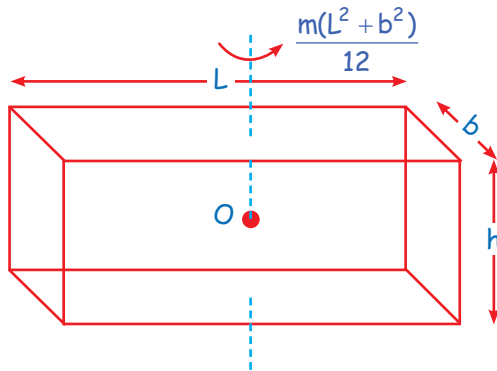


$$I_3 = \frac{mL^2}{3} = I_4$$

$$I_1 = \frac{mL^2}{12} = I_2$$

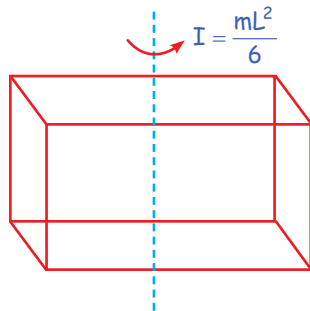
$$I_0 = \frac{mL^2}{12} + \frac{mL^2}{12} = \boxed{\frac{mL^2}{6}}$$

★ MOI of a cuboid about centre

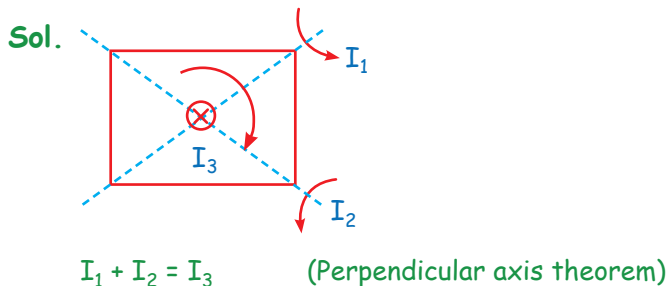
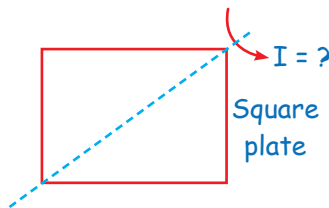


$$I_0 = \frac{mL^2}{12} + \frac{mb^2}{12}$$

★ MOI of a cube of mass m and side L about its centre



Q. Calculate MOI of a square plate about diagonal.



We Know ,

Apply perpendicular axis theorem for diagonals

$$I + I = \frac{mL^2}{6}$$

$$[I_1 = I_2 = I]$$

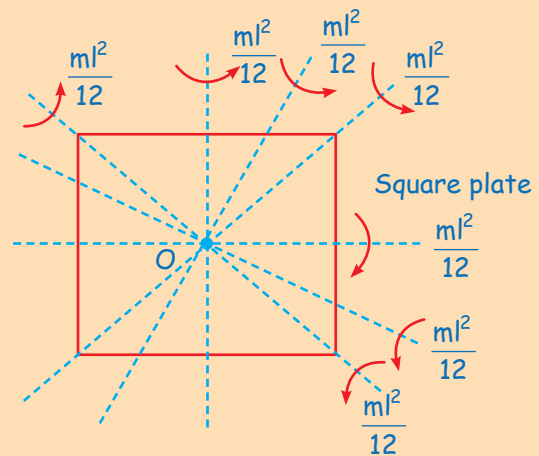
$$I = \frac{mL^2}{12}$$



काम का डब्बा

MOI about any axis passing through centre of square plate in plane of square plate is same

$$= \frac{mL^2}{12}$$

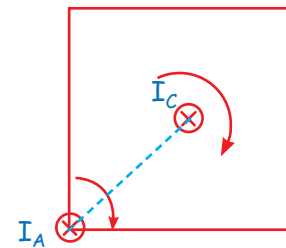


Note:

Square plate ke **plane** मे **centre** से pass होने वाली

har axis के about MOI $\frac{mL^2}{12}$ होगा।

Q. Find MOI I_A & I_C about axis as shown in figure for square plate of mass m and side length L

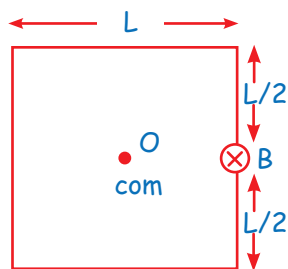


Sol. $I_A = I_C + md^2$

$$\text{We Know, } I_C = \frac{mL^2}{6}$$

$$= \frac{mL^2}{6} + m\left(\frac{L}{\sqrt{2}}\right)^2 = \frac{mL^2}{6} + \frac{mL^2}{2} = \boxed{\frac{2}{3}mL^2}$$

Q. In above question find I_B (square plate M, L)

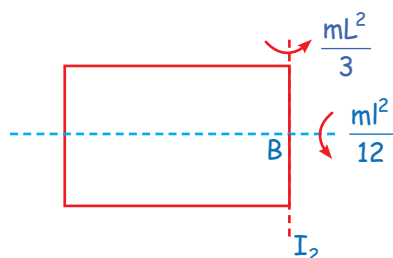


Sol. (M-I) $I_B = I_C + md^2$ [Parallel axis theorem]

$$I_B = \frac{mL^2}{6} + m\left(\frac{L}{2}\right)^2$$

$$\text{Or } I_B = \frac{5}{12} mL^2$$

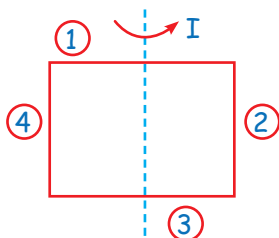
(M-II) From perpendicular axis theorem.



$$I_B = I_2 + I_1$$

$$I_B = \frac{mL^2}{3} + \frac{mL^2}{12} = \frac{5mL^2}{12}$$

Q. Four rods each of mass m and length L form square. Find MOI, I about the axis as shown in the figure.

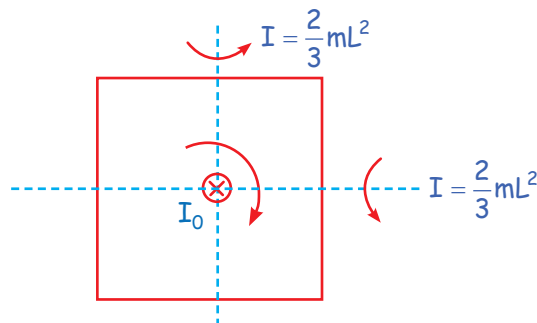


Sol. (a) $I = I_1 + I_2 + I_3 + I_4$

$$I = \frac{mL^2}{12} + m\left(\frac{L}{2}\right)^2 + \frac{mL^2}{12} + m\left(\frac{L}{2}\right)^2$$

$$I = \frac{2}{3} mL^2$$

Q. Four rods each of mass m and length L form square. Find MOI (I_0) about the axis as shown in the figure.

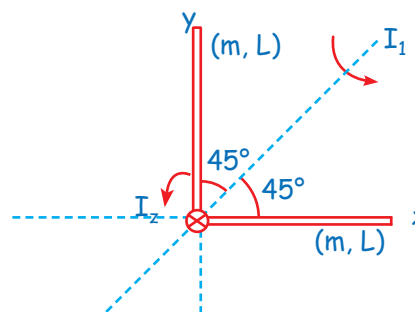


Sol. From perpendicular axis theorem

$$I_0 = I_1 + I_2$$

$$I_0 = \frac{2}{3} mL^2 + \frac{2}{3} mL^2 = \frac{4}{3} mL^2$$

Q. Two rods (m, L) are kept as shown in the figure. Find I_1 and I_2 ?



$$\text{Sol. } (I_x) = \frac{mL^2}{3} + 0 = \frac{mL^2}{3}$$

$$(I_y) = 0 + \frac{mL^2}{3} = \frac{mL^2}{3}$$

$$I_z = I_x + I_y$$

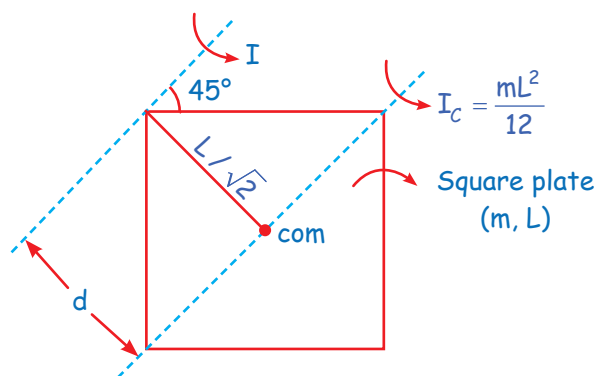
$$(I_z) = \frac{mL^2}{3} + \frac{mL^2}{3} = \frac{2mL^2}{3}$$

$$I_1 = ?$$

$I_1 \rightarrow$ Axis passing through origin and lies in the xy -plane.

$$I_1 = \frac{mL^2 \sin^2 45^\circ}{3} + \frac{mL^2 \sin^2 45^\circ}{3} = \frac{mL^2}{3}$$

Q. Calculate MOI(I) of square plate about an axis as shown in figure.

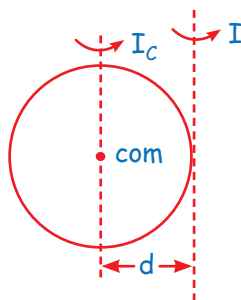


Sol. Using parallel axis theorem

$$I = I_c + md^2$$

$$= \frac{mL^2}{12} + m\left(\frac{L}{\sqrt{2}}\right)^2 = \frac{7}{12}mL^2$$

Q. Find MOI (I) of a solid sphere (m, R) about an axis as shown in figure.

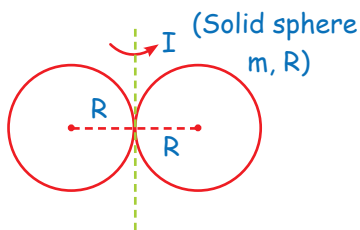


Sol. From parallel axis theorem

$$I = I_c + md^2$$

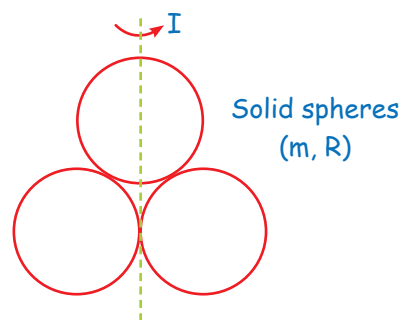
$$= \frac{2}{5}mR^2 + mR^2 = \frac{7}{5}mR^2$$

Q. Find I about an axis as shown in figure



Sol. $I = \frac{7}{5}mR^2 + \frac{7}{5}mR^2 = \frac{14}{5}mR^2$

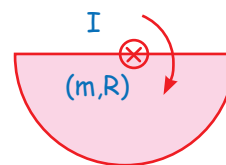
Q. Find I about axis as shown in figure



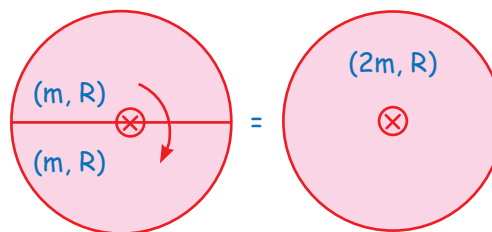
Sol. $I = \frac{2}{5}mR^2 + \left(\frac{2}{5}mR^2 + mR^2\right) \times 2$

Now you can solve.

Q. Find MOI of a semi circular disc of mass m and radius R about axis perpendicular to its plane and passing through centre.



Sol.



Complete disc will have mass 2m, if MOI of semi disc is I, then

Moment of inertia of complete disc is given by

$$I + I = (\text{mass of complete disc}) \frac{R^2}{2}$$

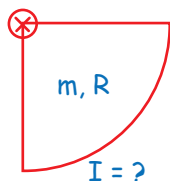
$$2I = \frac{2mR^2}{2}$$

$$I = \frac{mR^2}{2}$$

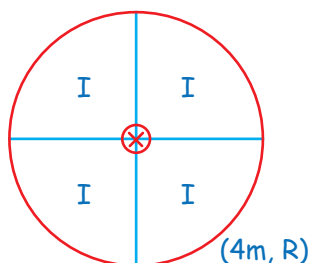
यूँ समझ लो कि अपने मन से हमने एक identical same mass की half disc assume करली जिससे एक 2m mass की disc complete बन गई।



- Q. Find MOI of quarter disc of mass m and radius R about axis perpendicular to its plane and passing through centre.



Sol.

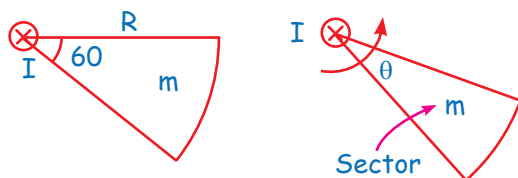


Same method से हमने तीन quarter disc और रख कर $4m$ mass की एक complete disc बना ली जिसका net MOI $4I$ बन गया

$$4I = (4m) \frac{R^2}{2}$$

$$I = \frac{mR^2}{2}$$

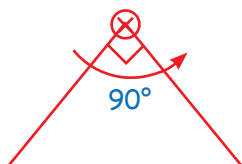
- Q. Find MOI, I of a sector of circle having mass m , radius ' R ' about axis shown in figure.



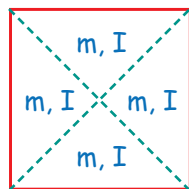
Sol. $6I = \frac{(6m)R^2}{2}$

$$I = \frac{mR^2}{2}$$

- Q. Find MOI, I of right angle triangular lamina about the axis shown?



Sol.



(four triangles makes square of mass $4m$)

$$I_{\text{square}} = \frac{M_{\text{total}} L^2}{6}$$

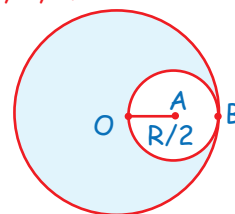
$$4I = \frac{4m \times L^2}{6}$$

$$I = \frac{mL^2}{6}$$

अब नीचे मैं ऐसा बेशरम ques. add कर रहा हूँ जो पहले advance में आया था और JEE mains में almost हर साल पुछा जाता है इसे अच्छे से practice कर लेना।

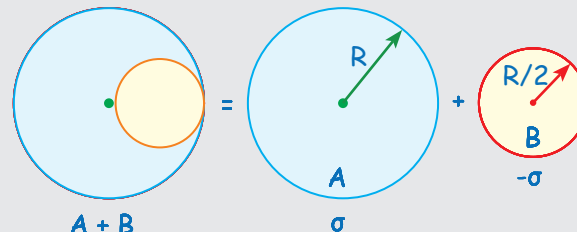


- Q. From a uniform disc of mass M and radius R , another disc of radius $R/2$ is removed as shown in fig. Find the MOI of the remaining part about an axis perpendicular to the disc and passing through O , A , B .



Sol.

अरे याद है ना COM में भी ऐसे ques किए थे।



मान लो एक -ve mass की disc B, badi wali disc A के upar raki hai to ye ques wali body 2 disc se milkar bani hai

$$m_A = \sigma \pi R^2 = m (\text{let})$$

$$m_B = -\sigma \pi \left(\frac{R}{2}\right)^2 = \frac{-m}{4}$$

$$I_O = \left(\frac{mR^2}{2} \right) - \left[\frac{\frac{m}{4} \left(\frac{R}{2}\right)^2}{2} + \frac{m}{4} \left(\frac{R}{2}\right)^2 \right]$$

bigger disc small disc

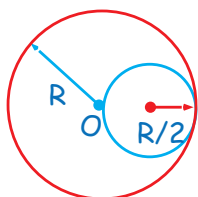
$$I_A = \left[\frac{mR^2}{2} + m \left(\frac{R}{2}\right)^2 \right] - \left[\frac{\frac{m}{4} \left(\frac{R}{2}\right)^2}{2} \right]$$

bigger disc smaller disc

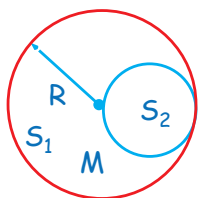
$$I_B = \left[\frac{mR^2}{2} + mR^2 \right] - \left[\frac{\frac{m}{4} \left(\frac{R}{2} \right)^2}{2} + \frac{m}{4} \left(\frac{R}{2} \right)^2 \right]$$

bigger disc smaller disc

Q. A hole of radius $R/2$ is cut from a solid sphere of radius R . If the mass of the remaining of sphere is M , then find moment of inertia of the body about an axis through O .



Sol. $57/140 MR^2$



$$S_1 + S_2 = S$$

$$I_{S1} + I_{S2} = I_S$$

$$M_2 = \rho \frac{4}{3} \pi \frac{R^3}{8} = \frac{M}{7} \text{ (mass of hole)}$$

$$\text{where } \rho = \frac{M}{\frac{4}{3} \pi R^3 - \frac{4}{3} \pi \left(\frac{R}{2} \right)^3}$$

MOI of hole about 'O'

$$I_{S2} = \frac{2}{5} \frac{M}{7} \left(\frac{R}{2} \right)^2 + \frac{M}{7} \left(\frac{R}{2} \right)^2$$

$$= \frac{7}{5} \frac{M}{7} \frac{R^2}{4} = \frac{MR^2}{20}$$

$$I_S = \frac{2}{5} \left(M + \frac{M}{7} \right) R^2$$

$$= \frac{16M}{35} R^2$$

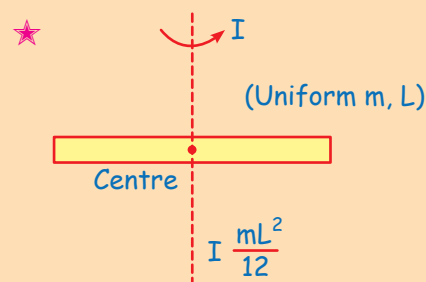
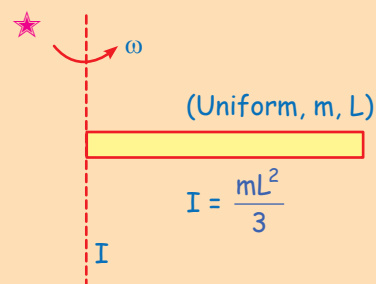
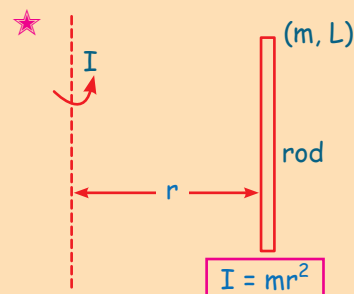
$$\therefore I_{S1} = I_S - I_{S2}$$

$$= \frac{16}{35} MR^2 - \frac{MR^2}{20}$$

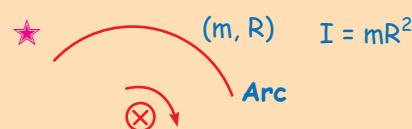
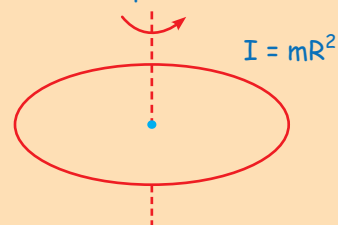
$$= \frac{57}{140} MR^2$$

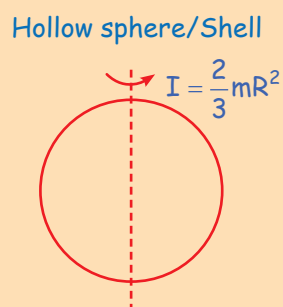
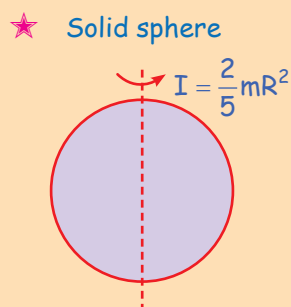
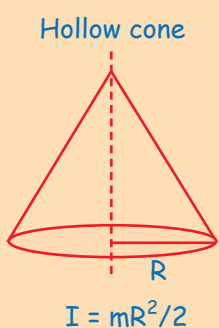
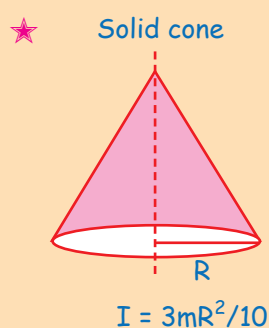
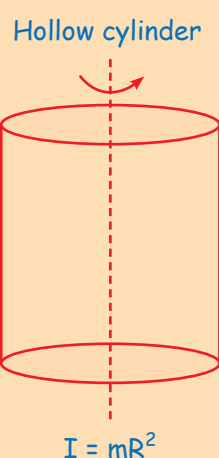
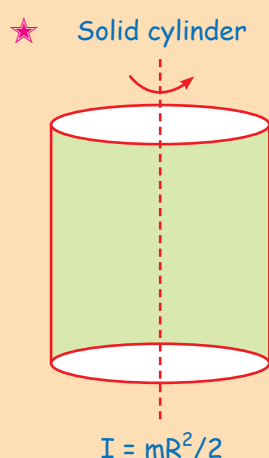
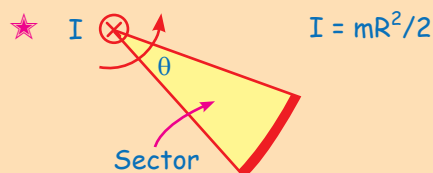
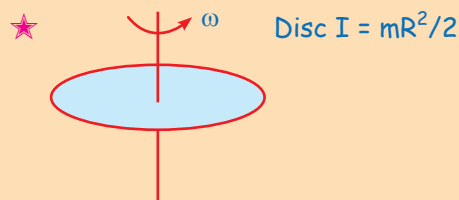


काम का डब्बा



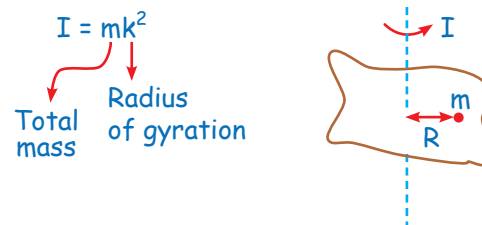
★ Ring, uniform or non-uniform both
Top view



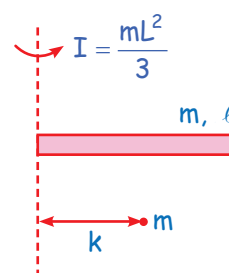


Radius of Gyration

It is the perpendicular distance from a given axis at which entire mass of object supposed to be concentrated (like point mass) without changing the MOI.



Q. Find radius of gyration of rod as shown in figure.

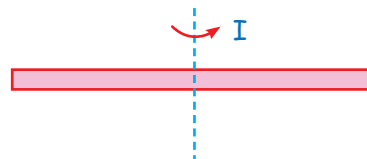


Sol. $I = mK^2$

$$\frac{mL^2}{3} = m \times K^2$$

$$K = \frac{L}{\sqrt{3}}, \text{ where } K = \text{Radius of gyration.}$$

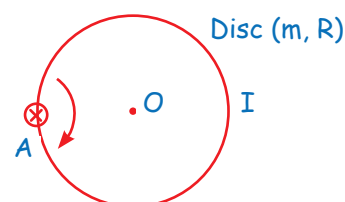
Q. Find radius of gyration.



Sol. $I = \frac{mL^2}{12} = mk^2$

$$k = \frac{L}{\sqrt{12}}$$

Q. Find radius of gyration about axis passing through A as shown in figure.



Sol. $I_A = \frac{mR^2}{2} + mR^2$ [Parallel axis theorem]

$$= \frac{3}{2}mR^2$$

$$\frac{3}{2}mR^2 = mk^2$$

$$k = \sqrt{\frac{3}{2}}R$$

Q. Find radius of gyration for rod (m, L) and mass 'm' as shown in figure.



Sol. Radius of gyration (k) $I = mk^2$

$$I_O = \frac{mL^2}{3} + mL^2 = \frac{4mL^2}{3}$$

$$I = (2m)k^2$$

$$k = \sqrt{\frac{2}{3}}L$$

Cross Product

अरे याद है ना vector में पढ़ा था..... cross product का अब यहाँ अच्छा use होगा so आगे बढ़ने से पहले cross product निकालना अच्छे से सीख लो



Cross product $\vec{C}$ of $\vec{A}$ and $\vec{B}$ is defined as:

★ $\vec{C} = \vec{A} \times \vec{B} = AB \sin \theta \hat{C}$

★ $|\vec{C}| = AB \sin \theta$

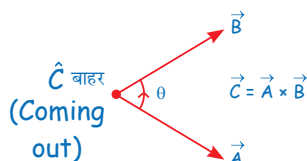
A → mag. of $\vec{A}$

B → mag. of $\vec{B}$

θ → Angle between $\vec{A}$ and $\vec{B}$

★ Direction of $\vec{A} \times \vec{B}$ is perpendicular to plane of $\vec{A}$ and $\vec{B}$. Its direction is given by right hand thumb rule.

★ $\hat{i} \times \hat{i} = 0 \quad \hat{j} \times \hat{j} = 0 \quad \hat{k} \times \hat{k} = 0$



Q. $\vec{A} = 4\hat{i} + 7\hat{j}$

$\vec{B} = 3\hat{i} + 5\hat{j}$

Sol. $\vec{A} \times \vec{B} = (4\hat{i} + 7\hat{j}) \times (3\hat{i} + 5\hat{j}) = 0 + 20\hat{k} - 21\hat{k} + 0 = -\hat{k}$

Q. $\vec{A} = 2\hat{i} + 3\hat{j} + 4\hat{k}$

$\vec{B} = 5\hat{i} + 6\hat{j} + 7\hat{k}$

Sol. $\vec{A} \times \vec{B} = (2\hat{i} + 3\hat{j} + 4\hat{k}) \times (5\hat{i} + 6\hat{j} + 7\hat{k})$
 $= 12\hat{k} - 14\hat{j} - 15\hat{k} + 0 + 21\hat{i} + 20\hat{j} - 24\hat{i} + 0$
 $= 3\hat{k} + 6\hat{j} - 3\hat{i}$

TORQUE

★ Torque of a force about a point is the moment of the force about that point.

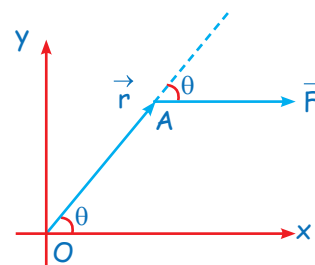
★ Vector quantity having unit Newton-meter (Nm).

Torque of Force $\vec{F}$ about 'O', is defined as

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\tau = rF \sin \theta$$

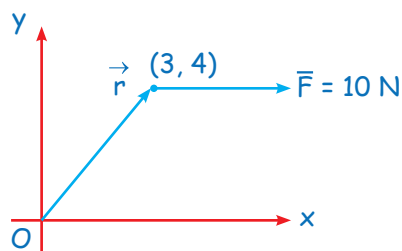
(magnitude)



SKC

$\vec{r}$ वो vector है जिसकी पूँछ वहाँ पर है जिसके about torque पूँछ रहा है और मुंडी वहाँ है जहाँ Force है।

Q. Find torque of Force $\vec{F}$ about 'O'



Sol. $\vec{r} = 3\hat{i} + 4\hat{j}$
 $\vec{F} = 10\hat{i}$
 $\vec{\tau}_0 = \vec{r} \times \vec{F} = (3\hat{i} + 4\hat{j}) \times 10\hat{i}$
 $\vec{\tau}_0 = 0 - 40\hat{k}$

CONCEPT

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\tau = rF\sin\theta$$

$$\tau = r(F\sin\theta)$$

$$\tau = rF_{\perp}$$

magnitude

or

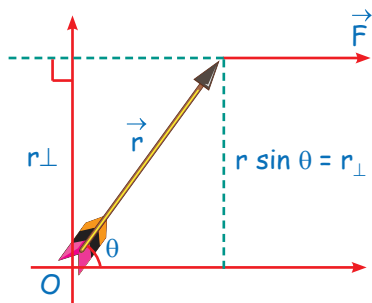
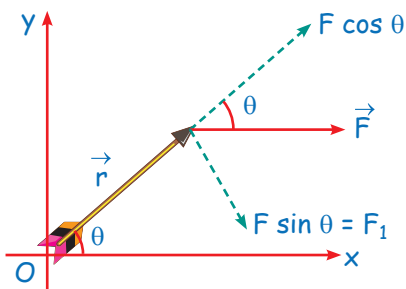
$$\tau = (r\sin\theta).F$$

$$\tau = r_{\perp}F$$

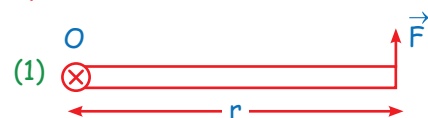
magnitude

$$\tau = rF_{\perp} = r_{\perp}F$$

magnitude



Q. Find τ abt O of force F.



$$\tau_0 = rF \text{ (Acw)}$$

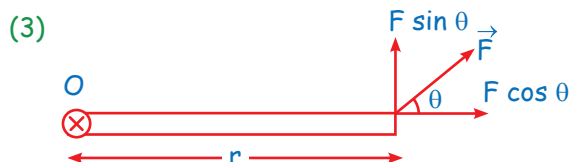


$$rF\sin\theta = 0$$

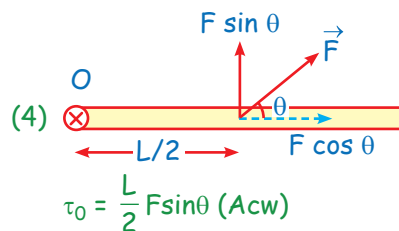
$$[\theta = 0]$$

$$\tau_0 = 0$$

Agar koi force F Axis se pass kar gaya to uska torque zero hoga

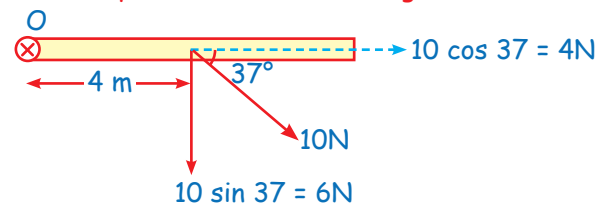


$$\tau_0 = rF\sin\theta \text{ (Acw)}$$



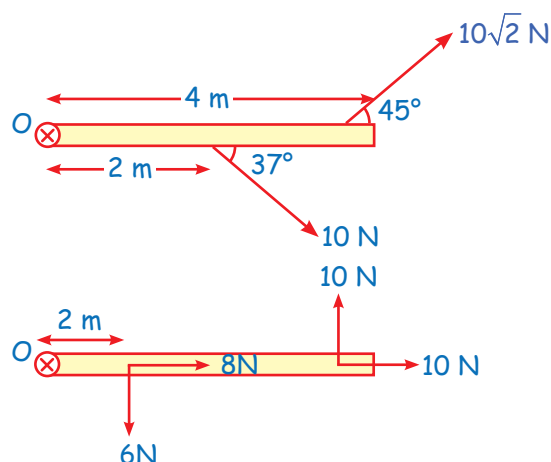
$$\tau_0 = \frac{L}{2} F\sin\theta \text{ (Acw)}$$

Q. Find torque of forces about hinge.



Sol. $\tau_0 = 4 \times 6 = 24 \text{ (Cw)}$ ($\tau = rF_{\perp}$)
 $\vec{\tau}_0 = 24(-\hat{k})$

Q. Find torque of forces shown about hinge.



Sol.

$$\begin{aligned} (\tau_0) &= -2 \times 6 + 10 \times 4 \\ &= -12 + 40 \\ &= +28 \text{ Acw} \end{aligned}$$

Note: $\text{Acw} = \hat{k}$ $\text{cw} = -\hat{k}$

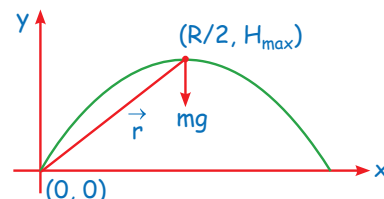
Q. Find torque by mg force about projection point (1) when particle at highest point, (2) just before it hits the ground (3) at time t.

Sol.

Method-1

(1) $\vec{r} = \frac{R}{2}\hat{i} + H_{\max}\hat{j}$

$$\vec{F} = mg(-\hat{j})$$



$$\vec{\tau} = \left(\frac{R}{2}\hat{i} + H_{\max}\hat{j} \right) \times [mg(-\hat{j})] = \frac{R}{2} mg(-\hat{k}) + 0$$

$$\vec{\tau} = \frac{mgR}{2}(-\hat{k}) = \text{cw}$$

- (2) Just before it hits the ground

$$\vec{r} = R\hat{i}$$

$$\vec{F} = -mg\hat{j}$$

$$\vec{\tau} = -mgR\hat{k}$$

- (3) At anytime 't'

$$\vec{r} = x\hat{i} + y\hat{j}$$

$$\vec{r} = (u\cos\theta)t\hat{i} + \left(usin\theta t - \frac{1}{2}gt^2\right)\hat{j}$$

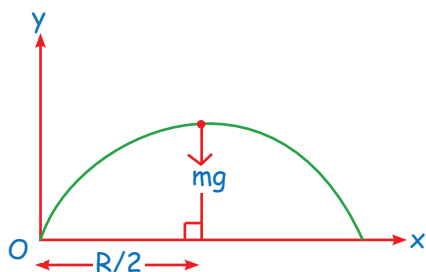
$$\vec{F} = mg(-\hat{j})$$

$$\vec{\tau} = \vec{r} \times \vec{F} = -mg u \cos\theta \cdot t (\hat{k})$$

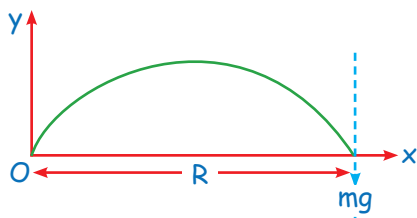
Method-2

- (1) $\tau = r_{\perp}F$ (magnitude)

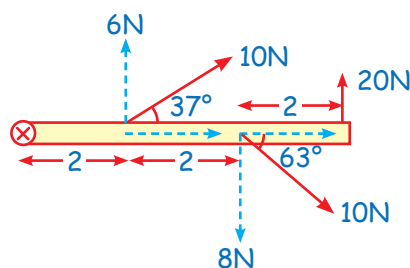
$$\tau = \frac{R}{2} mg \text{ (cw)}$$



- (2) $\tau_0 = r_{\perp}F$
 $\tau_0 = Rmg$

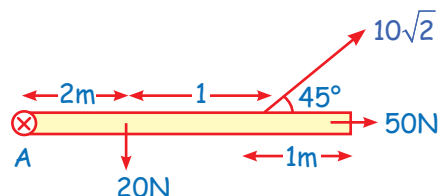


- Q. Find torque of forces about hinge.



Sol. $\vec{\tau}_0 = +6 \times 2 - 8 \times 4 + 20 \times 6$
 (ACW) (C.W) (ACW)
 $\vec{\tau}_0 = 100 \text{ ACW}$

- Q. Find torque of forces about hinge A.



Sol. $\tau_A = -(2 \times 20) + (3 \times 10) + 0 = -40 + 30 = -10 \text{ (Cw)}$

$$\vec{\tau}_A = 10 (-\hat{k})$$

★ If $\vec{F}_{\text{net}} = 0$ on body.

then $\vec{\tau}_{\text{net}}$ abt all point will be same.

EQUILIBRIUM

देख भाई equilibrium का नाम तुम पहले सुन चुके हो... NLM में जब हमने equilibrium पढ़ा था तो वहाँ $f_{\text{net}} = 0$ किया था means translation equilibrium उसी प्रकार if $\tau_{\text{net}} = 0$ it means rotational equilibrium....

Now we can say if a body is in equilibrium that means body is in translation equilibrium ($f_{\text{net}} = 0$) as well as rotational equilibrium ($\tau_{\text{net}} = 0$ about any point)



τ ki calculation ke liye mg ko com पर लगाओ।

- Q. A rod of mass m and length L is hanging by two strings attached at its ends as shown in figure.



Find tension in both string if the rod is in equilibrium.

Sol. Torque about C is zero.

$$\tau_C - T_1 \frac{L}{2} + 0 + T_2 \frac{L}{2} = 0$$

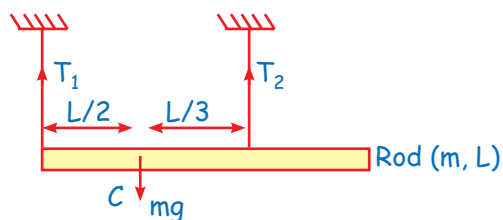
CW ACW

$$\Rightarrow T_1 = T_2$$

$$T_1 + T_2 = mg$$

$$\text{Hence } T_1 = T_2 = \frac{mg}{2}$$

- Q. A rod of mass m and length L is hanging by two strings as shown in figure.



Rod is in equilibrium, find tension in both string.

Sol. At equilibrium

$$T_1 + T_2 = mg \text{ ... (i) (Net force zero)}$$

$$\tau_c = 0 \text{ (Net torque zero)}$$

$$T_1 \frac{L}{2} - T_2 \frac{L}{3} = 0$$

$$T_1 \frac{L}{2} = T_2 \frac{L}{3}$$

$$T_2 = \frac{3}{2} T_1$$

... (ii)

$$\text{Ans. } T_1 = \frac{2}{5} mg$$

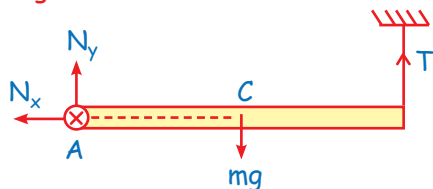
$$T_2 = \frac{3}{5} mg$$

- Q. A rod of mass m and length L is hanging at one end by string and at other end it is hinged as shown in figure.



Rod is in equilibrium, find Tension in string and hinge force.

Sol.



At equilibrium

$$N_y + T = mg$$

... (i)

$$N_x = 0$$

$$\text{Also } \tau_c = 0$$

Torque about c is zero

$$N_y \frac{L}{2} = \frac{TL}{2}$$

$$T = N_y$$

... (ii)

From equation (i) & (ii) we get,

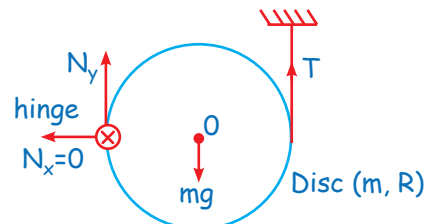
$$T = N_y = \frac{mg}{2}$$

$$\text{Or } \tau_A = 0$$

$$mg \frac{L}{2} = TL$$

$$T = \frac{mg}{2}$$

- Q. A disc of mass m and radius R is hanging as shown in figure. Find T and hinge force



Assume hinge force components be N_x & N_y

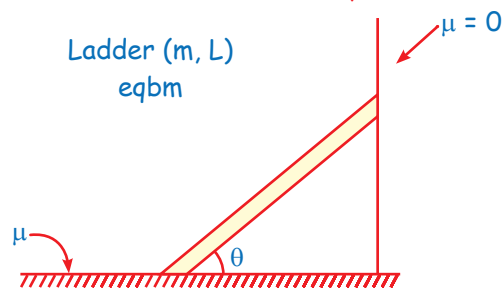
$$\text{Sol. } N_y + T = Mg$$

$$\tau_0 = 0$$

$$N_y R = TR$$

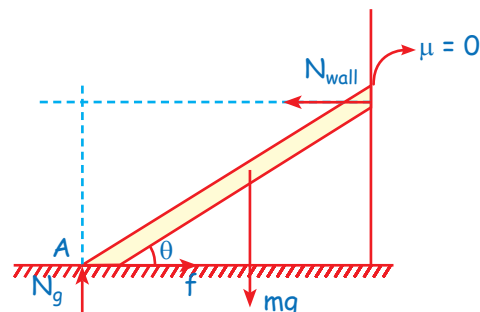
$$\Rightarrow N_y = T = \frac{mg}{2}$$

- Q. A ladder is resting as shown in figure. Wall is smooth, floor is rough coefficient of friction between ladder and floor is μ .



Find $\mu_{\min}$ so that ladder remain at rest.

Sol.



Torque about point A = 0

$$\tau_{\text{net}} = 0 \text{ (abt A)}$$

$$mg \frac{L}{2} \cos \theta = N_{\text{wall}} L \sin \theta$$

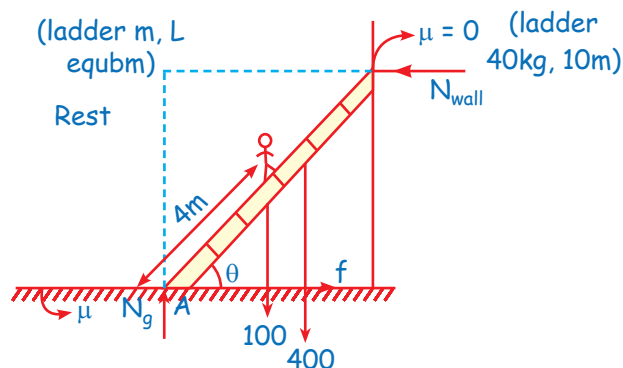
$$N_{\text{wall}} = \frac{mg \cot \theta}{2} = f$$

$$f = \mu N_g = \mu mg$$

$$\frac{mg \cot \theta}{2} = \mu mg$$

$$\mu_{\min} = \frac{\cot \theta}{2}$$

- Q. A man of mass 10 kg is climbing on ladder of mass 40 kg and length 10m as shown in figure. Man is standing at 4 m from end A of ladder. Wall is smooth floor is rough A boy (10 kg) is standing at 4m from lower end.



Given: $\theta = 37^\circ$

Find $\mu_{\min}$ so that ladder remains at rest

Sol. At equilibrium

$$N_g = 400 + 100 = 500 \text{ N}$$

$$N_{\text{wall}} = f$$

[Net torque about A = 0]

$$100 \times 4 \cos 37^\circ + 400 \times 5 \cos 37^\circ = N_{\text{wall}} \times 10 \sin 37^\circ$$

$$\Rightarrow 400 \times \frac{4}{5} + 2000 \times \frac{4}{5} = N_{\text{wall}} \times 6$$

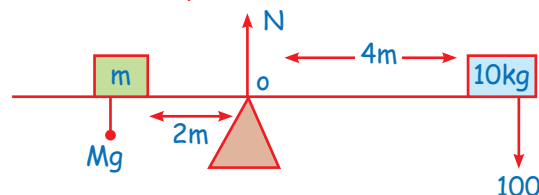
$$\Rightarrow N_{\text{wall}} = \frac{1600}{5} = 320 = f$$

$$f = \mu N$$

$$320 = \mu \times 500$$

$$\mu_{\min} = \frac{320}{500} = 0.64$$

- Q. On a seesaw weights are kept as shown in figure seesaw is in equilibrium.



Find m and normal force by seesaw and platform.

Sol. $\tau_0 = 0$ Torque about O is zero

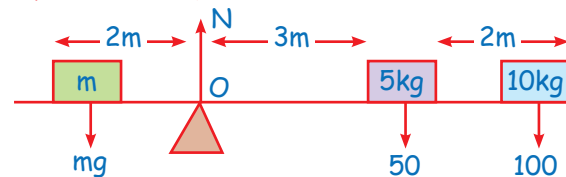
$$100 \times 4 = mg \times 2$$

$$m = 20 \text{ kg}$$

At equilibrium

$$N = mg + 100 = 200 + 100 = 300$$

- Q. System is in equilibrium. Find N



Sol. $\tau_0 = 0$, torque about O is 0

$$50 \times 3 + 100 \times 5 = mg \times 2$$

$$150 + 500 = m \times 20$$

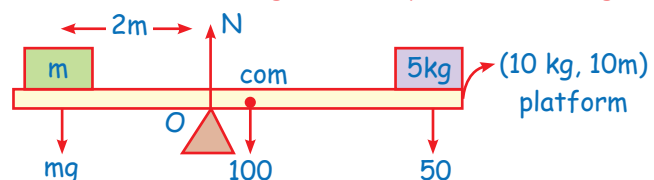
$$m = \frac{650}{20} = 32.5 \text{ kg}$$

(i)

Balancing forces we get,

$$N = mg + 50 + 100 = 325 + 150 = 475 \text{ Newton}$$

- Q. On a seesaw weights are kept as shown in figure.



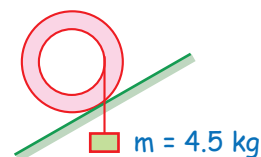
seesaw is in equilibrium. Find $m = ?$, $N = ?$

Sol. $\tau_0 = 0$, taking torque about O

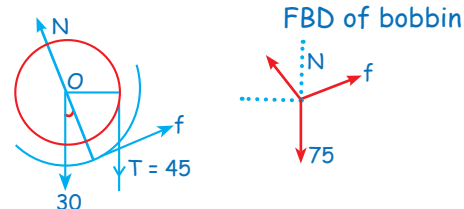
$$100 \times 3 + 50 \times 8 = mg \times 2$$

$$m = \frac{700}{2} = 35 \text{ Kg}$$

- Q. A 3.0 kg bobbin consists of a central cylinder of radius 5.0 cm and two end plates each of radius 6.0 cm. It is placed on a slotted incline, where friction is sufficient to prevent sliding. A block of mass 4.5 kg is suspended from a cord wound around the bobbin and passing through (the slot under the incline. If the bobbin is in static equilibrium, what is the angle of tilt of the incline? Bobbin mass $M = 3 \text{ kg}$



Sol.



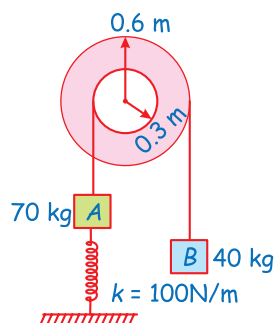
$$f = 75 \sin \theta$$

$$\tau \text{ about } O = 0$$

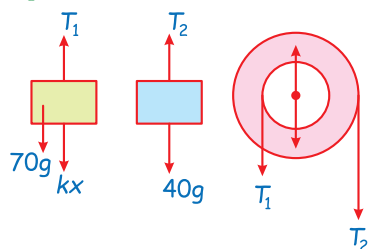
$$\therefore T \times 5 = f \times 6$$

$$45 \times 5 = 75 \times 6 \times \sin \theta \Rightarrow \theta = 30^\circ$$

Q. The figure shown is in equilibrium. Find out the extension in the spring (in cm).



Sol. [100.00]



$$T_1 = 70g + kx \quad \dots(i)$$

$$T_2 = 40g \quad \dots(ii)$$

$$T_1 (0.3) = T_2 (0.6) \quad \dots(iii)$$

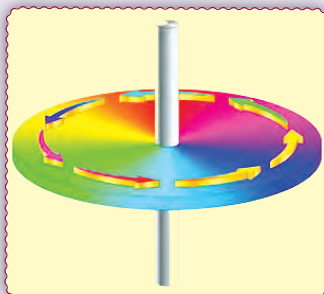
$$\Rightarrow 70g + kx = 80g$$

$$kx = 10g$$

$$\therefore x = 1\text{m}$$

FIXED AXIS ROTATION

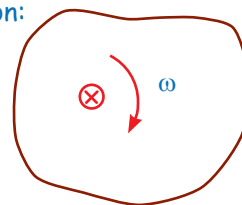
पहले इसकी feel लो की हमने किसी body को किसी fix point पर hinge कर दिया है और उसको rotate कर दिया है।



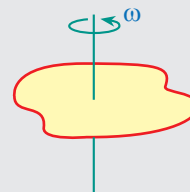
★ Kinetic Energy in Pure Rotation:

$$KE = \frac{1}{2} I \omega^2$$

I is the MOI about fix axis.



PROOF



$$K_i = \frac{1}{2} m_i v_i^2$$

where $v_i = r_i \omega$

The sum of such terms over the whole body gives

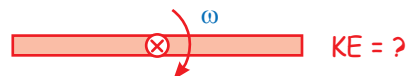
$$K = \sum \frac{1}{2} m_i \omega^2 r_i^2 = \sum \frac{1}{2} m_i \omega^2 r_i^2 = \frac{1}{2} (\sum m_i r_i^2) \omega^2$$

Q. A rod of mass m and length L is rotating with angular velocity ω about axis as shown in figure. Find kinetic energy of rod.



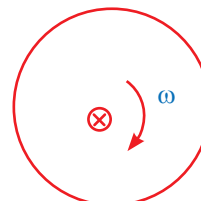
$$\text{Sol. } KE = \frac{1}{2} I \omega^2 = \frac{1}{2} \frac{mL^2}{3} \omega^2$$

Q. A rod of mass m and length L is rotating with angular velocity ω about axis as shown in figure. Find kinetic energy of rod.



$$\text{Sol. } KE = \frac{1}{2} \cdot \frac{mL^2}{12} \omega^2$$

Q. A disc of mass m and radius R is rotating with angular velocity ω about axis as shown in figure. Find kinetic energy of disc.



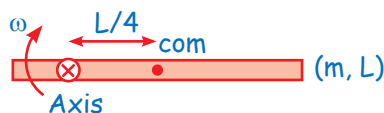
$$\text{Sol. } KE = \frac{1}{2} I \omega^2 = \frac{1}{2} \cdot \frac{mR^2}{2} \omega^2$$

- Q. Find K.E of rod & particle fixed on it rotating as shown in figure.



Sol. $KE = \frac{1}{2} I \omega^2 = \frac{1}{2} \left(\frac{mL^2}{3} + mL^2 \right) \omega^2$

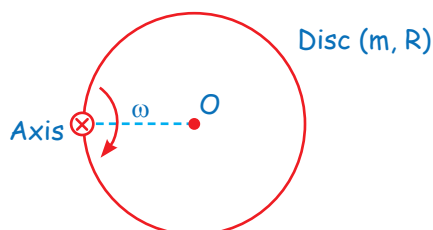
- Q. Find K.E



Sol. Moment of inertia about axis = $\frac{mL^2}{12} + m \left(\frac{L}{4} \right)^2$

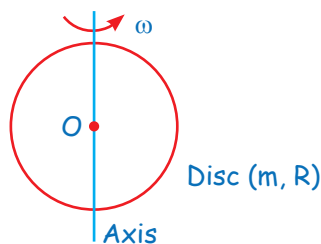
$KE = \frac{1}{2} I \omega^2 = \frac{1}{2} \left(\frac{mL^2}{12} + m \left(\frac{L}{4} \right)^2 \right) \omega^2$

- Q. Find K.E of disc.



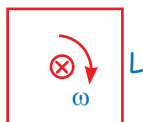
Sol. $KE = \frac{1}{2} I \omega^2 = \frac{1}{2} \left(\frac{mR^2}{2} + mR^2 \right) \omega^2$

- Q. Find K.E



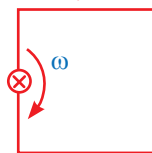
Sol. $KE = \frac{1}{2} I \omega^2 = \frac{1}{2} \left(\frac{mR^2}{2} \right) \omega^2$

- Q. Find K.E of square plate mass m side L.



Sol. $KE = \frac{1}{2} I \omega^2 = \frac{1}{2} \left(\frac{mL^2}{6} \right) \omega^2$

- Q. Find KE of square plate (m, L) if the axis of rotation is as shown.



Sol. $KE = \frac{1}{2} I \omega^2 = \frac{1}{2} \left(\frac{5}{12} mL^2 \right) \omega^2$

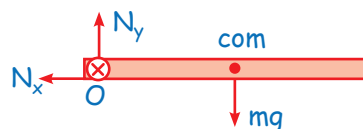
Fixed Axis Rotational Motion

$$\vec{\tau}_{\text{axis}} = I_{\text{axis}} \vec{\alpha} \text{ or } \tau = I\alpha$$

- Q. A rod of mass m and length L is hinged as shown in figure. At t = 0 is string is cut, find angular acceleration α just after the string is cut?



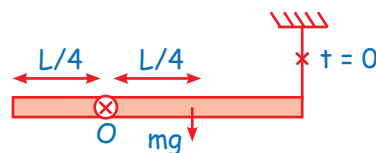
- Sol. Suppose hinged force has component N_x and N_y as shown in figure FBD of rod is



$\tau_{\text{Axis}} = \tau_0 = mg \cdot \frac{L}{2} = I\alpha = \frac{mL^2}{3} \alpha$

$\alpha = \frac{3g}{2L}$

- Q. A rod of mass m and length L is hinged as shown in figure. At t = 0 is string is cut, find angular acceleration α just after cut the string

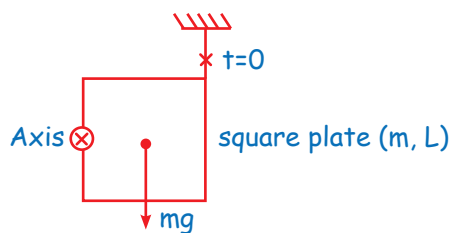


Sol. $\tau_0 = mg \frac{L}{2} = I\alpha$

$mg \frac{L}{2} = \left\{ \frac{mL^2}{12} + m \left(\frac{L}{4} \right)^2 \right\} \alpha$

$\alpha = \checkmark$

- Q. A square plate of mass m and side length L is hinged as shown in figure. At t = 0 is string is cut, find angular acceleration α just after the string is cut.

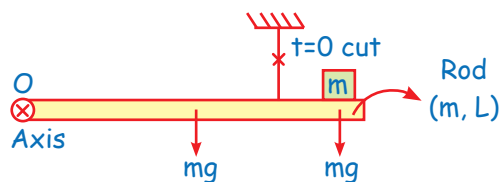


Sol. $\tau_0 = I_0 \alpha$

$$mg \frac{L}{2} = \left[\frac{mL^2}{6} + m \left(\frac{L}{2} \right)^2 \right] \alpha$$

$\alpha = \checkmark$

Q. Find α just after string is cut?

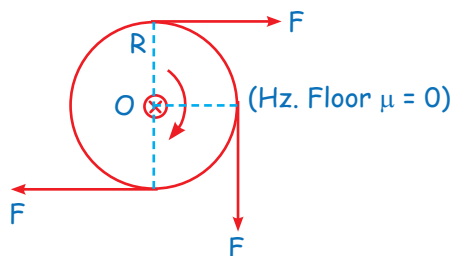


Sol. $\tau_0 = I \alpha$

$$mg \frac{L}{2} + mgL = \left(\frac{mL^2}{3} + mL^2 \right) \alpha$$

rod mass

Q. On a horizontal floor disc(m, R) can rotate about axis. Find α



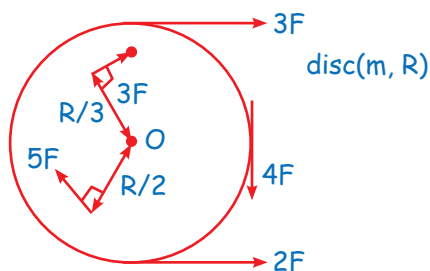
Sol. $\tau_0 = I_0 \alpha$

$$FR + FR + FR = \frac{mR^2}{2} \alpha$$

$$3FR = \frac{mR^2}{2} \alpha$$

$$\alpha = \frac{6F}{mR}$$

Q. Forces are acting on disc of mass m and radius R as shown in figure. Find angular acceleration α .



Sol. $\tau_0 = +3FR + 4FR + 5F \frac{R}{2} + 3F \frac{R}{3} - 2FR = \frac{mR^2}{2} \alpha$
 $\alpha = (\checkmark) (C\omega)$

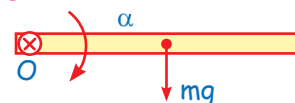
अब मैं rotation motion का one of the most important ques. add कर रहा हूँ जो कम से कम 10 बार इस chapter में use होगा और exams में भी बार-बार पूछा जाता है। इसमें हर एक चीज जो पूछी है उसके अच्छे से practice कर लेना।



Q. A rod of mass m and length L is hinged as shown in figure. At $t = 0$ string is cut.



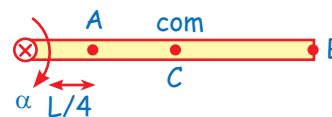
(a) Find angular acceleration just after ($t = 0$) string is cut?



Sol. Torque about O

$$\tau_0 = \frac{mgL}{2} = \frac{mL^2}{3} \alpha = \alpha = \frac{3g}{2L}$$

(b) Find linear acc. of points of A, B, C



Sol. $\omega = 0 \Rightarrow$ centripetal acc of each point = 0

Net acceleration = centripetal acceleration + tangential acceleration

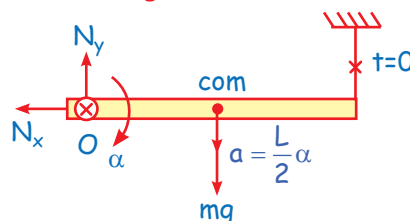
$$\vec{a}_{\text{net}} = \vec{a}_t \text{ (Only tangential)}$$

$$a_A = (a_t)_A = r\alpha = \frac{L}{4} \alpha$$

$$a_B = r_B \alpha = L\alpha$$

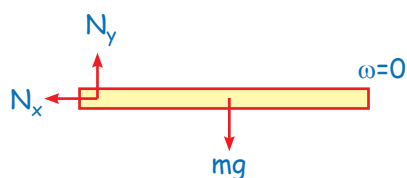
$$a_C = \frac{L}{2} \alpha$$

(c) Find the hinge force?



$$a_{\text{com}} = \frac{L}{2} \alpha = \frac{L}{2} \cdot \frac{3g}{2L} = \frac{3g}{4} \quad \left(\alpha = \frac{3g}{2L} \right)$$

$$F_{\text{net}} = ma_{\text{com}}$$



$$mg - N_y = ma_{com}$$

$$mg - N_y = m \frac{3g}{4}$$

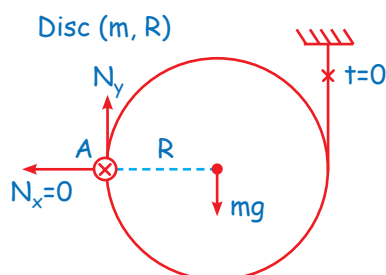
$$N_y = \frac{mg}{4}$$

$$(F_x)_{net} = m(a_{com})_x$$

$$N_x = m \times 0 \text{ (No motion along x)}$$

$$N_x = 0$$

Q. Find α , a_{com} , and hinge force at $t = 0$, as the string in just cut.



Sol. $\tau_A = I_A \alpha$

$$mgR = \left(\frac{mR^2}{2} + mR^2 \right) \alpha$$

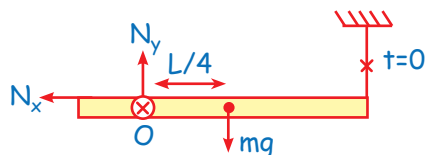
$$\alpha = \frac{2g}{3R}$$

$$a_{com} = R\alpha = 2g/3$$

$$Mg - N_y = ma_{com} \quad (\because F_{net} = m a_{com})$$

$$Mg - N_y = m \cdot \frac{2g}{3} \Rightarrow N_y = \frac{mg}{3}$$

Q. Find α , a_{com} , hinged force just after the string is cut?



Sol. $\tau_O = I_O \alpha$

$$Mg \frac{L}{4} = \left[\frac{ML^2}{12} + M \left(\frac{L}{4} \right)^2 \right] \alpha$$

$$\alpha = \checkmark$$

$$a_{com} = \frac{L}{4} \alpha$$

$$Mg - N_y = Ma_{com}$$

by substituting appropriate values we can get hinge force as we did in previous question.

PULLEY (ROTATION)



फिर आगई pulley ये तो हमे अच्छे से आता है अब तो top college पक्का



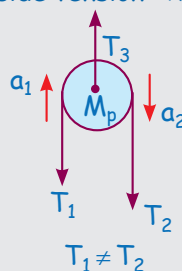
भाई ये NLM वाली pulley नहीं है



Ye Kab Hua?

NLM में हमने जो pulley पढ़ी थी वो ideal थी i.e., massless थी and no friction between string and pulley और दोनों side string में tension same थी

But अब हम असली वाली real pulley पढ़ेंगे जिसके पास mass भी है, string और pulley के बीच friction भी है और अब दोनों side tension अलग अलग है।



Torque applied by tension about O gives α to the pulley about O.

$$\tau = (T_2 - T_1)R = I\alpha$$

★ Friction is sufficient to avoid slipping.

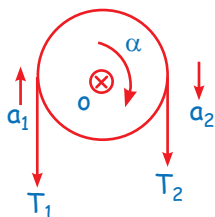
★ $a_1 = R\alpha$ (bcz no slipping between pulley and rope)

★ Similarly, $a_2 = R\alpha$

★ $\alpha \rightarrow \text{CW}$ (if)

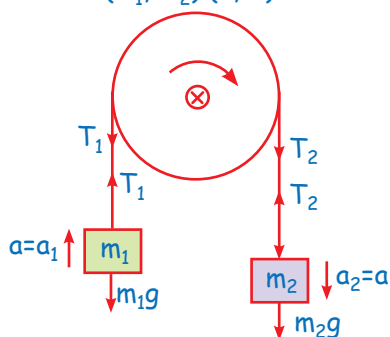
$$\tau_o = I_o \alpha$$

$$T_2R - T_1R = I_o \alpha$$



Q. Find acceleration of m_1 & m_2 if pulley has MOI I & radius R .

$(m_1, m_2) (I, R)$



$$\text{Sol. } a_2 = R\alpha \quad \dots(1)$$

$$a_1 = a_2 = a = R\alpha \text{ (let)}$$

$$a_1 = R\alpha \quad \dots(2)$$

$$T_1 - m_1g = m_1a_1 \quad \dots(3)$$

$$m_2g - T_2 = m_2a_2 \quad \dots(4)$$

$$\tau_o = I\alpha$$

$$T_2R - T_1R = I\alpha \quad \dots(5)$$

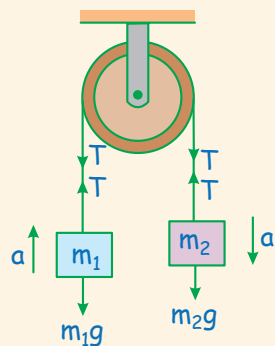
Now you can solve all the eqn. मुझे पता है इनको solve करना आसान काम नहीं है but tension मत ले इसकी भी अपने पास इलाज है अभी तु concept पर focus कर



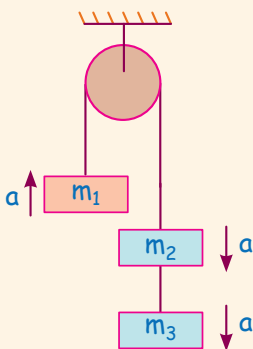
अब तुम्हारे लिए एक मस्त वाला surprise है जिसको set करने में बहुत मेहनत लगी यार.... अगर मजा अजाए तो मुझे insta पर tag कर saleem.nitt.

NLM Question We Already Solved

Pulley is ideal, massless and friction between rope and pulley is zero.



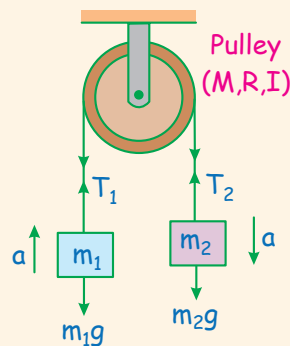
$$a = \frac{m_2g - m_1g}{m_1 + m_2}$$



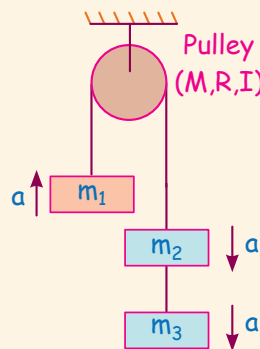
$$a = \frac{m_3g + m_2g - m_1g}{m_1 + m_2 + m_3}$$

Rotation Wale Question

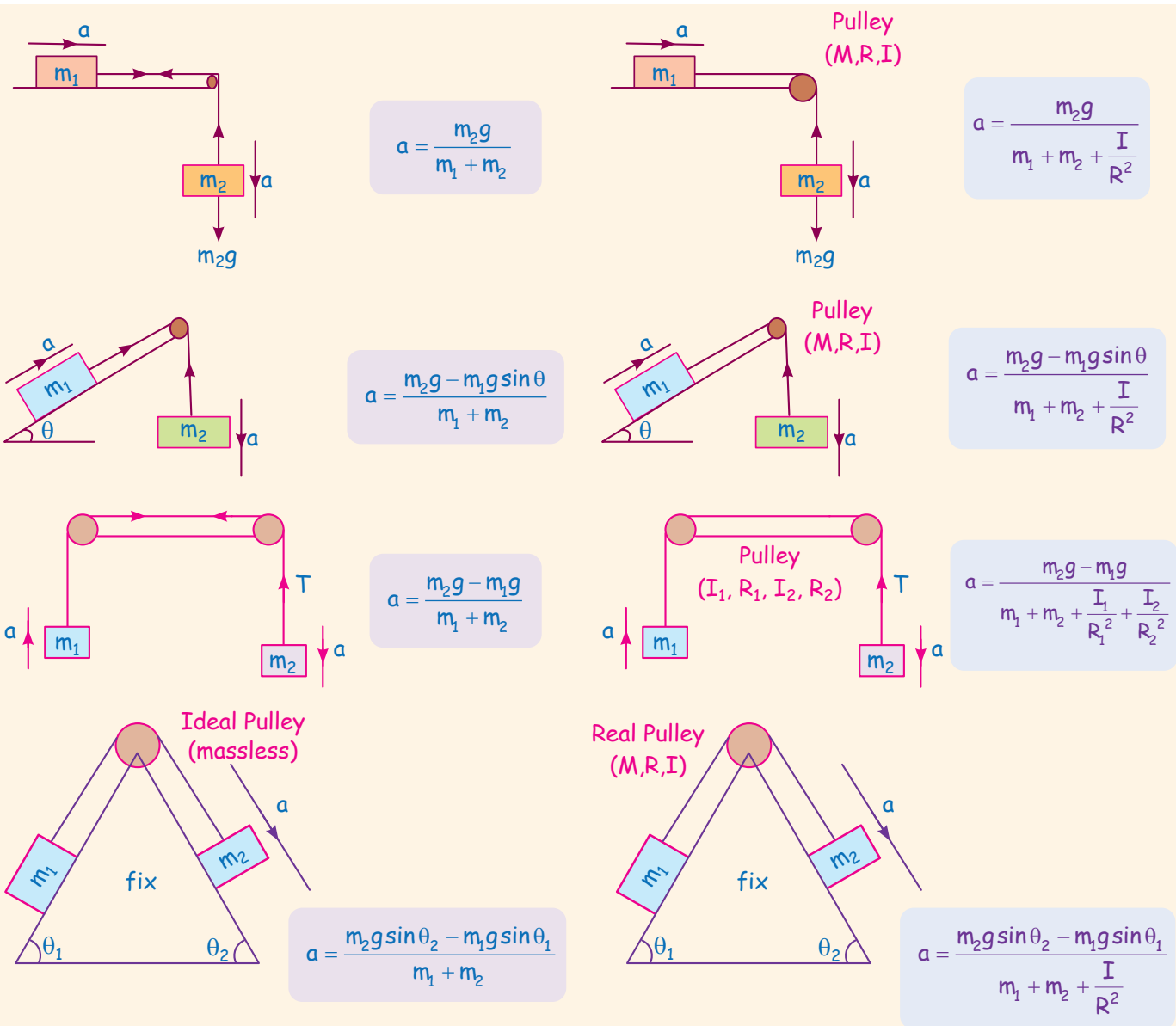
Pulley is real having mass M , radius r and moment of inertia I . friction is sufficient to avoid slipping between pulley and rope.



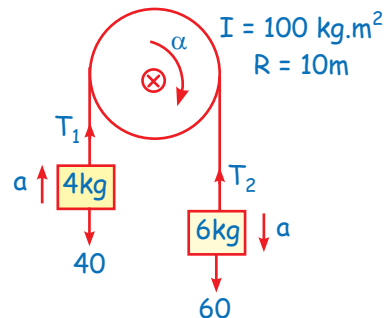
$$a = \frac{m_2g - m_1g}{m_1 + m_2 + \frac{I}{R^2}}$$



$$a = \frac{m_3g + m_2g - m_1g}{m_1 + m_2 + m_3 + \frac{I}{R^2}}$$



Q. Find acceleration of blocks in following?



Sol. $a = 10\alpha$

$$T_1 - 40 = 4a \dots (1)$$

$$60 - T_2 = 6a \dots (2)$$

$$T_2 - T_1 = a \dots (3)$$

$$a = \frac{20}{11}$$

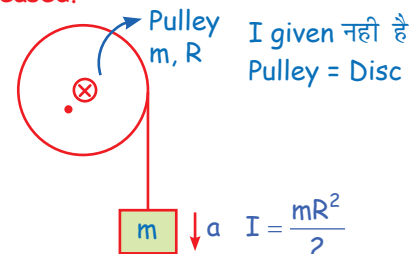
$$T_2 R - T_1 R = I\alpha$$

$$T_2 R - T_1 R = I \frac{a}{R}$$

Trick:

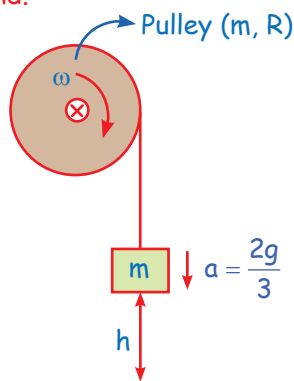
$$a = \frac{m_2 g - m_1 g}{m_1 + m_2 + \frac{I}{R^2}} = \frac{60 - 40}{4 + 6 + \frac{100}{10^2}} = \frac{20}{11}$$

Q. Find acceleration a of block when system is released.

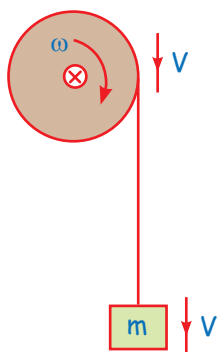


Sol. $a = \frac{mg - 0}{0 + m + \frac{I}{R^2}} = \frac{mg}{m + \frac{mR^2}{2R^2}} = \frac{2g}{3}$

- Q. Find acceleration of block when dropped at height h above ground & velocity with which it hit ground.



Sol. (1) $a = \frac{m \cdot g - 0}{0 + m + \frac{mR^2}{2R^2}} = \frac{2g}{3}$



(2) Speed of block just before it hits the ground

M-1: Apply $V^2 = u^2 + 2as$

$$V^2 = 0 + 2 \times \frac{2g}{3} h$$

$$V = \sqrt{\frac{4gh}{3}}$$

$$V = R\omega \quad \omega = \frac{V}{R}$$

M-2: WET (Work Energy Theory)

$$W_{\text{hinge}} + (WD)_g + (WD)_f + (WD)_T = \Delta KE$$

$$0 + mgh + 0 + 0 = \left(\frac{1}{2} mV^2 + \frac{1}{2} I\omega^2 \right) - 0$$

$$V = (R\omega)$$

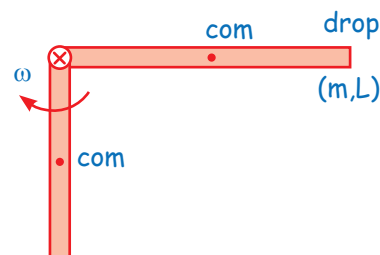
$$mgh = \frac{1}{2} mV^2 + \frac{1}{2} \cdot \frac{mR^2}{2} \cdot \frac{V^2}{R^2}$$

$$mgh = \frac{3}{4} mV^2$$

$$v = \sqrt{\frac{4gh}{3}}$$



- Q. Rod dropped from horizontal position



Find ω of rod when rod reaches at lowest point

Sol. $w_g + w_{\text{hinged}} = \Delta K.E = k_f - k_i$ [WET]
(rod)

$$mg \frac{L}{2} + 0 = \frac{1}{2} I\omega^2 - 0$$

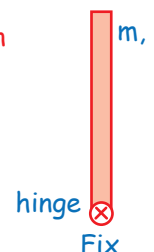
$$mg \frac{L}{2} = \frac{1}{2} \frac{mL^2}{3} \omega^2$$

$$\omega = \sqrt{\frac{3g}{L}}$$

- Q. Rod dropped from vertical position

(1) Find ω of rod become horizontal.

(2) ω of rod when it reaches at lowest point.



Sol. (1)

$$W_g + W_{\text{hinge}} = \Delta KE$$

$$mg \frac{L}{2} + 0 = \frac{1}{2} \frac{mL^2}{3} \omega^2 = 0$$

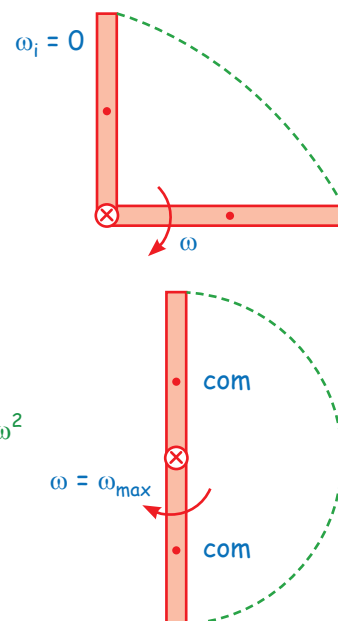
$$\omega = \sqrt{\frac{3g}{L}}$$

(2)

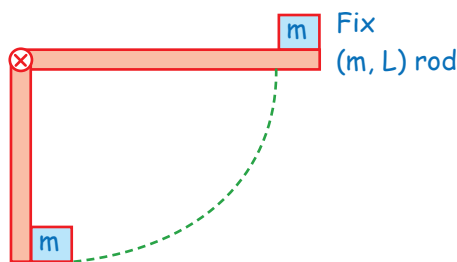
$$W_g + W_{\text{hinges}} = \Delta K.E$$

$$mgL + 0 = \frac{1}{2} \frac{mL^2}{3} \times \omega^2$$

$$\omega = \sqrt{\frac{6g}{L}}$$



- Q. Block is fix to the rod and dropped from horizontal position as shown in fig. Find ω when rod becomes vertical.



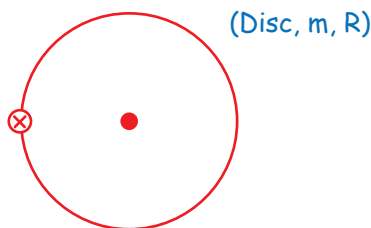
From WET

Sol. $W_g + W_{\text{hinge}} = 0 = \Delta K.E = \frac{1}{2} I \omega^2 - 0$

$$mg \frac{L}{2} + mgL = \frac{1}{2} \left[\frac{mL^2}{3} + mL^2 \right] \omega^2$$

$\omega = \checkmark$

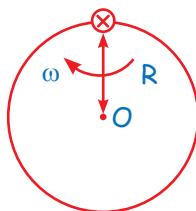
- Q. Disc can move about hinge in vertical plane about axis as shown in figure, dropped from here. Find ω when it reaches lowest position



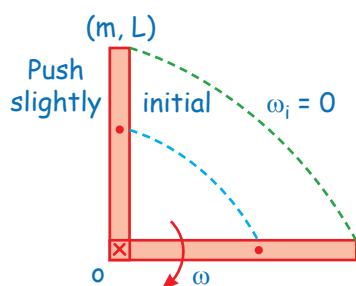
Sol. $mgR + 0 = \frac{1}{2} \left[\frac{MR^2}{2} + MR^2 \right] \omega^2$

$$mgR = \frac{3}{4} MR^2 \omega^2$$

$$\omega = \sqrt{\frac{4g}{3R}}$$



- Q. Vertical rods is pushed slightly.



Find angular velocity ω , angular acceleration α , acceleration of centre of mass a_{com} and hinged forced when rod become horizontal.

Sol.

1. $\omega = ?$

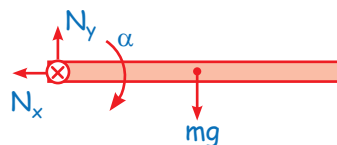
Apply WET

$$W_g + W_{\text{hinged}} = \Delta K.E = k_f - k_i$$

$$+ mg \frac{L}{2} + 0 = \frac{1}{2} \frac{mL^2}{3} \omega^2 - 0$$

$$\omega = \sqrt{\frac{3g}{L}}$$

2. $\alpha = ?$



$$\tau_0 = I_0 \alpha$$

$$mg \frac{L}{2} = \frac{mL^2}{3} \alpha \quad \alpha = \frac{3g}{2L}$$

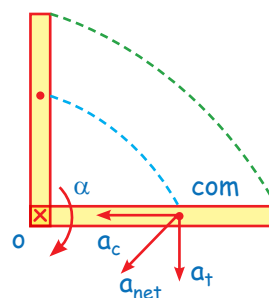
SKC



याद है ये question पहले भी किया था जब horizontal rod को drop किया था तब at $t = 0$, $\omega = 0$ but now जब rod horizontal हुई तब ω है so centripetal acc भी होगा और α है इसलिए a_t भी होगा।



3.



$$a_c = r\omega^2$$

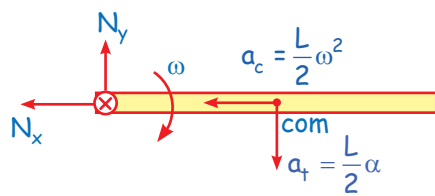
$$a_t = r\alpha$$

$$a_{\text{net}} = \sqrt{a_t^2 + a_c^2}$$

$$(a_{\text{com}})_{\text{tangential}} = r\alpha = \frac{L}{2} \alpha$$

$$(a_{\text{com}})_{\text{centripetal}} = r\omega^2 = \frac{L}{2} \omega^2$$

4. Find hinge force at horizontal position



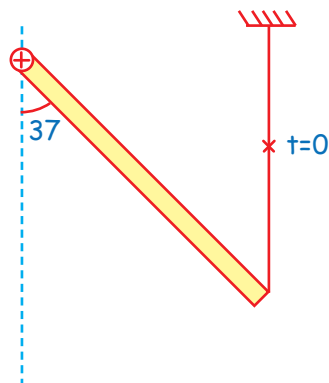
in y direction $(F_{\text{net}})_{\text{ext}} = m(a_{\text{com}})_y$

$$Mg - N_y = \frac{mL}{2}\alpha$$

in x direction $N_x = \frac{mL}{2}\omega^2$

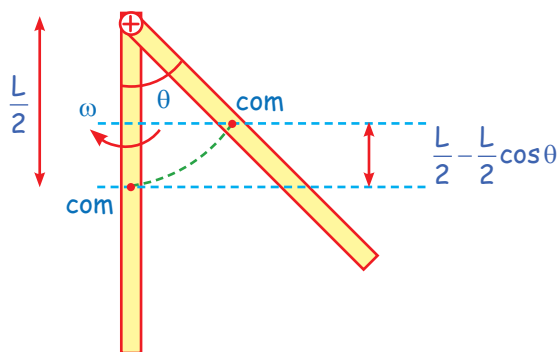
Net hinged force = $\sqrt{N_x^2 + N_y^2}$

Q. String is cut at $t = 0$



Find angular velocity ω of rod when rod reaches at lowest point.

Sol. Centre of mass moves as shown in figure.



From work energy theorem

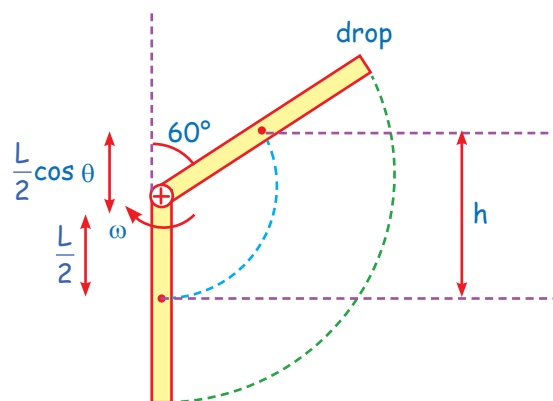
$W_g + W_{\text{hinged}} = \Delta KE$ [WET]

$$mgh + 0 = \frac{1}{2} \frac{mL^2}{3} \omega^2 - 0$$

$$h = \frac{L}{2} - \frac{L}{2} \cos \theta$$

$$\omega = \sqrt{\frac{3g}{5L}}$$

Q. Find angular velocity when rod becomes vertical, as shown in figure.

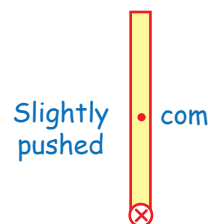


Sol. $W_g + W_{\text{hinged}} = \Delta KE$ [WET]

$$Mg \left(\frac{L}{2} \cos \theta + \frac{L}{2} \right) + 0 = \frac{1}{2} \frac{mL^2}{3} \omega^2 - 0$$

$$\omega = \sqrt{\frac{9g}{2L}}$$

Q. When rod reaches at lowest point. Find ω , α , a_{com} , hinged force.



Sol. From work energy theorem,

$W_g + W_{\text{Hinge}} = \Delta KE$

$$mgL + 0 = \frac{1}{2} \frac{mL^2}{3} \omega^2 - 0$$

$$\omega = \sqrt{\frac{6g}{L}}$$

torque about hinge $(\tau_0) = 0 = I\alpha$

acceleration of centre of mass

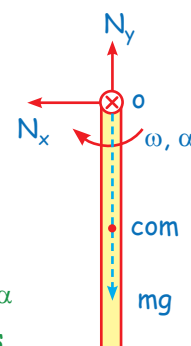
$$a_{\text{com}} = 3g$$

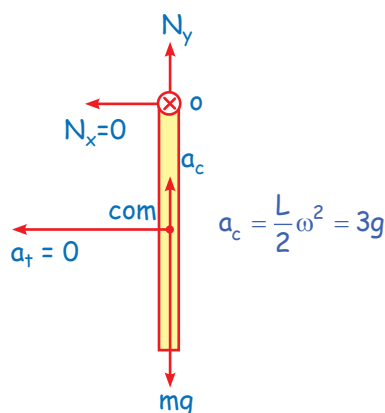
tangential $a_t = \frac{L}{2}\alpha$

$$\alpha = 0, a_t = 0$$

centripetal acceleration $a_c = \frac{L}{2}\omega^2$

$$= \frac{L}{2} \cdot \frac{6g}{L} = 3g$$



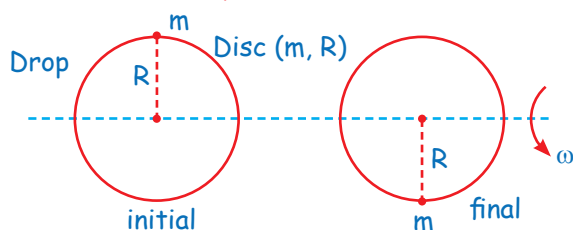


$$N_y - mg = m3g$$

$$N_y = 4mg; N_x = 0$$

$$\text{Net hinge force} = 4mg$$

- Q. Disc rotates about its diameter with particle of mass 'm' fix on it. Dropped from initial position find ω in final position.



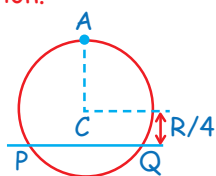
Sol. (1) $W_g + W_{\text{hinge}} = \Delta KE = KE_f - KE_i$

$$Mg(2R) + 0 + 0 = \frac{1}{2} \left(\frac{MR^2}{4} + MR^2 \right) \omega^2$$

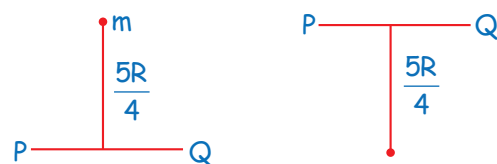
$$\omega = \sqrt{\frac{16g}{5R}}$$

(2) Speed of particle
at final position = $R\omega$

- Q. A uniform circular disc has radius R and mass m. A particle also of mass m is fixed at a point A on the edge of the disc as in figure. The disc can rotate freely about a fixed horizontal chord PQ that is at a distance R/4 from the centre C of the disc. The line AC is perpendicular to PQ. Initially the disc is held vertical with the point A at its highest position. It is then allowed to fall so that it starts rotating about PQ. Find the linear speed of the particle at it reaches its lowest position.



Sol. $\sqrt{5gR}$



$$KE_f = \frac{1}{2} I \omega^2; KE_i = 0; W_g = mg \left(\frac{2R}{4} \right) + mg2 \times \left(\frac{5R}{4} \right)$$

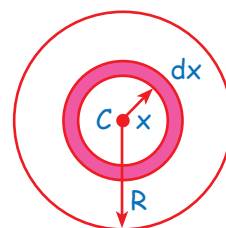
$$I = m \left(\frac{5R}{4} \right)^2 + \left(\frac{MR^2}{4} + \frac{MR^2}{16} \right) = MR^2 \left(\frac{25}{16} + \frac{1}{4} + \frac{1}{16} \right)$$

$$W_g = KE_f - KE_i$$

$$\text{We get } v = \left(\frac{5R}{4} \right) \omega$$

- Q. A uniform disc of radius R and mass M is spinned to an angular speed ω_0 in its own plane about its centre and then placed on a rough horizontal surface such that plane of the disc is parallel to the horizontal plane. If co-efficient of friction between the disc and the surface is μ then how long will it take for the disc to come to stop.

- Sol. Consider a differential circular strip of the disc of radius x and thickness dx. Mass of this strip is $dm = 2\rho\pi x dx$.



where $\rho = \frac{M}{\pi R^2}$. Frictional force on this strip is

along the tangent and is equal to $dF = \mu\rho 2\pi x dx g$

Torque on the strip due to frictional force is equal to $d\tau = \mu\rho g 2\pi x^2 dx$, disc is supposed to be the combination of number of such strips hence torque on the disc is given by

$$\tau = \int d\tau = \mu\rho g 2\pi \int_0^R x^2 dx = \mu\rho g 2\pi \frac{R^3}{3}$$

$$\Rightarrow \tau = \mu Mg(2/3)R \Rightarrow \alpha = \frac{2\mu MgR}{3 \left(\frac{MR^2}{2} \right)} = \frac{4}{3} \frac{\mu g}{R}$$

The α is opposite to the ω

$$\therefore \omega(t) = \omega_0 + \alpha t$$

$$\Rightarrow 0 = \omega_0 - \frac{4\mu g}{3R} t \Rightarrow t = \frac{3\omega_0 R}{4\mu g}$$

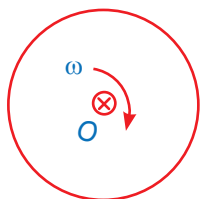
ANGULAR MOMENTUM ($\vec{L}$)

- ★ It is the total amount of rotational motion contained in a system.
- ★ It is a moment of momentum.
- ★ Angular momentum of a moving particle $\vec{V}$ about a point P is given by $\vec{L} = \vec{r} \times m\vec{V} = \vec{r} \times \vec{p}$
 $\vec{r}$ is the position vector of the particle relative to the origin (or axis of rotation).
- ★ The angular momentum L of a rotating object is defined as the product of the moment of inertia I and the angular velocity ω :

$$L = I\omega$$

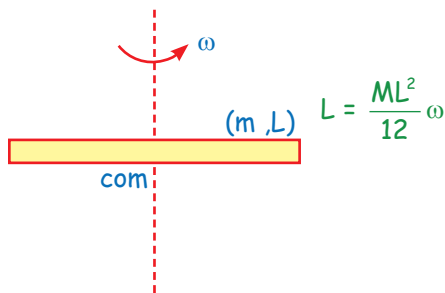
Q. Find Angular momentum in following cases:

- (1) (Disc, m, R)



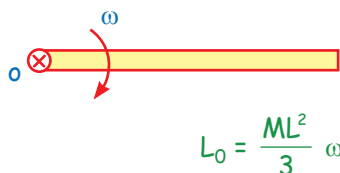
$$\begin{aligned}\vec{L}_O &= I\omega \\ &= \frac{MR^2}{2} \omega (-\hat{k})\end{aligned}$$

- (2)



$$L = \frac{ML^2}{12} \omega$$

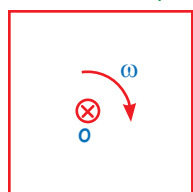
- (3)



$$L_O = \frac{ML^2}{3} \omega$$

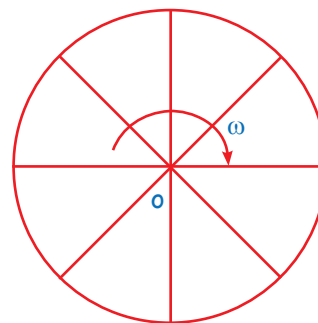
(square plate m, L)

- (4)



$$L_O = \frac{ML^2}{6} \omega$$

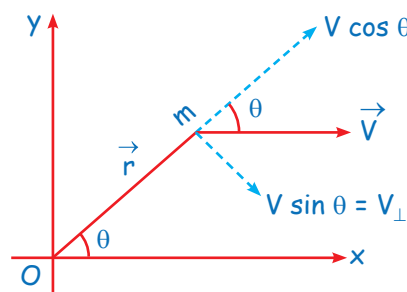
Q. Find K.E & Angular momentum about axis passing through O and perpendicular to the plane arrangement of ring of mass m and radius R and 8 rods of mass m each and length R rotating as shown in figure.



$$\text{Sol. } L = \left(MR^2 + \frac{MR^2}{3} \times 8 \right) \omega$$

$$KE = \frac{1}{2} \left(MR^2 + \frac{MR^2}{3} \times 8 \right) \omega^2$$

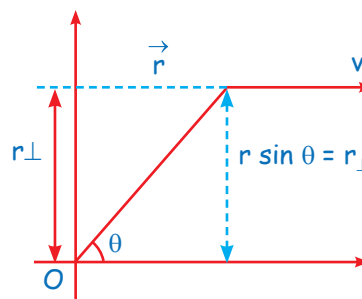
ANGULAR MOMENTUM OF PARTICLE



$$\vec{L} \text{ of particle about } O = \vec{r} \times m\vec{V}$$

$$L = r m v \sin \theta \text{ (magnitude)}$$

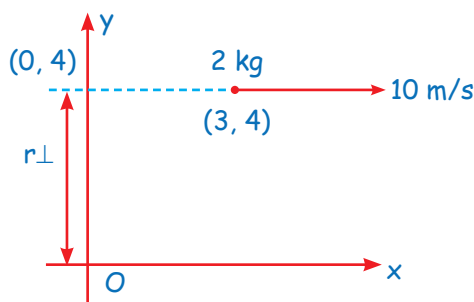
$$L = r m v_{\perp}$$



$$L = m v (r \sin \theta)$$

$$L = m v r_{\perp}$$

Q. Find angular momentum about 'O'.



Sol. $L_0 = mvr_{\perp} = 2 \times 10 \times 4 = 80 \text{ (c.w.)} = -80\hat{k}$

Or

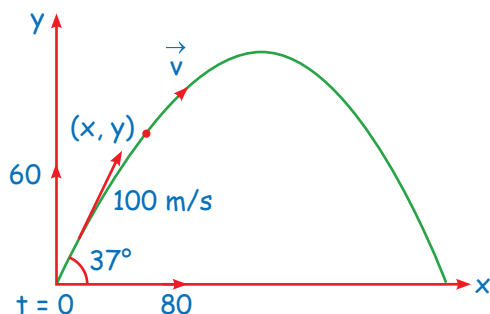
$$\vec{L}_0 = \vec{r} \times m\vec{v}$$

$$\vec{L} = (3\hat{i} + 4\hat{j}) \times (20\hat{i})$$

$$\vec{L} = -80\hat{k}$$

JEE Mains Favourite question

Q. Calculate angular momentum 2s after body of mass 2kg is projected at 100 m/s.



Sol. Given, mass = 2 kg, initial velocity = $80\hat{i} + 60\hat{j}$

velocity at $t = 2$, $\vec{v} = 80\hat{i} + 40\hat{j}$

Position $\vec{r}$ at $t = 2$ s

$$x = 80 \times 2 = 160$$

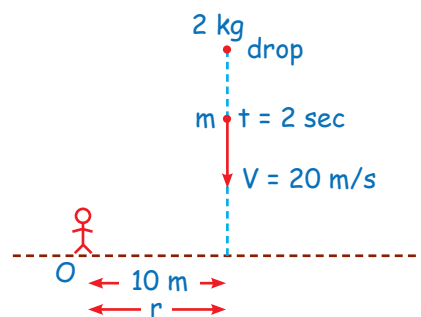
$$y = 60 \times 2 - \frac{1}{2} \times 10 \times 2^2 \Rightarrow y = 100$$

$$(x, y) = (160, 100) \Rightarrow \vec{r} = (160\hat{i} + 100\hat{j})$$

Angular momentum of particle at $t = 2$ sec abt projection point

$$= \vec{r} \times m\vec{v} = (160\hat{i} + 100\hat{j}) \times (160\hat{i} + 80\hat{j})$$

Q. A particle of mass 2 kg is dropped $t = 0$ as shown in figure. Find angular momentum of particle about O at $t = 2$ sec.

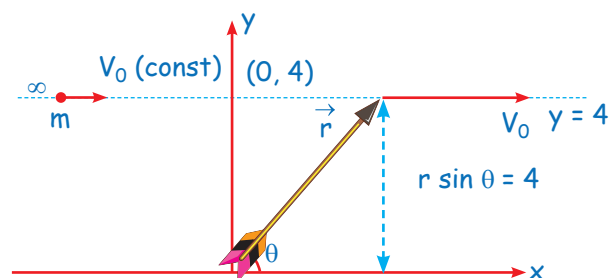


Sol. Velocity at $t = 2$ s = $0 + 10 \times 2 = 20 \text{ m/s}$

Angular momentum L of particle about O = mvr

$$= 2 \times 20 \times 10 = 400 \text{ (cw)} = 400(-\hat{k})$$

Q. Particle is moving on st line $y = 4$ parallel to x-axis with constant velocity V_0 from $-\infty$ to $+\infty$.



Choose correct option about Angular momentum of particle abt 'O' is

- (1) zero
- (2) magnitude constant, direction constant
- (3) first increase then decrease
- (4) first increase then increase
- (5) magnitude constant direction change.

Sol. $L_0 = (r \cos \theta \hat{i} + r \sin \theta \hat{j}) \times mV_0 \hat{i}$

$$= mr \sin \theta V_0 (-\hat{k})$$

$$= mV_0 (r \sin \theta) (-\hat{k}) = mV_0 4(-\hat{k})$$

2nd option is correct.

ANGULAR MOMENTUM CONSERVATION

In the absence of external torques, the total angular momentum of a system remains constant.

$$\vec{\tau} = \frac{d\vec{L}}{dt}$$



काम का डब्बा

★ जैसा हमने COM में पढ़ा था कि अगर system पर net external force zero है तो system का momentum conserve होगा उसी प्रकार जिस point के about net torque zero है उस point के about angular momentum conserve होगा

$$(\vec{F}_{\text{net}})_{\text{ext}} = \frac{d\vec{p}}{dt}$$

$$\text{If } (F_{\text{net}})_{\text{ext}} = 0$$

$P \rightarrow$ conserve

$$P_i = P_f$$

$$\vec{\tau}_{\text{ext}} = \frac{d\vec{L}}{dt}$$

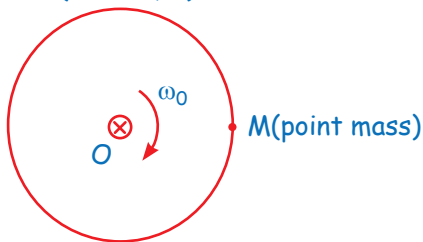
$$\text{If } (\tau_{\text{net}})_{\text{ext}} = 0$$

$\vec{L} =$ Angular momentum will be conserve

$$\vec{L}_i = \vec{L}_f$$

Q. A point mass M is generally placed on the periphery of disc at a distance R from centre such that both rotate together without slipping, find ω after चिपका-चिपकी.

(Disc M, R)



Sol. M-1: Net torque about 'O' is zero so angular momentum will be conserved about O

$$L_i = L_f \text{ (abt 'O')}$$

$$I\omega_0 + 0 = I\omega + MvR$$

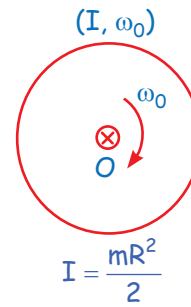
Disc particle disc particle

$$\frac{MR^2}{2}\omega_0 + 0 = \frac{MR^2}{2}\omega + M.R.\omega.R$$

$$\frac{MR^2}{2}\omega_0 = \left(\frac{MR^2}{2} + MR^2\right)\omega$$

Now solve and get

Q. An identical disc is placed on disc rotating with angular speed (ω_0) such that after some time both rotate together without slipping find ω_f .



$$I = \frac{mR^2}{2}$$

Sol. Net torque is 0 so angular momentum will be conserved

$$L_i = L_f \text{ (abt 'O')}$$

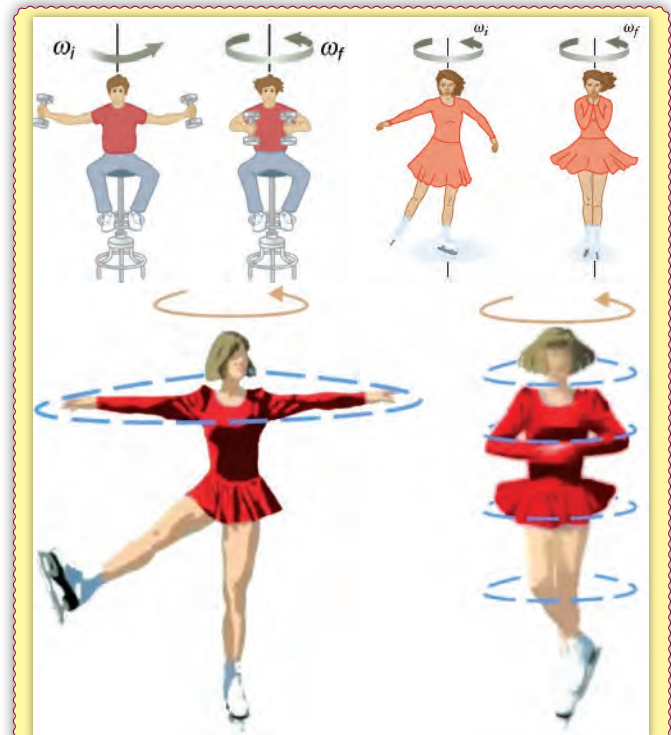
$$I\omega_0 + 0 = I\omega + I\omega$$

$$\omega = \frac{\omega_0}{2}$$

Work done by friction

$$(W)_{\text{friction}} = k_f - k_i = \left(\frac{1}{2}I\omega^2 \times 2\right) - \frac{1}{2}I\omega_0^2$$

$$= \frac{I\omega_0^2}{4} - \frac{I\omega_0^2}{2} = \frac{-I\omega_0^2}{4}$$

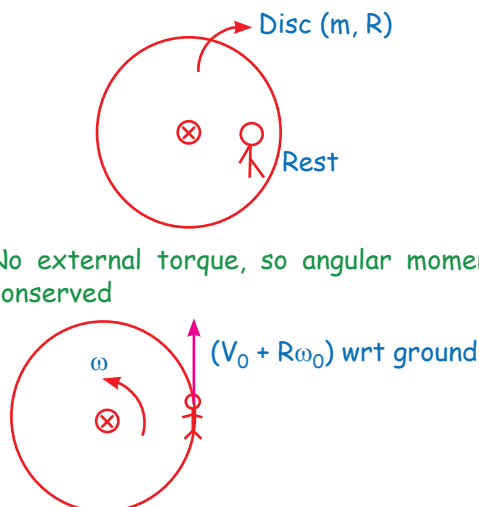


भाई तुझे जिस figure से समझना है समझले here angular momentum is conserve $\Rightarrow I\omega$ constant

I decreases by folding hand $\Rightarrow \omega$ increases but kinetic energy increases bcz $K.E = L^2/2I$

- Q. Man start walking along periphery on disc so that its speed on disc is V_0 wrt disc in Anticlockwise sense. Find ω of disc.

Sol. No external torque, so angular momentum is conserved



$$L_i = L_f$$

$$0 + 0 = \frac{MR^2}{2} \omega + m(V_0 + R\omega)R$$

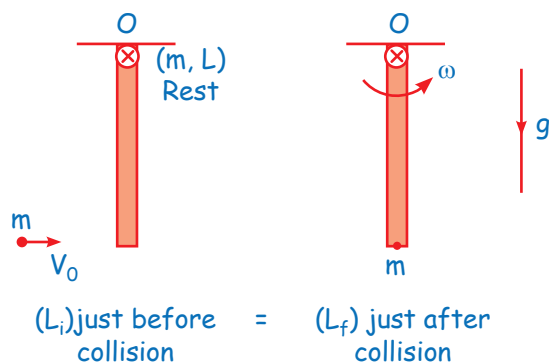
$$-mV_0R = \left(mR^2 + \frac{mR^2}{2} \right) \omega$$

$$\omega = \frac{-2V_0}{3R}$$

अगला question बहुत important article है जिसे advance ने भी दो से तीन बार पूछा है... इस तरह के सारे सवालो मे कुछ नहीं करना बस hinge point के about before and after collision angular momentum of system conserve कर देना



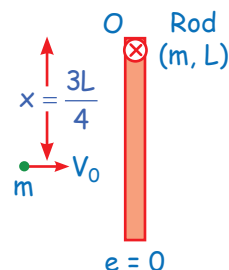
- Q. A rod of mass m and length L is hinged at O as shown in figure. Particle moving at velocity V_0 sticks to end of rod after collision find combined angular velocity.



Sol. $L_i = L_f$ - (Angular momentum about O is conserved)

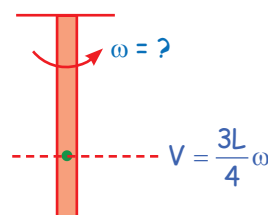
$$mV_0L + 0 = m \cdot L \omega \cdot 2 + \frac{mL^2}{3} \omega$$

- Q. If a particle of mass m collides with rod as shown in figure and sticks to rod. Find ω



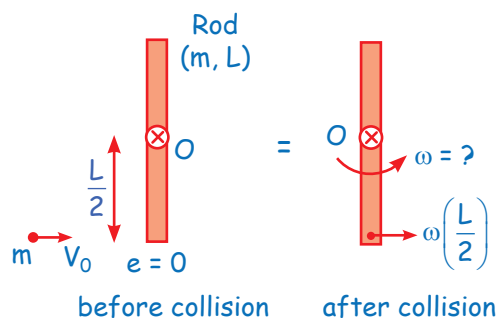
Sol. $L_i = L_f$ (about O)

$$mV_0x + 0 = (mx^2)\omega + \frac{mL^2}{3} \omega$$



$$mV_0 \frac{3L}{4} + 0 = m \cdot \frac{3L}{4} \omega \cdot \frac{3L}{4} + \frac{mL^2}{3} \omega$$

- Q. A rod of mass m and length L is hinged at O as shown in figure. Particle moving at velocity V_0 sticks to rod after collision find combined angular velocity.

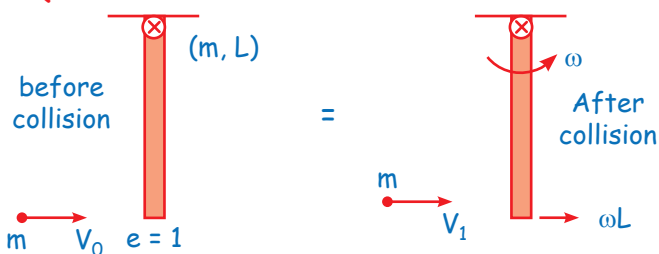


Sol. Net torque is 0 so angular momentum will be conserved

$$L_i = L_f$$

$$mV_0 \frac{L}{2} + 0 = \frac{mL}{2} \omega \cdot \frac{L}{2} + \frac{mL^2}{12} \omega$$

Q. Find ω after collision.



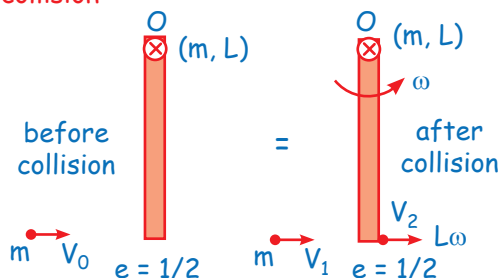
Sol. $L_i = L_f$

$$mV_0L + 0 = mV_1L + \frac{mL^2}{3}\omega \quad \dots(1)$$

$$e = 1 = \frac{\omega L - V_1}{V_0} \quad \dots(2)$$

Now you can solve and get V_1 and ω .

Q. Coefficient of restitution = $1/2$. Find ω after collision



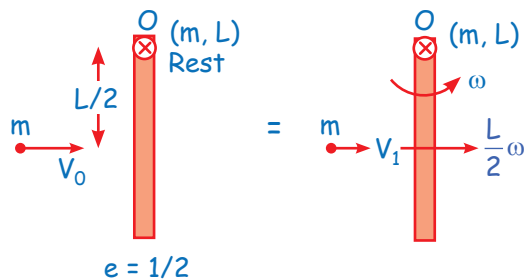
Sol. $L_i = L_f$ (abt O)

$$mV_0L = mV_1L + \frac{mL^2}{3}\omega \quad \dots(1)$$

$$e = \frac{1}{2} = \frac{V_2 - V_1}{U_1 - U_2} = \frac{L\omega - V_1}{V_0} \quad \dots(2)$$

Now you can solve both eqn and get the answer.

Q.



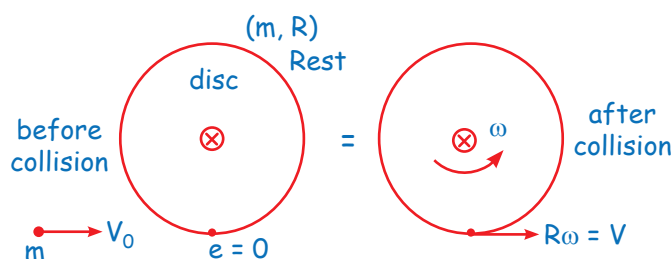
Sol. $L_i = L_f$ (abt 'O')

$$mV_0 \frac{L}{2} + 0 = mV_1 \frac{L}{2} + \frac{mL^2}{3}\omega \quad \dots(1)$$

$$e = \frac{1}{2} = \frac{\frac{L}{2}\omega - V_1}{V_0} \quad \dots(2)$$

= ✓

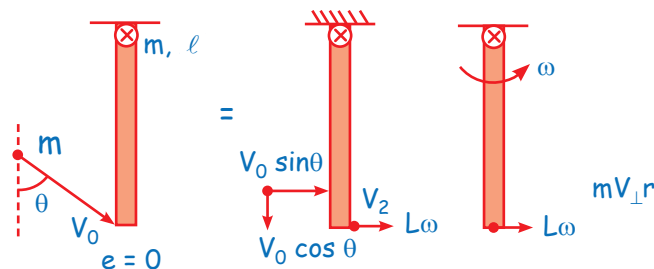
Q. A particle of mass m collides with the disc of mass m and radius R as shown in figure and sticks to it. Find ω



Sol. $L_i = L_f$ (abt 'O')

$$mV_0R + 0 = m.R\omega.R + \frac{mR^2}{2}\omega$$

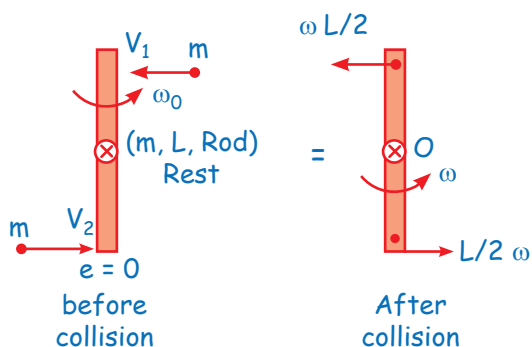
Q. Find ω , if particle hits rod end at an angle θ with vertical.



Sol. $L_i = L_f$

$$mV_0 \sin \theta + 0 = m.L\omega.r + \frac{mL^2}{3}\omega$$

Q. Find ω , if two particles collide with rod and stick as shown in figure.

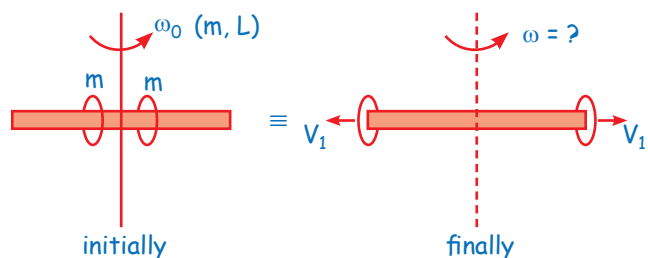


Sol. $L_i = L_f$ (abt O)

$$mV_1 \frac{L}{2} + mV_2 \frac{L}{2} = \frac{mL}{2}\omega \frac{L}{2} + \frac{mL}{2}\omega \frac{L}{2} + \frac{mL^2}{12}\omega$$

$P_1 \quad P_2 \quad P_1 \quad P_2 \quad \text{rod}$

Q. Rings slips outwards as shown in figure. Find ω



Consider ring as point mass $\mu = 0$ everywhere.

Sol. Net torque is zero so angular momentum will be conserved.

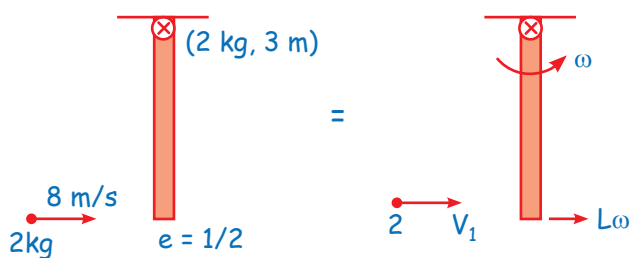
$$L_i = L_f$$

$$0 + 0 + \frac{mL^2}{12}\omega_0 = \frac{mL}{2}\omega \cdot \frac{L}{2} + \frac{mL}{2}\omega \cdot \frac{L}{2} + \frac{mL^2}{12}\omega$$

$$K.E_i = K.E_f$$

$$\frac{1}{2} \frac{mL^2}{12} \omega_0^2 + 0 + 0 = \frac{1}{2} \frac{mL^2}{12} \omega^2 + \left[\frac{1}{2} m v_1^2 + \frac{1}{2} m (R\omega)^2 \right] \times 2$$

Q. A particle is moving on the frictionless horizontal floor and its collide at the rod at one end as shown in fig. and data. Find impulse imparted by particle on the rod.



Sol. $L_i = L_f$ $I = \frac{mL^2}{3} = 6$

$$2 \times 8 + 3 \times 0 = 2 \times V_1 \times 3 + 6\omega$$

$$8 = V_1 + \omega \quad \dots(1)$$

$$e = \frac{1}{2} = \frac{L\omega - V_1}{8} = \frac{3\omega - V_1}{8}$$

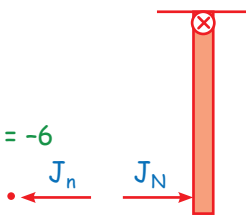
$$4 = 3\omega - V_1 \quad \dots(2)$$

Solving we get,

$$\omega = 3, V_1 = 5$$

$$(\Delta P) \text{ particle} = 2 \times 5 - 2 \times 8 = -6$$

$$\vec{J}_N = -6 \text{ NS}$$



- (1) Impulse imparted by the rod on the particle
= change in momentum of particle = -6 (6 पीछे)
- (2) Impulse imparted by particle on the rod
= 6 (आगे) = +6i

ऐसा Jhapad जो नचा दे उसके effect ko 'angular impulse' कहते हैं।

$m \rightarrow \text{MOI}$

$v \rightarrow w$

$a \rightarrow \alpha$

$F \rightarrow \vec{\tau} = \vec{r} \times \vec{F}$

$P \rightarrow \vec{L} = \vec{r} \times \vec{P}$

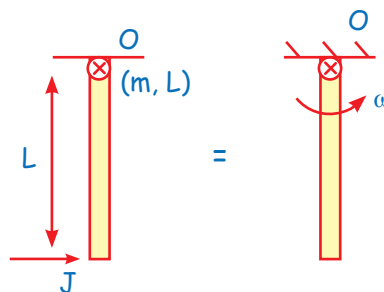
$J \rightarrow \vec{H} = \vec{r} \times \vec{J}$



ANGULAR IMPULSE ($\vec{H}$)

- ★ It is the moment of impulse
- ★ The net angular impulse acting on a rigid body is equal to the change in angular momentum of the body.
- ★ $\vec{H} = \int \vec{\tau}_{ext} dt$
- ★ $\therefore \vec{H} = \Delta \vec{L} = \vec{L}_f - \vec{L}_i$
- ★ $\vec{H} = \vec{r} \times \vec{J}$
- $\vec{H} = \vec{r} \times \vec{J}$ Impulse = ΔP
- $\vec{H} = \int \vec{\tau} \cdot dt = \Delta \vec{L}$ $\vec{J} = \int \vec{F} dt = \Delta \vec{P}$
- ★ Angular impulse = change in angular momentum.

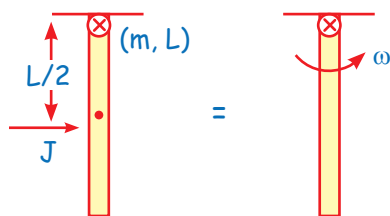
Q.



Sol. $L = J \times L = I \cdot \omega$

$$J \times L = \frac{mL^2}{3} \omega - 0 \Rightarrow \omega = \frac{3J}{mL}$$

Q. Impulse is given to rod. Find ω .



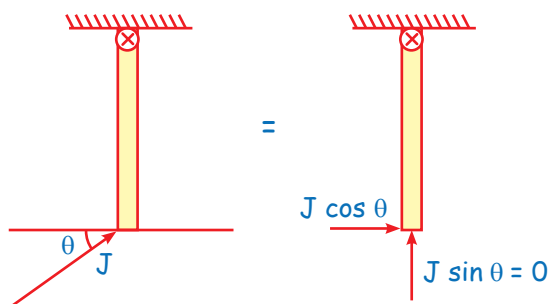
Sol. $\vec{H} = \Delta \vec{L} = \vec{L}_f - \vec{L}_i$

Angular impulse = $I\Delta\omega$

$$J \cdot \frac{L}{2} = \frac{mL^2}{3} \omega - 0$$

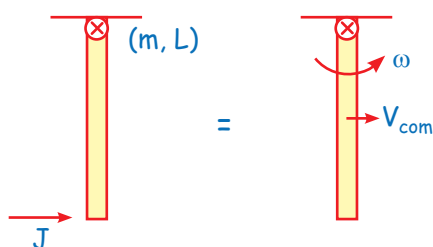
$$\omega = \frac{3J}{2mL}$$

Q. Impulse is given at an angle θ as shown find ω ?



Sol. $J \cos \theta \times L = \frac{mL^2}{3} \omega - 0$

Q. Find (1) ω and (2) impulse by hinge



Sol. (1) $JL = \frac{mL^2}{3} \omega$

$$\omega = \frac{3J}{mL}$$

$$V_{com} = \frac{L}{2} \omega$$

$$(2) J_H = \frac{L}{2} \cdot \frac{3J}{mL} = \frac{3J}{2m}$$

$$J + J_H = mV_{com} - 0$$

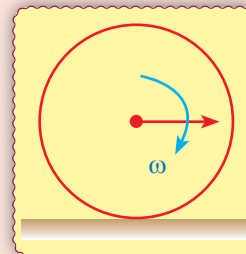
$$J + J_H = m \cdot \frac{3J}{mL}$$

$$J_H = \frac{J}{2}$$

	Quantity	Linear	Rotational
1.	Inertia	m	$\sum m_i r_i^2$ or $\int r^2 dm$
2.	Newton's Second Law	$\vec{F}_{ext} = m\vec{a}$ $\vec{F}_{ext} = \frac{d\vec{p}}{dt}$	$\vec{\tau}_{ext} = I\vec{\alpha}$ $\vec{\tau}_{ext} = \frac{d\vec{L}}{dt}$
3.	Work	$W_{lin} = \int \vec{F} \cdot d\vec{s}$	$W_{rot} = \int \vec{\tau} \cdot d\vec{\theta}$
4.	Kinetic Energy	$K_{lin} = \frac{1}{2}mv^2$	$K_{rot} = \frac{1}{2}I\omega^2$
5.	Work Energy Theorem	$W_{lin} = \Delta K_{lin}$	$W_{rot} = \Delta K_{rot}$
6.	Impulse	$\vec{I} = \int \vec{F}_{ext} dt$	$\vec{J} = \int \vec{\tau}_{ext} dt$
7.	Momentum	$\vec{p} = m\vec{v}$	$\vec{L} = I\vec{\omega}$
8.	Impulse Momentum Theorem	$\vec{I} = \Delta \vec{p}$	$\vec{J} = \Delta \vec{L}$
9.	Power	$P = \vec{F} \cdot \vec{v}$	$P = \vec{\tau} \cdot \vec{\omega}$

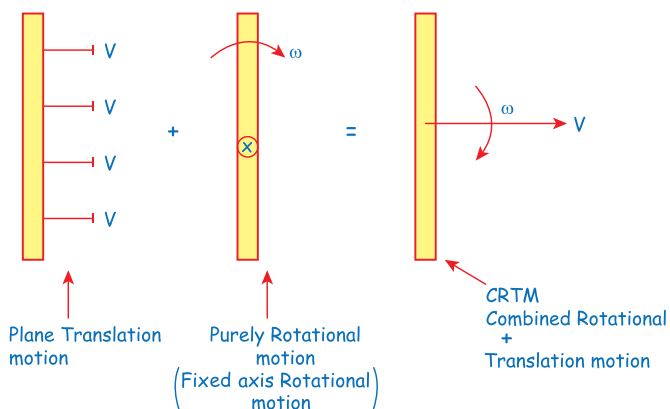
COMBINED ROTATIONAL + TRANSLATION MOTION (CRTM)

चलो अब शुरू करते हैं rotation का सबसे फोड़ article CRTM or CTRM... ऐसे समझ जैसे चलती cycle/car का पहिया (wheel)... जो घूम भी रहा है और चल भी रहा है।



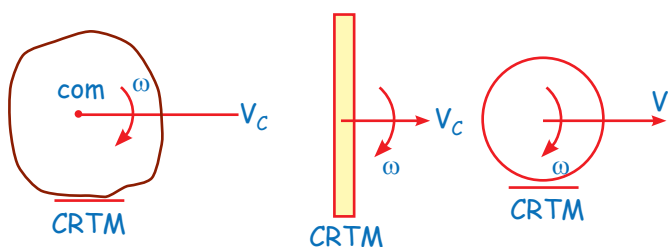
भाई तू cycle के wheel पर focus कर





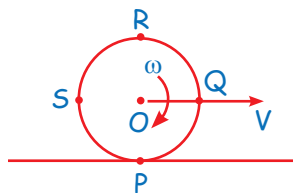
★ CRTM refers to the motion of an body that is both translating (moving in space) and rotating (spinning about an axis).

★ CRTM can be assumed as combination of translation motion of COM and rotation about an axis through COM.



CRTM में ये बहुत जरूरी है की हमें किसी point की velocity और acceleration निकालना आना चाहिए चलो पहले इसकी practice करते हैं।

Q. Disc is rotating with angular velocity ω & has translational velocity V .



Find the velocity of points P, Q, R, S, O in ground frame.

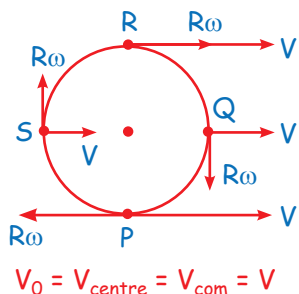
Sol. $\vec{V}_P = \vec{V}_{P/COM} + \vec{V}_{COM}$

$$\vec{V}_R = \vec{V}_{top} = \vec{V}\hat{i} + R\omega\hat{i}$$

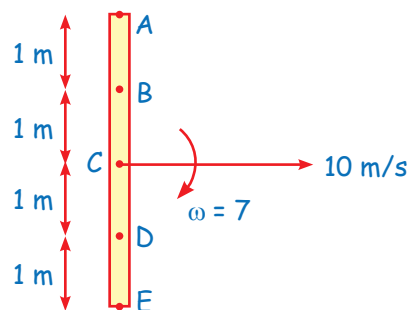
$$\vec{V}_Q = \vec{V}\hat{i} - R\omega\hat{j}$$

$$\vec{V}_P = \vec{V}\hat{i} - R\omega\hat{i}$$

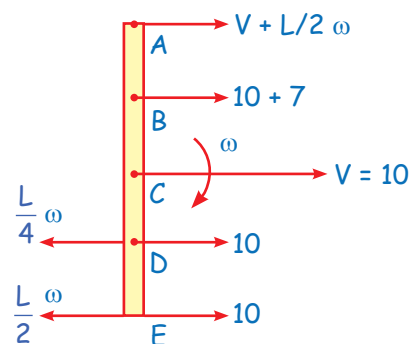
$$\vec{V}_S = \vec{V}\hat{i} + R\omega\hat{j}$$



Q. Find velocity of points on rod which has translational velocity 10m/s & rotational velocity $\omega = 7$



Sol.



$$\vec{V}_A = \left(V + \frac{L}{2} \omega \right) \hat{i} = 10 + 2 \times 10 = 24\hat{i}$$

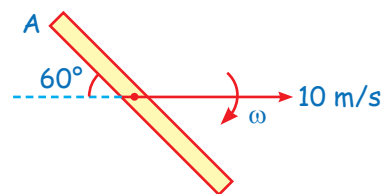
$$\vec{V}_B = V\hat{i} + \frac{L}{4} \omega\hat{i} = 10\hat{i} + 7\hat{i} = 17\hat{i}$$

$$\vec{V}_D = 10\hat{i} - 7\hat{i} = 3\hat{i}$$

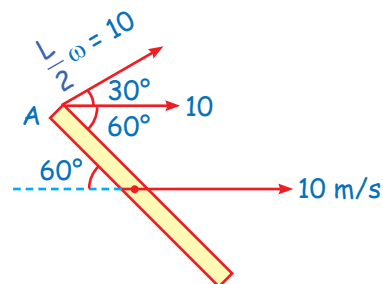
$$\vec{V}_E = 10\hat{i} - 14\hat{i} = -4\hat{i}$$

$$\boxed{V_C = 10\hat{i}}$$

Q. A rod of length L is under (RTM as shown in figure) $L = 4$, $\omega = 5$ rad/sec. Find V_A . Velocity of centre of mass.



Sol.

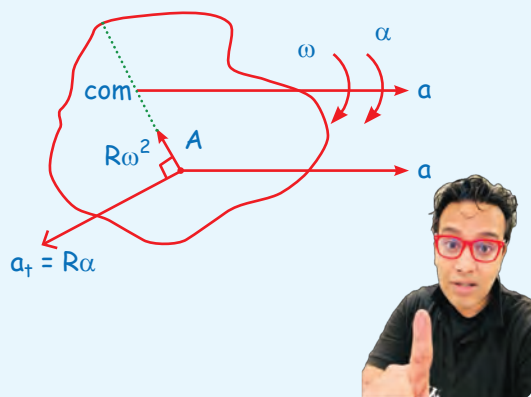


$$V_A = \sqrt{10^2 + 10^2 + 2 \times 10 \times 10 \cos 30^\circ}$$

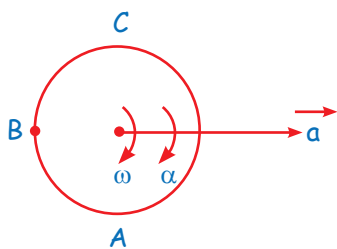
$$\vec{V}_A = 10\hat{i} + (10\cos 30^\circ\hat{i} + 10\sin 30^\circ\hat{j})$$

$$\vec{V}_A = (10 + 5\sqrt{3})\hat{i} + 5\hat{j}$$

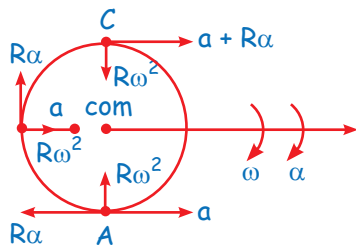
अब हम CRTM मे किसी point का acceleration निकालना सीखेंगे वैसे तो we can say $\vec{a}_p = \vec{a}_{p/com} + \vec{a}_{com}$ but ये ध्यान रखना की यहाँ तीन term आएंगे।



Q. Write acceleration of points on disc as shown in figure.



Sol. $\vec{a}_C = a\hat{i} + R\alpha\hat{i} - R\omega^2\hat{j}$

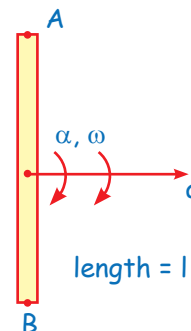


$$\vec{a}_A = a\hat{i} - R\alpha\hat{i} + R\omega^2\hat{j}$$

$$\vec{a}_B = a\hat{i} - R\alpha\hat{j} + R\omega^2\hat{i}$$

$$\vec{a}_D = a\hat{i} + R\alpha\hat{j} - R\omega^2\hat{i}$$

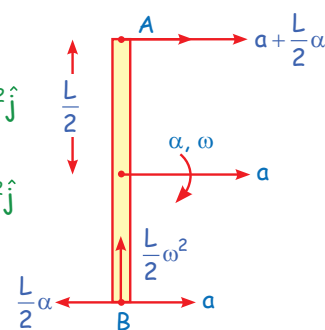
Q. Find acceleration of points A & B for the rod as shown in figure.



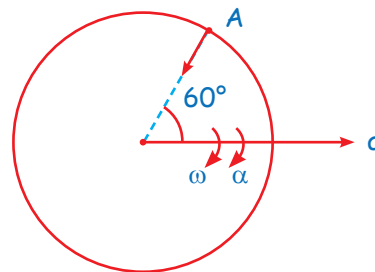
Sol. $\vec{a}_A = a\hat{i} + R\alpha\hat{i} - R\omega^2\hat{j}$

$$\vec{a}_A = a\hat{i} + \frac{L}{2}\alpha\hat{i} - \frac{L}{2}\omega^2\hat{j}$$

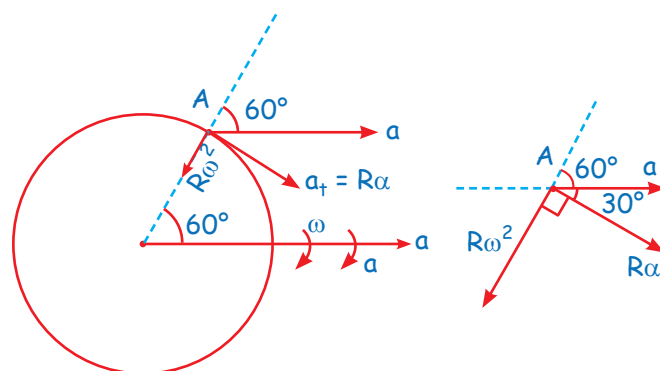
$$\vec{a}_B = a\hat{i} + \frac{L}{2}\alpha\hat{i} - \frac{L}{2}\omega^2\hat{j}$$



Q. Find acceleration of point A.



Sol.



$$\vec{a}_A = (a + R\alpha\cos 30^\circ - R\omega^2\cos 60^\circ)\hat{i} + (R\alpha\sin 30^\circ + R\omega^2\sin 60^\circ)(-\hat{j})$$

$$\vec{a}_A = \left(a + \frac{R\alpha\sqrt{3}}{2} - \frac{R\omega^2}{2}\right)\hat{i} - \left(\frac{R\alpha}{2} + \frac{R\omega^2\sqrt{3}}{2}\right)\hat{j}$$

K.E in CRTM

Total kinetic energy of a body performing CRTM is

$$\text{given by } KE = \frac{1}{2}mV_C^2 + \frac{1}{2}I_C\omega^2$$

Here

V_C is speed of COM

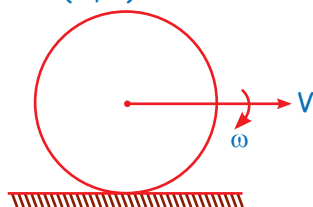
I_C is MOI of body about COM

ये formula गलती से भी मत भूलना



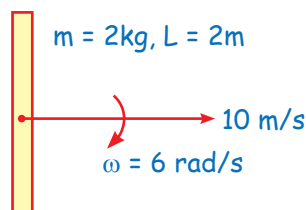
Q. Find K.E of disc under CRTM.

Disc (m, R)



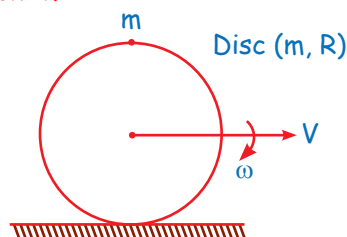
$$\begin{aligned} \text{Sol. } KE &= \frac{1}{2}mV^2 + \frac{1}{2}I\omega^2 \\ &= \frac{1}{2}mV^2 + \frac{1}{2} \cdot \frac{mR^2}{2} \cdot \omega^2 \end{aligned}$$

Q. Find K.E of rod as shown in figure.



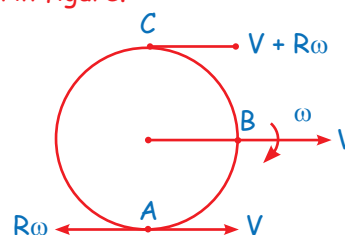
$$\begin{aligned} \text{Sol. } KE &= \frac{1}{2}mV^2 + \frac{1}{2}I\omega^2 \\ &= \frac{1}{2} \times 2 \times 10^2 + \frac{1}{2} \times 2 \times \frac{(2)^2}{12} \times (6)^2 \\ &= 100 + 12 = 112 \text{ J} \end{aligned}$$

Q. Find K.E of disc (m, R) and particle of mass (m) fixed on it.



$$\begin{aligned} \text{Sol. } KE_{\text{Disc}} + KE_{\text{mass}} \\ &= \left(\frac{1}{2}mV^2 + \frac{1}{2}I\omega^2 \right) + \left(\frac{1}{2}m(V + R\omega)^2 \right) \end{aligned}$$

Q. If $V = R\omega$, find velocity of points A, B and C as shown in figure.



$$\text{Sol. } \vec{V}_A = (V - R\omega)\hat{i}$$

$$\text{If } V = R\omega, \vec{V}_A = 0$$

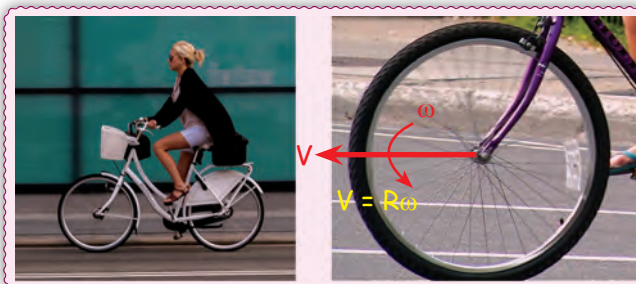
$$\vec{V}_C = (V + R\omega)\hat{i} = 2V\hat{i}$$

$$\vec{V}_B = V\hat{i} - R\omega\hat{j} = V\hat{i} - V\hat{j}$$

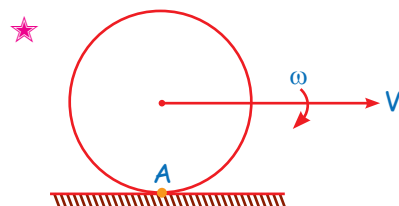
$$\downarrow \quad \quad \quad \downarrow$$

$$V\sqrt{2}$$

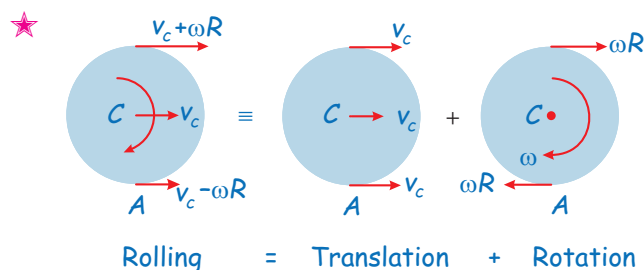
Pure Rolling & Rolling without Slipping



★ The point of the object in contact with the surface has zero velocity relative to the surface.

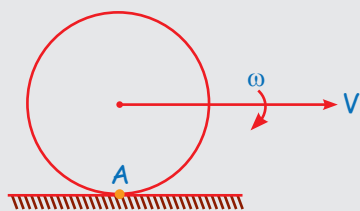


$$V = R\omega \text{ (Condition for rolling)}$$



★ The total kinetic energy of the wheel is the sum of the translational and rotational kinetic energies:

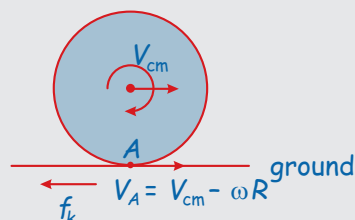
$$K_{\text{total}} = \frac{1}{2}Mv_{\text{cm}}^2 + \frac{1}{2}I\omega^2$$



$$V = R\omega \text{ (Condition for rolling)}$$

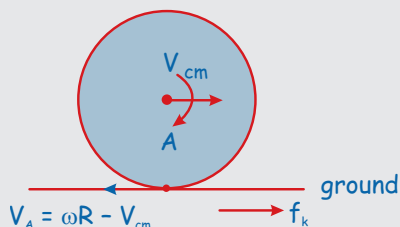
If $V_{cm} > \omega R$ or $V_{cm} < \omega R$ then Slipping (no pure rolling)

Case-I: $V_{cm} > \omega R$



Note: The friction will be kinetic in nature & its magnitude can be determined using $f_k = \mu_k N$. Direction will be backwards (because point of contact A is moving forward w.r.t. ground). The effect of f_k is to decrease V_{cm} and increase ω until $V_{cm} = \omega R$ and $V_A = 0$

Case-II: $V_{cm} < \omega R$



Point of contact A moves backwards. So kinetic friction acts in forward direction. The effect of the friction is to increase V_{cm} and decrease ω until $V_{cm} = \omega R$ and $V_A = 0$

SKC

अगर slipping हुई तो friction kinetic लगेगा
अगर Pure Rolling है तो अगर लगा तो f हमेशा static ही लगेगा।



Q. Write equation for pure rolling in following question:

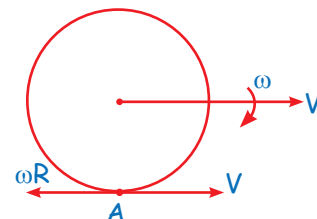
Pure Rolling : Contact point must be at rest wrt contact surface (point)

1. A is point of contact

$$V_A = 0 \text{ (Rolling)}$$

$$V - R\omega = 0$$

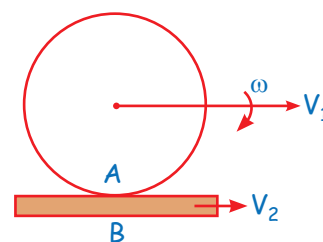
$$V = R\omega \text{ Pure Rolling}$$



2.

$$\vec{V}_A = \vec{V}_B$$

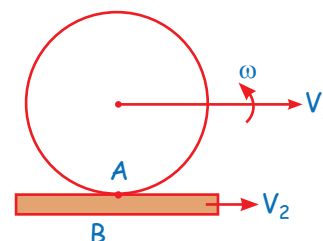
$$V_1 - R\omega = V_2$$



3.

$$V_1 + R\omega = V_2$$

$$V_A = V_B$$



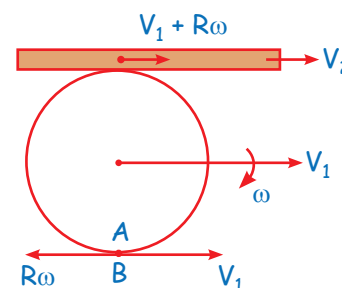
4.

$$V_A = 0$$

$$V_1 - R\omega = 0$$

$$V_1 = R\omega \quad \dots(1)$$

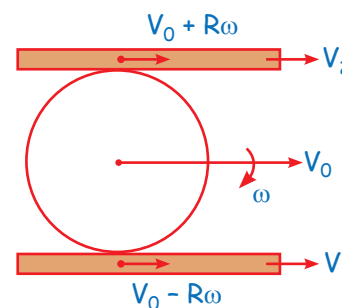
$$V_1 + R\omega = V_2 \quad \dots(2)$$



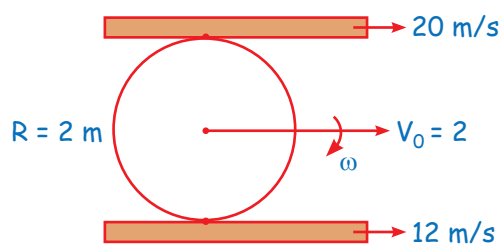
5.

$$V_0 + R\omega = V_2 \quad \dots(1)$$

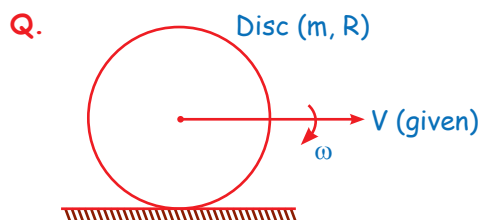
$$V_0 - R\omega = V_1 \quad \dots(2)$$



Q. If there is no slipping anywhere find ω in following figure.



Sol. $20 = V_0 + R\omega$ (Rolling condition for upper part)
 $12 = V_0 - R\omega$ (Rolling condition for lower part)
 Put $R = 2$ solve and get the $V_0 = 16$, $\omega = 2$

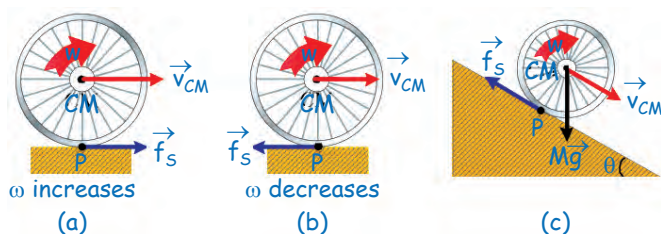


Disc is under pure rolling with translation as shown in figure. Find its KE?

Sol. $V = R\omega$, $\omega = V/R$

$$KE = \frac{1}{2}mV^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}mV^2 + \frac{1}{2} \frac{mR^2}{2} \left(\frac{V}{R}\right)^2$$

$$= \frac{1}{2}mV^2 + \frac{1}{4}mV^2 = \frac{3}{4}mV^2$$



Q. Friction is sufficient for pure rolling. Find acceleration of com.

Sol. $F - f = Ma$... (1)

$$\tau = I\alpha \text{ (about com)}$$

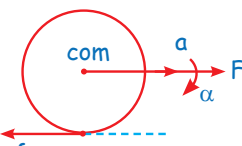
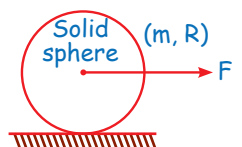
$$0 + fR = I\alpha = \frac{2}{5}mR^2\alpha \dots (2)$$

$$a = R\alpha \dots (3)$$

Now you can solve above eqn and get

$$a = \frac{5F}{7M}, \quad f = \frac{2F}{7}$$

$$\alpha = \frac{a}{R} \Rightarrow \alpha = \frac{5F}{7MR}$$



SKC

Rolling के question को तोड़ने के तीन हथियार follow this

1. $F = ma$

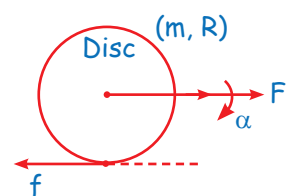
2. $\tau_{com} = I_{com}\alpha$

3. $a = R\alpha$

Pure rolling me if friction लगा तो static f_s लगेगा।



Q. A constant force F is applied as shown in figure. If friction is sufficient for pure rolling. Find acceleration of com.



Sol. $F - f = ma$... (1)

$$0 + fR = \frac{mR^2}{2}\alpha \dots (2)$$

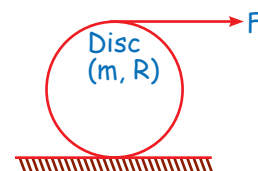
$$a = R\alpha \dots (3)$$

$$fR = \frac{mR^2}{2} \frac{a}{R}$$

$$f = \frac{ma}{2} \Rightarrow f = \frac{F}{3}$$

$$a = \frac{2F}{3} \quad F = \frac{3ma}{2}$$

Q. A constant force F is applied as shown in figure. If friction is sufficient for pure rolling. Find acceleration of com and α .



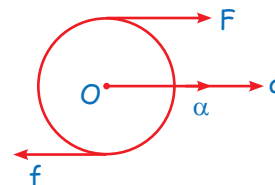
Sol. $a = \frac{3F}{4m}$, $\alpha = \frac{3F}{4mR}$, $f = \frac{-F}{3}$

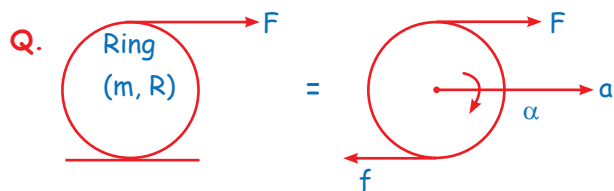
$$F - f = ma \dots (1)$$

$$\tau_0 = F.R + fR = \frac{mR^2}{2} \cdot \alpha$$

$$FR + fR = \frac{mR^2}{2} \cdot \frac{a}{R}$$

$$F + f = \frac{ma}{2} \dots (2)$$





Friction is sufficient for pure rolling. Find a , α , f .

Sol. $F - f = ma$... (1)

$\tau_0 = I\alpha$ $a = R\alpha$

$FR + fR = MR^2\alpha$

$FR + fR = mR^2 \cdot \frac{a}{R}$

$F + f = ma$... (2)

Add (1) and (2)

$F = ma$ so $f = 0$

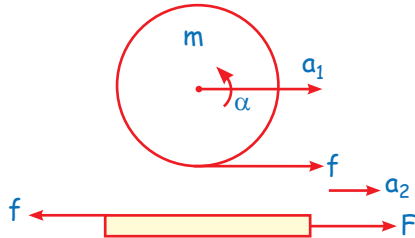
$a = \frac{F}{m}$, $\Rightarrow \alpha = \frac{F}{mR}$



Friction is sufficient for pure rolling between solid sphere and platform.

Find acceleration of platform and sphere and α .

Sol.



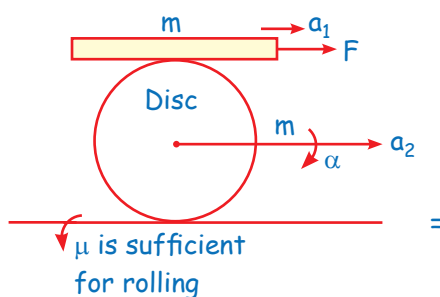
$F - f = ma_2$

$f = ma_1$

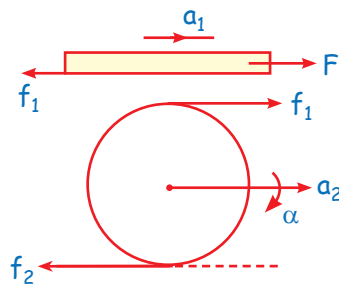
$f \cdot R = \frac{2}{5} MR^2\alpha$

$a_1 + R\alpha = a_2$

Q. Force F is applied on upper plank. Find acceleration of plank and disc.



Sol.



$a_2 = R\alpha$... (1)

$a_2 + R\alpha = a_1$... (2)

$F - f = ma_1$... (3)

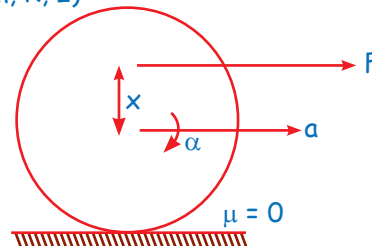
$F_1 - f_2 = ma_2$... (4)

$F_1R + f_2R = \frac{MR^2}{2}\alpha$... (5)

Solve above eqⁿs to get a_1 & a_2

Q. Find the value of x so that given body rolls without slipping on the smooth surface.

(m, R, I)



Sol. $f = 0$

$F = ma$ & $a = R\alpha$

$\tau_0 = F \cdot x = I\alpha$

$(ma)x = \frac{Ia}{R}$

$x = \frac{I}{mR}$

देखा..... देखा तुमने.... यहाँ बिना friction के भी rolling हो रही है



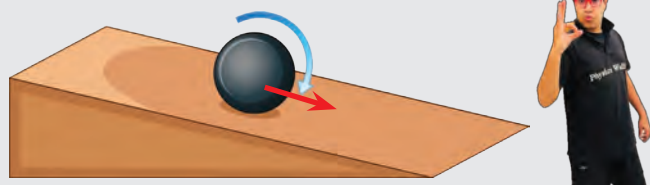
So, now find the value x in above question if body is ring and disc

For Ring $x = \frac{I}{mR} = \frac{mR^2}{mR}$, $x = R$

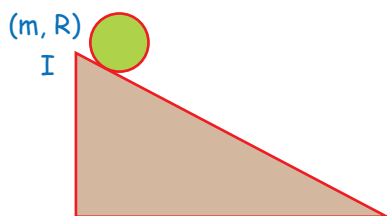
For Disc $x = \frac{I}{mR} = \frac{mR^2/2}{mR} = \frac{R}{2}$

Rolling on Inclined Plane

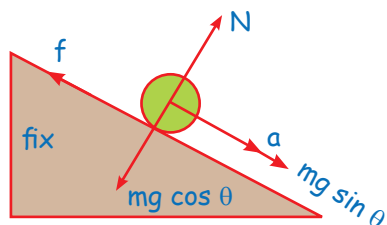
This is very very very important article for Mains/NEET/Advance, Mains में तो हर साल इस पर सवाल आते हैं जो directly formula based होते हैं इसकी feel के लिए सोचो अगर मैं किसी rough incline से किसी sphere को drop करूँ



Q. Suppose from a rough inclined a sphere/ring/disc/etc is dropped such that friction is sufficient for pure rolling. Find a_{com} , α , f .



Sol.



$$mg \sin \theta - f = ma \quad \dots(1)$$

$$0 + fR = I\alpha \quad \dots(2)$$

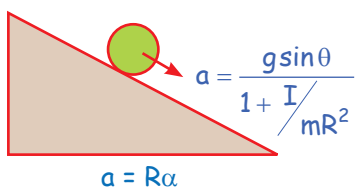
$$a = R\alpha \quad \dots(2)$$

$$fR = \frac{Ia}{R} \Rightarrow f = \frac{Ia}{R^2} \text{ substitute in } \dots(1)$$

$$mg \sin \theta = ma + \frac{Ia}{R^2}$$

$$a = \frac{g \sin \theta}{1 + \frac{I}{mR^2}}$$

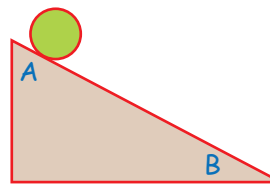
$$f = \frac{Ia}{R^2} = \frac{mg \sin \theta}{m \left(\frac{R^2}{I} + \frac{1}{m} \right)}$$



Rotation

Q. If ring, disc, hollow cylinder, solid cylinder, solid sphere, hollow sphere of same mass and radius. R is release from top of incline plane such that friction is sufficient for pure rolling.

$t_A - t_B \Rightarrow ?$ who will reach 'B' first



Sol.

$$a = \frac{g \sin \theta}{1 + \frac{I}{mR^2}}$$

★ जिसका $I \uparrow$, $a \downarrow$, $v_f \downarrow$, $\omega_f \downarrow$, Time $\uparrow$, बाद में पहुँचेगा
Translation K.E $\downarrow$, Rotational K.E $\uparrow$

MOI order

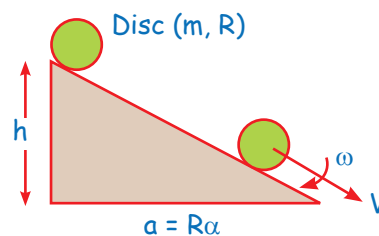
Ring = Hollow cylinder > Hollow sphere > Disc
= Solid cylinder > Solid sphere

Ring takes max time & solid sphere least

Ring and Hollow cylinder $\Rightarrow$ सबसे last में

Solid sphere $\Rightarrow$ सबसे पहले

Q.

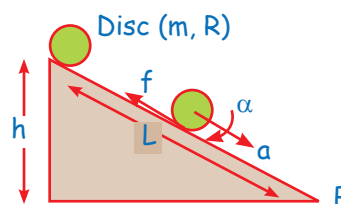


Friction is sufficient for pure rolling. Disc is released as shown in diagram.

Find :

1. v , ω when disc reaches at lowest point.
2. a_{com}
3. α
4. Translation KE when reaches at lowest point.
5. Rotation K:E when reaches lowest point.
6. Total KE when reaches lower p.t.

Sol.



$$1. a = \frac{g \sin \theta}{1 + \frac{I}{mR^2}} = \frac{g \sin \theta}{1 + \frac{mR^2}{2mR^2}} = \frac{2}{3} g \sin \theta$$

$$2. a = R\alpha, \alpha = \frac{a}{R} = \frac{2g \sin \theta}{3R}$$

$$3. V^2 = u^2 + 2aS$$

$$V^2 = 0 + \frac{2 \times 2g \sin \theta}{3} (L)$$

$$4. V = R\omega \Rightarrow \omega = \frac{V}{R}$$

$$\text{Translation K.E at B} = \frac{1}{2} mV^2$$

$$\text{Rotational K.E} = \frac{1}{2} I\omega^2 = \frac{1}{2} \cdot \frac{mR^2}{2} \left(\frac{V}{R} \right)^2 = \frac{1}{4} mV^2$$

$$\text{Total KE} = \text{Translation KE} + \text{Rot. KE}$$

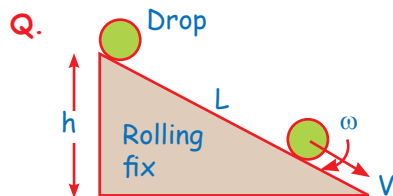
$$= \frac{1}{2} mV^2 + \frac{1}{4} mV^2 = \frac{3}{4} mV^2$$

$$AB = L$$

$$AB = S = ut + \frac{1}{2} at^2$$

$$L = 0 + \frac{1}{2} \cdot a \cdot t^2$$

$$t = \sqrt{\frac{2L}{2g \sin \theta}} \quad \text{Time to reach ground.}$$



Sol. $w_g + w_N + w_f = \Delta \text{K.E (WET)}$

$$mgh + 0 + 0 = \frac{1}{2} mV^2 + \frac{1}{2} I\omega^2 - 0$$

$$mgh = \frac{1}{2} mV^2 + \frac{1}{2} \cdot \frac{mR^2}{2} \left(\frac{V}{R} \right)^2$$

$$V = R\omega$$

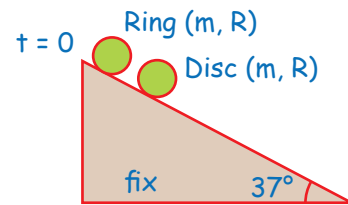
$$\omega = \frac{V}{R}$$

$$mgh = \frac{1}{2} mV^2 + \frac{mV^2}{4} = \frac{3}{4} mV^2$$

$$V = \sqrt{\frac{4}{3} mgh}$$

$$h = L \sin \theta$$

Q. Find a_{ring} , a_{disc} , a_{rel} , and separation between ring and disc at $t = 6$ sec. if both are released from same point.



Sol. $a = \frac{g \sin \theta}{1 + \frac{I}{mR^2}}$

$$a_{\text{ring}} = \frac{g \sin \theta}{1 + \frac{mR^2}{mR^2}} = \frac{2g \sin \theta}{2} = \frac{6}{2} = 3$$

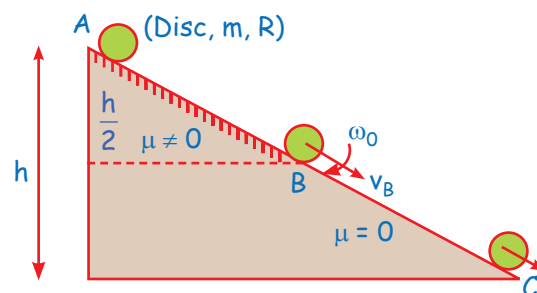
$$a_{\text{rel}} = 4 - 3 = 1$$

$$S_{\text{rel}} = U_{\text{rel}} t + \frac{1}{2} a_{\text{rel}} t^2$$

$$\text{Separation} = S_{\text{rel}} = 0 t + \frac{1}{2} \times 1 \times 6^2 = 18$$

Q. A disc is released from top of a incline plane which is half rough and half smooth. A $\rightarrow$ B friction is sufficient for pure rolling and B $\rightarrow$ C surface is smooth.

Find V_B , ω_B , V_C , ω_C



Sol. A $\rightarrow$ B (WET)

$$w_g + w_N + w_f = \Delta \text{KE}$$

$$\frac{mgh}{2} + 0 + 0 = \frac{1}{2} mV^2 + \frac{1}{2} \cdot \frac{mR^2}{2} \left(\frac{V}{R} \right)^2$$

$$\frac{mgh}{2} = \frac{3}{4} mV^2$$

$$V = \sqrt{\frac{2}{3} gh} \quad \omega_0 = \frac{V}{R} = \frac{1}{R} \sqrt{\frac{2gh}{3}} = \omega_B$$

B $\rightarrow$ C (WET)

$$f = 0$$

$$\alpha = 0, \omega = \text{const} = \omega_0$$

$$mg \frac{h}{2} + 0 + 0 = (KE)_C - (KE)_B$$

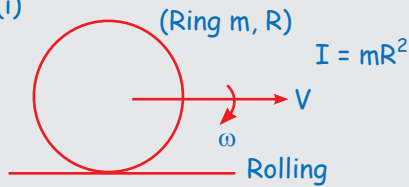
$$mg \frac{h}{2} = \left(\frac{1}{2} m V_C^2 + \frac{1}{2} I_C \omega^2 \right) - \left(\frac{1}{2} m V_B^2 + \frac{1}{2} I_B \omega^2 \right)$$

$$mg \frac{h}{2} = \frac{1}{2} m V_C^2 - \frac{1}{2} m \frac{2gh}{3}$$

$$V_C = \sqrt{\frac{5}{3} gh}$$

#Ratio of Rotational KE to Translation KE in Rolling

(i)

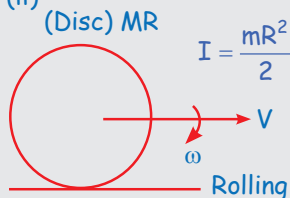


$$\text{Rotation KE} = \frac{1}{2} m R^2 \frac{V^2}{R^2} = \frac{1}{2} m V^2$$

$$\text{Translation KE} = \frac{1}{2} m V^2$$

$$\text{Ratio} = 1$$

(ii)



$$\frac{\text{Rot KE}}{\text{Tran KE}} = \frac{\frac{1}{2} m R^2 \frac{V^2}{R^2}}{\frac{1}{2} m V^2} = \frac{1}{2}$$

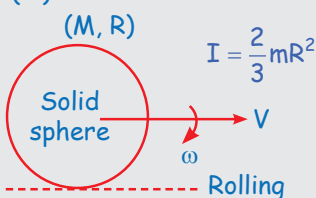
$$I = \left(\frac{1}{2} \right) m R^2$$

Q. Solid cylinder rolling purely find ratio of Rot KE to the trans KE

$$I = \left(\frac{1}{2} \right) m R^2 \left(\frac{\text{Rot KE}}{\text{Tran KE}} \right)$$

$$\text{Sol. } \frac{1}{2}$$

(iii)



$$\frac{\text{Rot KE}}{\text{Tran KE}} = \frac{\frac{1}{2} \left(\frac{2}{5} m R^2 \right) \left(\frac{V}{R} \right)^2}{\frac{1}{2} m V^2} = \frac{2}{5}$$

$$I = \left(\frac{2}{5} \right) m R^2$$

Angular Momentum in CRTM ($\vec{L}$)

If a rigid body is performing CRTM then angular momentum of the body about a general point is given by relation

$$\vec{L}_{\text{body about P}} = I_C \vec{\omega} + \vec{r} \times m \vec{V}_C$$

$$\vec{L}_{\text{body about P}} = I_C \vec{\omega} + \vec{r} \times \vec{P}$$

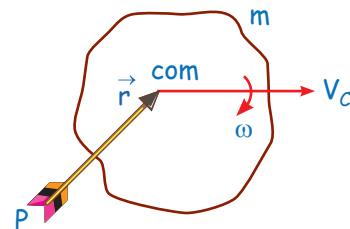
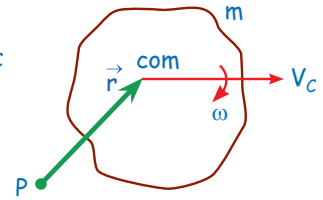
$m \rightarrow$ Mass of the body

$\vec{V}_C \rightarrow$ Velocity of COM

$\vec{P} \rightarrow$ Momentum of the body

$I_C \rightarrow$ MOI of body about centre of mass

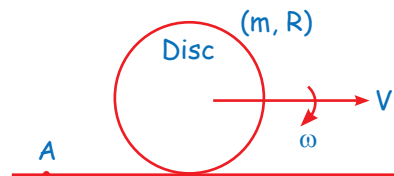
$\vec{r} \rightarrow$ Position vector of COM w.r.t P



भाई CRTM किसी general point P के about angular momentum लिखने में $\vec{r}$ बहुत important होता है जिसका head COM पर होता है और tail P point पर होती है और मैं चाहता हूँ ये vector आपको अलग चमके इसलिए मैं इसे green color के vector से show कर रहा हूँ ताकि revision में आपका समय बचे.... अबे तेरी हर genuine problem का मैंने खयाल रखा है।



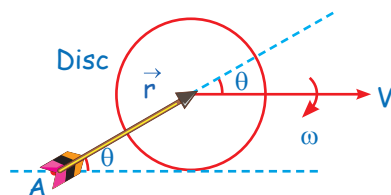
Q. Write the angular momentum of disc about A.



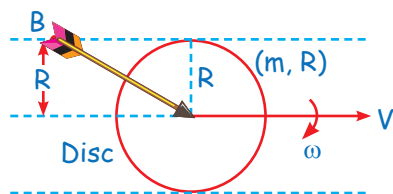
$$\text{Sol. } \vec{L}_{\text{Disc about A}} = I_C \vec{\omega} + (\vec{r} \times m \vec{V}_C)$$

$$= \frac{m R^2}{2} \cdot \omega + m V R$$

(C.W) (C.W)



Q. Find angular momentum about B.



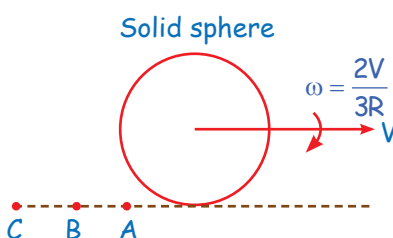
Sol. $\vec{L}_{\text{Disc about B}} \Rightarrow I_C \omega + \vec{r} \times m \vec{V}_C$

$$= \frac{mR^2}{2} \omega - mVR$$

(CW) (ACW)

CW $\rightarrow +ve$ ACW $\rightarrow -ve$

Q. Find angular momentum of sphere about A, B & C.



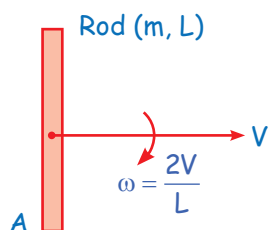
Sol. $\vec{L}_A = I_C \omega + \vec{r} \times m \vec{V}_C$

$$= \frac{2}{5} mR^2 \cdot \frac{2V}{3R} + MVR$$

(C.W) (C.W)

$\vec{L}_{\text{abt}} = \vec{L}_{\text{abt}} = \vec{L}_{\text{abt}}$
 A B C
 Magnitude wise

Q. Find angular momentum of rod about A (ground point) and about COM



Sol. (1) $\vec{L}_{\text{rod abt A}} = I_C \omega + \vec{r} \times m \vec{V}_C$

$$= \frac{mL^2}{12} \cdot \frac{2V}{L} + m \cdot \frac{V \cdot L}{2}$$

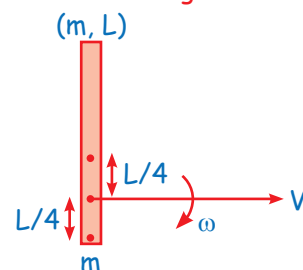
(C.W) (C.W)

(2) $\vec{L}_{\text{rod abt com}} = I_C \omega + \vec{r} \times m \vec{V}_C$

$$= \frac{mL^2}{12} \cdot \frac{2V}{L} + 0$$

(C.W)

Q. In following question a particle of mass m is fixed at one end of the rod of same mass m and length ℓ as shown in figure.



(a) Find angular momentum of system about com of system.

Sol. $\vec{L} = I_C \omega + \vec{r} \times m \vec{V}_C$ com par puch rahe hai isliye

$$= I_C \omega + 0$$

com of system के about

$$I_C = \left(\frac{mL^2}{12} + m \left(\frac{L}{4} \right)^2 \right) + m \left(\frac{L}{4} \right)^2 = \frac{5}{24} mL^2$$

(rod) (mass)

$$\vec{L} = \left(\frac{5}{24} mL^2 \right) \omega$$

(b) Find angular momentum of system about point A (ground point).

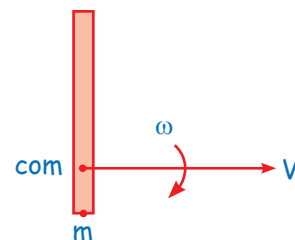
Sol. $\vec{L} = I_C \omega + \vec{r} \times m \vec{V}_C$

$$= \frac{5}{24} mL^2 \omega + 2m \cdot V \cdot \frac{L}{4}$$

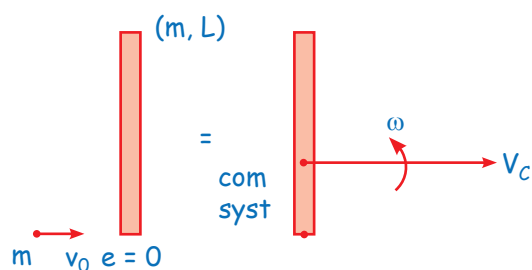
(C.W) (C.W)

= ✓

$$I_C = \frac{5}{24} mL^2 \text{ (same रहेगा)}$$



- Q. Rod lie on smooth horizontal surface at rest. A particle moving as shown with v_0 velocity sticks to end of rod. Find ω of the rod after collision and V_{com} .



Sol. $P_i = P_f$ (momentum conservation)

$$mV_0 + 0 = 2mV_c \quad V_c = \frac{V_0}{2} \text{ (const.)}$$

$L_i = L_f$ (about COM)

$$mV_0 \frac{L}{4} + 0 = I_c \omega + (\vec{r} \times m\vec{V}_c) \quad \text{zero}$$

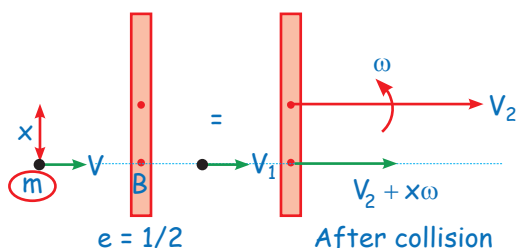
(ACW)

$$mV_0 \frac{L}{4} = \frac{5}{24} mL^2 \omega$$

$$\omega = \frac{6 V_0}{5 L}$$

- Q. Repeat the above problem if $e = 1/2$ and particle collide at a distance x from COM of rod.

Sol.



$P_i = P_f$

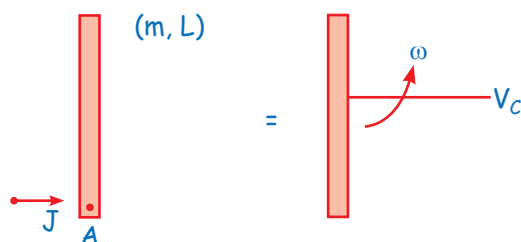
$$mV_0 + 0 = mV_1 + mV_2$$

$$e = \frac{1}{2} = \frac{V_2 + x\omega - 0}{V_0}$$

$L_i = L_f$ (about B on ground, ये point particle की velocity वाली line पर आगया)

$$0 + 0 = 0 + \left(\frac{mL^2}{12} \omega - mV_2 x \right) \quad \text{rod}$$

- Q. Rod lie on smooth horizontal floor.



A sharp impulse is given at end A.

Find V_{com} , ω just after impulse.

Sol. Impulse = Change in momentum

$$J = mV_c - 0$$

$$V_c = \frac{J}{M}$$

Angular impulse = $\vec{L}_f - \vec{L}_i$ (about com)

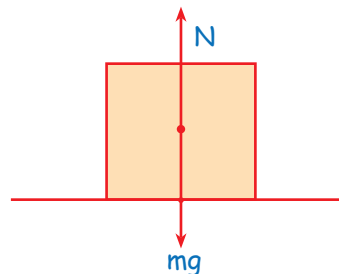
$$\frac{JL}{2} = \frac{mL^2}{12} \omega + 0$$

$$\omega = \frac{6J}{mL}$$

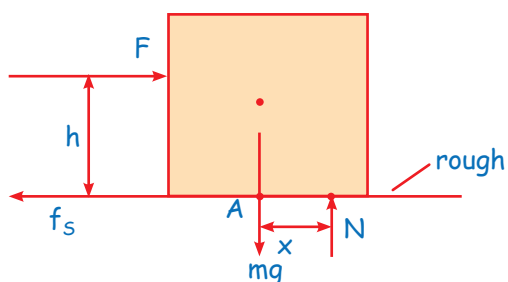
जैसे Impulse होने पर momentum बदलता था वैसे ही Angular impulse होने पर angular momentum बदलना है।

TOPPLING

Suppose we have a block (cube, L) of mass m is in equilibrium on rough surface. $N = mg$



Suppose friction is sufficient enough to prevent the sliding of block. Now in above case force F is applied as shown in figure and It's value increases slowly slowly from zero. We observe block still is in equilibrium.



At equilibrium

$$F = f_s \text{ and } N = mg$$

Torque about A = 0

$$\tau_A = 0$$

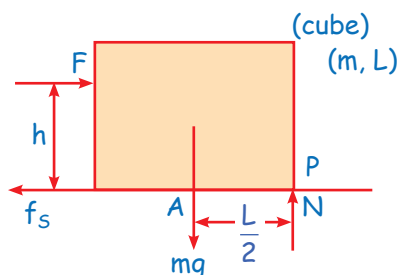
$$-Fh + Nx = 0$$

$$Fh = Nx \Rightarrow x = \frac{Fh}{N}$$

If F increases x also increase. Normal can shift maximum to the edge.

Condition for Toppling

Maximum F can be found just before toppling



if N passes through point 'P'

$$\tau_A = 0$$

$$Fh = N \times \frac{L}{2} \quad \left(x = \frac{L}{2} \right)$$

$$Fh = mg \frac{L}{2}$$

$$F = \frac{mgL}{2h}$$

You can also verify this result by putting torque about point P = 0.

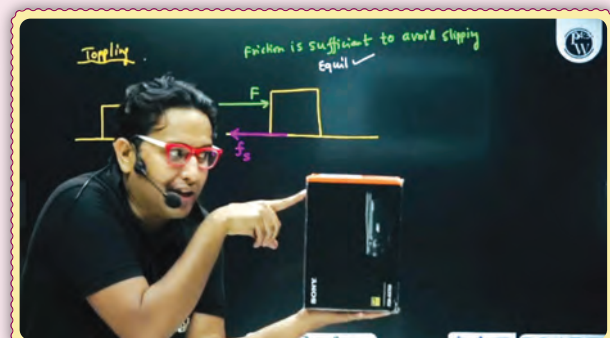
$$0 + 0 - mg \frac{L}{2} + Fh = 0$$

$$F = \frac{mgL}{2h}$$

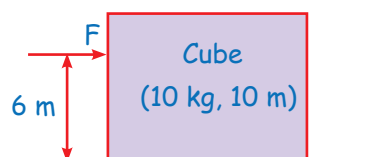
Here

$$F_{\min} \text{ to side } (F_{\min})_{\text{side}} = \mu_s mg$$

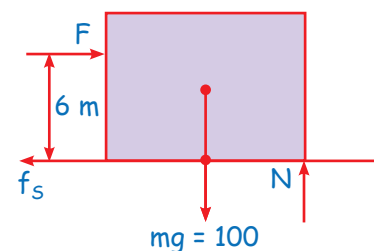
$$F_{\min} \text{ to topple } (F_{\min})_{\text{topple}} = \frac{mgL}{2h}$$



Q. Friction is sufficient to avoid slipping, find $F_{\min}$ so that body topple.



Sol.



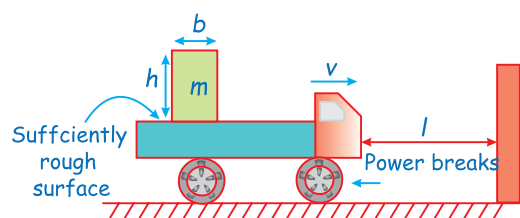
$$\tau_A = 0$$

$$Fh = mg \frac{L}{2}$$

$$F \times 6 = 100 \times \frac{10}{2}$$

$$F = \frac{1000}{12} \text{ N}$$

- Q. Find minimum value of l so that truck can avoid the dead end, without toppling the block kept on it.



Sol. If the truck decelerates at a rate a then in the frame of the truck, pseudo force will act on the block in the forward direction at its center of mass.

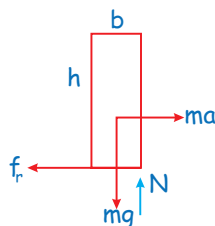
$$ma \frac{h}{2} \leq mg \frac{b}{2}$$

$$a \leq \frac{b}{h} g$$

Final velocity of truck is zero. So that

$$0 = v^2 - 2 \left(\frac{b}{h} g \right) l$$

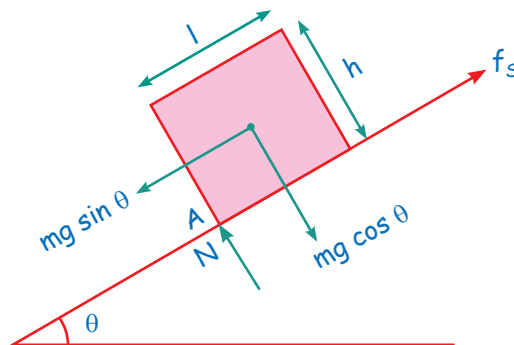
$$l = \frac{h}{2b} \frac{v^2}{g}$$



TOPPLING ON INCLINED PLANE



Let a block is placed on a inclined plane whose $\theta = 0$ and we increase the angle of the inclined plane θ slowly slowly from zero. Suppose friction is sufficient to avoid slipping then at an angle θ block will just about to topple. At this angle



$$\tau_A = 0 \text{ Torque about A} = 0$$

$$mg \sin \theta \frac{h}{2} = mg \cos \theta \frac{l}{2}$$

$$\tan \theta = \frac{l}{h}$$

$$\theta_{\text{topple}} = \tan^{-1} \left(\frac{l}{h} \right)$$

हमने देखा if friction is large enough to avoid slipping

$$\theta_{\text{topple}} = \tan^{-1} \left(\frac{l}{h} \right)$$

और NLM में हमने पढ़ा था $\theta_{\text{slide}} = \tan^{-1} (\mu_s)$

Body will slide first

If $\theta_{\text{topple}} > \theta_{\text{slide}}$

$$\tan^{-1} \frac{l}{h} > \tan^{-1} \mu$$

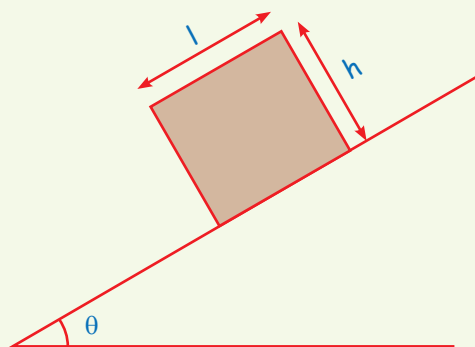
$$\frac{l}{h} > \mu$$

Body will topple first

If $\theta_{\text{topple}} < \theta_{\text{slide}}$

$$\tan^{-1} \frac{l}{h} < \tan^{-1} \mu$$

$$\frac{l}{h} < \mu$$



अब मैं आपके concept को और strong करने के लिए कुछ और important selected question with hint and sol. attach कर रहा हूँ इन्हें visualise करके खुद से solve करना

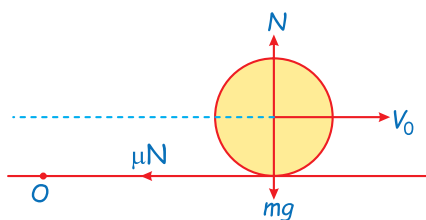


Q. A bowling ball of mass 'm' is thrown on a horizontal surface in such a way that it slides with speed V_0 initially without rolling.

(a) Find the velocity of the ball when it starts rolling on the surface.

(b) How much kinetic energy is lost before it starts rolling. (The co-efficient of kinetic friction is μ)

Sol.



From the free body diagram it is clear that $N = mg$.

From eqns. of motion we have

$$m a_{cm} = -\mu mg \quad \dots (i)$$

$$I_{cm} \alpha = \mu mg R \quad \dots (ii)$$

$R =$ radius of the ball

$$\Rightarrow a_{cm} = -\mu g$$

$$\Rightarrow \alpha = \frac{\mu mg R}{I_{cm}} = \frac{\mu mg R}{\frac{2}{5} m R^2} = \frac{5\mu g}{2R}$$

After time 't', the velocity V_f of center of ball and angular velocity ω_f will be

$$V_f = V_0 - \mu g t \quad \dots (iii)$$

$$\omega_f = 0 + \frac{5\mu g}{2R} t \quad \dots (iv)$$

If it starts rolling after time t then

$$V_f - \omega_f R = 0 \quad \dots (v)$$

After solving eqns (iii), (iv) and (v) we have

$$(a) V_f = \frac{5}{7} V_0 \text{ and } \omega_f = \frac{5V_0}{7R}$$

(b) The change in kinetic energy is

$$\begin{aligned} \Delta KE &= \left(\frac{1}{2} m V_{cm}^2 + \frac{1}{2} \times \frac{2m}{5} \omega^2 R^2 \right) - \left(\frac{1}{2} m V_0^2 \right) \\ &= \left(\frac{1}{2} m \left(\frac{5}{7} V_0 \right)^2 + \frac{m}{5} \left(\frac{5 V_0}{7 R} \right)^2 R^2 \right) - \frac{1}{2} m V_0^2 = -\frac{m V_0^2}{7} \end{aligned}$$

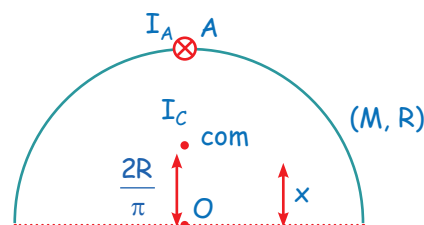
Alternative method: Angular momentum of the cylinder about point O is conserved (see fig.). Since $N = mg$ the torque due to these forces cancels out. The torque due to friction is zero since its moment arm is zero about O. Hence

$$L_i = L_f$$

$$m V_0 R = m V_f R + I_{cm} \omega_f = \frac{7}{5} m R V_f$$

$$\Rightarrow V_f = \frac{5}{7} V_0$$

Q. Find MOI of a semiconductor ring of mass m and radius R about an axis passing through A perpendicular to plane.



Sol. From parallel axis theorem

$$I_O = I_C + m x^2$$

$$m R^2 = I_C + m x^2 \quad [\because I_O = m R^2]$$

$$I_C = m R^2 - m x^2$$

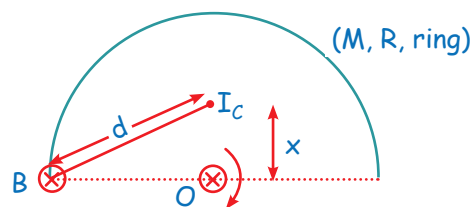
Again from parallel axis theorem,

$$I_A = I_C + m d^2$$

$$I_A = (m R^2 - m x^2) + m (R - x)^2$$

$$\text{Put } x = 2R/\pi$$

Q. Find MOI about B is I_B .



Sol. $I_O = m R^2$

$$d = \sqrt{R^2 + x^2}$$

$$I_O = I_C + m d^2 \quad [\text{Parallel axis theorem}]$$

$$m R^2 = I_C + m x^2$$

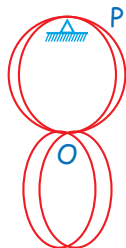
$$I_C = m R^2 - m x^2 \quad \dots (i)$$

$$I_B = I_C + m d^2 \quad [\text{Parallel axis theorem}]$$

$$= m R^2 - m x^2 + m (R^2 + x^2)$$

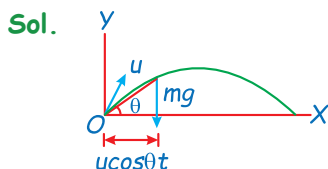
$$I_B = 2m R^2$$

- Q. Two identical rings each of mass m with their planes mutually perpendicular, radius R are welded at their point of contact O . If the system is free to rotate about an axis passing through the point P perpendicular to the plane of the paper, then the moment of inertia of the system about this axis is equal to



Sol. $11.5 mR^2$

- Q. Find torque due to gravity at any time t about point of projection O , if a body is projected with velocity u at an angle θ with the horizontal.

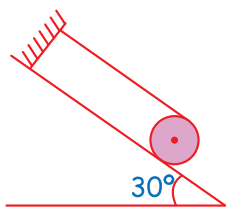


At every instant force mg is in downward direction and perpendicular distance will be displacement along x -axis i.e. $x = (u \cos \theta) t$

$\tau_O = mg u \cos \theta t$, it is increasing with time.

- Q. A thin hoop of weight 500 N and radius 1 m rests on a rough inclined plane as shown in the figure.

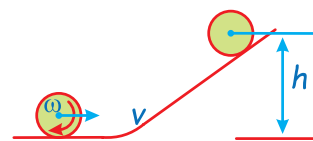
The minimum coefficient of friction needed for this configuration is:



Sol. $\frac{1}{2\sqrt{3}}$

- Q. A ball of radius R and mass m is rolling without slipping on a horizontal surface with velocity of its centre of mass v . It then rolls without slipping up a hill to a height h before momentarily coming to rest. Find h .

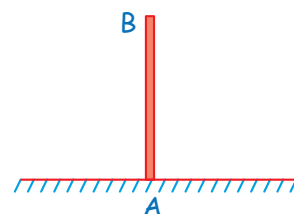
Sol. $\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = mgh$



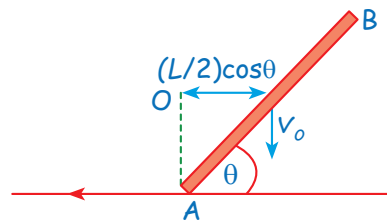
$$v = \omega R, I = \frac{2}{5}mR^2$$

$$h = \frac{7v_{CM}^2}{10g}$$

- Q. A uniform rod AB of length L and mass m is held vertically on smooth horizontal surface. At $t = 0$, it is slightly disturbed so that it starts slipping on the smooth surface. Find the angular velocity ω and velocity v_o of center of mass when rod makes an angle θ with the horizontal.



- Sol. Since external force, namely, mg and N act in vertical direction only, the center of mass will move in the vertical direction. End A of the rod can only slip in the horizontal direction (see fig. below). The instantaneous axis rotation will be at point O when rod makes an angle θ with the horizontal.



Conserving mechanical energy, we obtain

$$(mg L / 2)(1 - \sin \theta) = \frac{1}{2}I_o \omega^2$$

in which $I_o = I_{cm} + md^2$

$$= \frac{mL^2}{12} + \frac{mL^2}{4} \cos^2 \theta$$

$$\Rightarrow \omega = \sqrt{\left(\frac{1 - \sin \theta}{1 + 3 \cos^2 \theta} \right) \frac{12g}{L}}$$

The velocity of the center of mass is given by.

$$v_o = \omega(L/2) \cos \theta$$

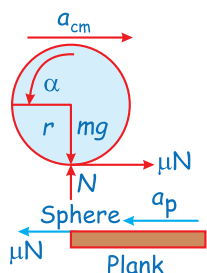
$$= \sqrt{\frac{12(1 - \sin \theta)gL}{1 + 3 \cos^2 \theta}} \frac{\cos \theta}{2}$$

- Q.** A plank of mass M rests on a smooth horizontal plane. A sphere of mass m and radius r placed on the rough upper surface of the plank and the plank is suddenly given a velocity v in the direction of its length. Find the time after which the sphere begins pure rolling, if the coefficient of friction between the plank and the sphere is μ and the plank is sufficiently long.

Sol. Let t be the time after which slipping between the sphere and plank ceases.

For the sphere, $N = mg$, $\mu N = ma_{cm}$

$$\Rightarrow a_{cm} = \mu g$$



$$\tau = I\alpha \Rightarrow \mu mgr = (2/5)mr^2\alpha$$

$$\Rightarrow \alpha = \frac{5\mu g}{2r}$$

For the plank,

$$\mu N = Ma_p \Rightarrow a_p = \frac{\mu mg}{M}$$

After time t ,

$$\text{velocity of plank, } v_p = v - \frac{\mu mg}{M}t$$

$$\text{velocity of sphere, } v_{cm} = \mu gt$$

$$\text{angular velocity of sphere, } \omega = \frac{5\mu g}{2r}t$$

For no slipping, the point of contact of sphere should have same speed as that of plank,

$$\Rightarrow v_{cm} + \omega r = v_p$$

Substituting and solving

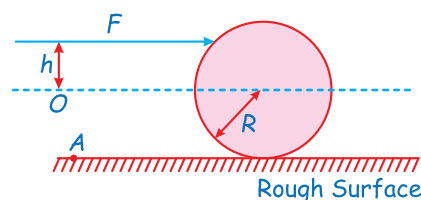
$$\mu gt + \left(\frac{5\mu g}{2r}t\right)r = v - \frac{\mu mg}{M}t$$

$$\Rightarrow t = \frac{2v}{g(2\mu + 5\mu + 2\mu m/M)} = \frac{2v}{\mu g[7 + (2m/M)]}$$

- Q.** A billiards ball, initially at rest is given a sharp impulse by a cue. The cue is held a distance h above center line (see fig.). The ball leaves the cue with a speed v_0 and because of its forward motion finally acquires a speed of $(9/7)v_0$.

Show that $h = (4/5)R$

where R is the radius of ball



Sol. F acts for time Δt . Then

$$V_{cm} = v_0 = \frac{F\Delta t}{M} \quad \& \quad \omega = \frac{F\Delta th}{(2/5)MR^2}$$

$$\Rightarrow \omega = \frac{5v_0 h}{2R^2}$$

After the application of impulse, angular momentum about A will be conserved.

$$\Rightarrow L_i = L_f$$

$$\Rightarrow Mv_0 R + I_{cm}\omega = Mv_f R + I_{cm}\frac{v_f}{R}$$

$$\Rightarrow Mv_0 R + \frac{2}{5}MR^2 \times \frac{5v_0 h}{2R^2} = \left(MR + \frac{2}{5}\frac{MR^2}{R}\right)\frac{9}{7}v_0$$

$$\Rightarrow MR + Mh = \frac{9}{5}MR \Rightarrow h = \frac{4}{5}R$$

- Q.** A cube of side length a moves on smooth horizontal surface (see fig.). At point P , it collides inelastically with a small obstacle. As a result it starts rotating about it. Find the minimum velocity v that should be imparted to the block so that it topples over.



Sol. During collision angular momentum about P would be conserved (Why?). Energy will not be conserved as collision is not elastic. Let ω_0 be the angular velocity of cube just after collision. Conserving angular momentum about P , we obtain

$$mv\frac{a}{2} = I_P\omega_0$$

$$\text{Where } I_P = I_{cm} + md^2$$

$$= \left(m\left(\frac{a^2}{12} + \frac{a^2}{12}\right)\right) + ma^2/2 = \frac{2}{3}ma^2$$

$$\Rightarrow v = \frac{4}{3}a\omega$$

If the cube manages to rotate by $\pi/4$, then it will topple over. Applying energy conservation after the collision, we have

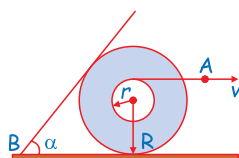
$$\left[0 - \frac{1}{2} I_p \omega^2\right] + mg \left[\frac{a}{\sqrt{2}} - \frac{a}{2}\right] = 0$$

$$\Rightarrow \omega^2 = \frac{2mga}{I_p \sqrt{2}} \left[1 - \frac{1}{\sqrt{2}}\right] \Rightarrow \omega^2 = \frac{3}{2} (g/a)(\sqrt{2} - 1)$$

Substituting it in $v = (4/3)a\omega$, gives

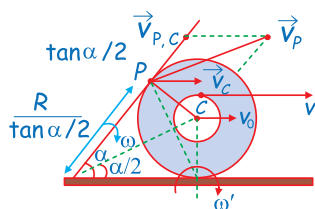
$$v = \sqrt{\frac{8}{3}ga(\sqrt{2} - 1)}$$

- Q.** A bobbin rolls without slipping over a horizontal surface so that the velocity v of the end of the thread (point A) is



directed along the horizontal. A board hinged at B leans against the bobbin as shown in figure. The inner and outer radii of the bobbin are r and R respectively. Determine the angular velocity ω of the board as a function of angle α .

- Sol.** Let P be the point of contact of the board and the bobbin at a certain instant. As the bobbin is in pure rolling, the velocity of point P ,
 $\vec{v}_P = \vec{v}_{\text{rotation}} + \vec{v}_{\text{translation}} = \vec{v}_{P,C} + \vec{v}_C$



If the angular velocity of the board at this instant is ω , the linear velocity of the point of

the board in contact with the bobbin is $\frac{\omega R}{\tan \frac{\alpha}{2}}$

normal to the board.

If the board is to remain in contact with the bobbin, their normal velocities should be same, i.e.,

$$\frac{\omega R}{\tan \frac{\alpha}{2}} = v_0 \sin \alpha \quad \dots (i)$$

Comparing angular speed about instantaneous centre of rotation,

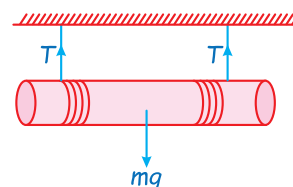
$$\frac{v_0}{R} = \frac{v}{R+r} = \text{angular velocity of bobbin, } \omega$$

$$\therefore v_0 = \frac{vR}{R+r} \quad \dots (ii)$$

$$\text{Now Eq.(i) becomes } \omega R = \frac{vR}{R+r} \sin \alpha \tan \frac{\alpha}{2}$$

$$\text{or } \omega = \frac{2v \sin^2 \frac{\alpha}{2} \cos \frac{\alpha}{2}}{(R+r)}$$

- Q.** A cylinder of mass m is suspended through two strings wrapped around it as shown in figure. Find (a) the tension T in the string and (b) the speed of the cylinder as it falls through a distance h .



- Sol.** The portion of the strings between the ceiling and the cylinder is at rest. Hence the points of the cylinder where the strings leave it are at rest. The cylinder is thus rolling without slipping on the strings. Suppose the centre of the cylinder falls with an acceleration a . The angular acceleration of the cylinder about its axis is $\alpha = a/R$, as the cylinder does not slip over the strings.

The equation of motion for the centre of mass of the cylinder is

$$mg - 2T = ma \quad \dots (i)$$

and for the motion about the centre of mass, it is

$$2Tr = I\alpha$$

$$2Tr = \frac{1}{2} mr^2 \frac{a}{r} \text{ or } 2T = \frac{ma}{2} \quad \dots (ii)$$

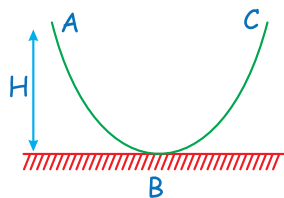
$$\text{From (i) and (ii), } a = \frac{2}{3}g \text{ and } T = \frac{mg}{6}.$$

As the centre of the cylinder starts moving from rest, the velocity after it has fallen through a distance h is given by

$$v^2 = 2\left(\frac{2}{3}g\right)h \text{ or } v = \sqrt{\frac{4gh}{3}}.$$

- Q.** A sphere of mass m and radius r is released from rest at point A on a track in vertical plane. The track is rough enough to support rolling between A and B and from B onwards

it is smooth. The maximum height attained by sphere from ground on its journey from B onwards is



Sol. At B

$$mgH = \frac{1}{2} m v_{cm}^2 + \frac{1}{2} I_{cm} \omega^2$$

$$= \frac{1}{2} m \omega^2 r^2 + \frac{1}{2} \times \frac{2}{5} m r^2 \omega^2 = \frac{7}{10} m r^2 \omega^2$$

$$\Rightarrow \omega^2 = \frac{10gH}{7r^2}$$

At highest point along curve BC,

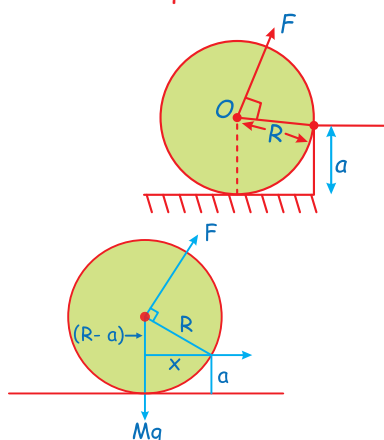
$$mgH = mgh + \frac{1}{2} I_{cm} \omega^2$$

$$\Rightarrow mgH = mgh + \frac{1}{2} \times \frac{2}{5} m r^2 \times \frac{10gH}{7r^2}$$

$$\Rightarrow h = H \left(1 - \frac{2}{7} \right) = \frac{5H}{7}$$

Q. A uniform cylinder of mass M and radius R is to be pulled over a step of height a ($a < R$) by applying a force F at its centre ' O ' perpendicular to the plane through the axes of the cylinder on the edge of the step (see figure). The minimum value of F required is

Sol.

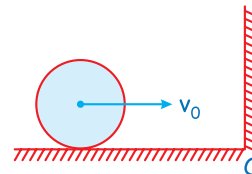


To step up, $F \times R \geq Mg \times x$

$$\Rightarrow F_{\min} = \frac{Mg}{R} \times \sqrt{R^2 - (R-a)^2}$$

$$= Mg \sqrt{1 - \left(\frac{R-a}{R} \right)^2}$$

Q. Consider a sphere of mass ' m ' radius ' R ' doing pure rolling motion on a rough surface having velocity $\vec{v}_0$ as shown in the figure. It makes an elastic impact with the smooth wall and moves back and starts pure rolling after some time again.



(a) Change in angular momentum about ' O ' in the entire motion equals $2mv_0 R$ in magnitude.

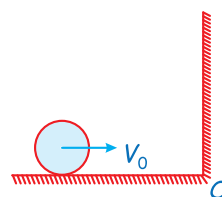
(b) Moment of impulse provided by wall during impact about O equals $2mv_0 R$ in magnitude

(c) Final velocity of ball will be $\frac{3}{7} \vec{v}_0$

(d) Final velocity of ball will be $-\frac{3}{7} \vec{v}_0$

Sol. (a, b, d)

(a) Change in angular momentum.



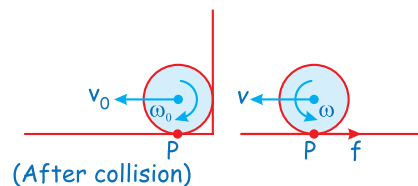
$$= L_f - L_i$$

$$= (I\omega - mv_0 R) - (I\omega + mv_0 R) = -2mv_0 R$$

(b) Impulse = Change in momentum

$$= -2mv_0 R$$

(d)



Conserving angular momentum about the point P

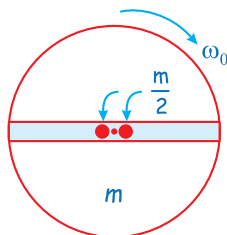
$$I\omega_0 - mv_0 R = I\omega + mvR$$

$$\Rightarrow \frac{2}{5} m R^2 \times \frac{v_0}{R} - mv_0 R$$

$$= \frac{2}{5} m R^2 \times \frac{v}{R} + mvR$$

$$\Rightarrow v = -\frac{3v_0}{7}$$

Comprehension (Q. (a) to (b)): A uniform disc of mass 'm' and radius R is free to rotate in horizontal plane about a vertical smooth fixed axis passing through its centre. There is a smooth groove along the diameter of the disc and two small balls of mass $\frac{m}{2}$ each are placed in it on either side of the centre of the disc as shown in figure. The disc is given initial angular velocity ω_0 and released.



(a) The angular speed of the disc when the balls reach the end of the disc is:

Sol. Let the angular speed of disc when the balls reach the end be ω . From conservation of angular momentum

$$\frac{1}{2}mR^2\omega_0 = \frac{1}{2}mR^2\omega + \frac{m}{2}R^2\omega + \frac{m}{2}R^2\omega$$

$$\text{or } \omega = \frac{\omega_0}{3}$$

(b) The speed of each ball relative to ground just after they leave the disc is:

Sol. The angular speed of the disc just after the balls leave the disc is

$$\omega = \frac{\omega_0}{3}$$

Let the speed of each ball just after they leave the disc be v.

From conservation of energy

$$\frac{1}{2}\left(\frac{1}{2}mR^2\right)\omega_0^2 = \frac{1}{2}\left(\frac{1}{2}mR^2\right)\omega^2 + \frac{1}{2}\left(\frac{m}{2}\right)v^2 + \frac{1}{2}\left(\frac{m}{2}\right)v^2$$

Solving we get

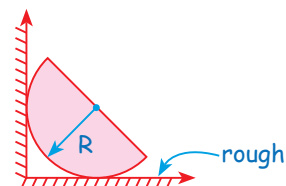
$$v = \frac{2R\omega_0}{3}$$

(c) The net work done by forces exerted by disc on one of the ball for the duration ball remains on the disc is

Sol. Work done by all forces equal $K_f - K_i = \frac{1}{2}\left(\frac{m}{2}\right)v^2$

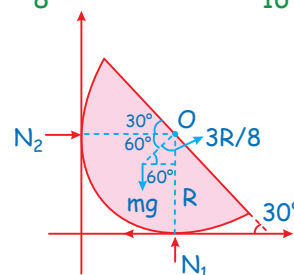
$$= \frac{mR^2\omega_0^2}{9}$$

Q. A solid hemisphere rests in equilibrium on a rough ground and against a smooth wall. The curved surface touches the wall and the ground. The angle of inclination of the circular base to the horizontal is 30° , the minimum coefficient of friction required between ground and hemisphere is μ . Find the value of 32μ .



Sol. [6.00] Torque about O,

$$fR = mg \times \frac{3R}{8} \cos 60^\circ \Rightarrow f = \frac{3mg}{16}$$



$$N_2 = f$$

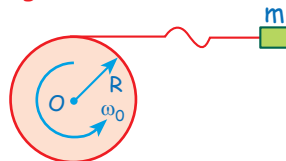
$$N_1 = mg$$

$$f = \frac{3mg}{16}$$

$$f \leq \mu N_1$$

$$\mu \geq \frac{3}{16} \Rightarrow 32\mu = 6$$

Q. Disc of mass $M = 10$ kg and radius $R = 50$ cm is revolving about fixed vertical axis on smooth horizontal surface with angular velocity $\omega_0 = 10$ rad/sec. A block of mass $m = 5$ kg is connected to disc through slack string as shown in the diagram. Find the angular velocity (in rad/s) when string become taut.



Sol. Conserving angular momentum about O

$$\frac{1}{2} \times M \times R^2 \omega_0 = \frac{1}{2} M R^2 \omega + m v R$$

$$v = R\omega$$

$$\Rightarrow \frac{1}{2} M \omega_0 = \left(\frac{1}{2} M + m \right) \omega$$

$$\omega = \frac{M \omega_0}{M + 2m} = \frac{10 \times 10}{10 + 10} = 5 \text{ rad/s}$$